WiL Proposal

Vision-Based Navigation Pipeline for AMRs in Dynamic Warehouses

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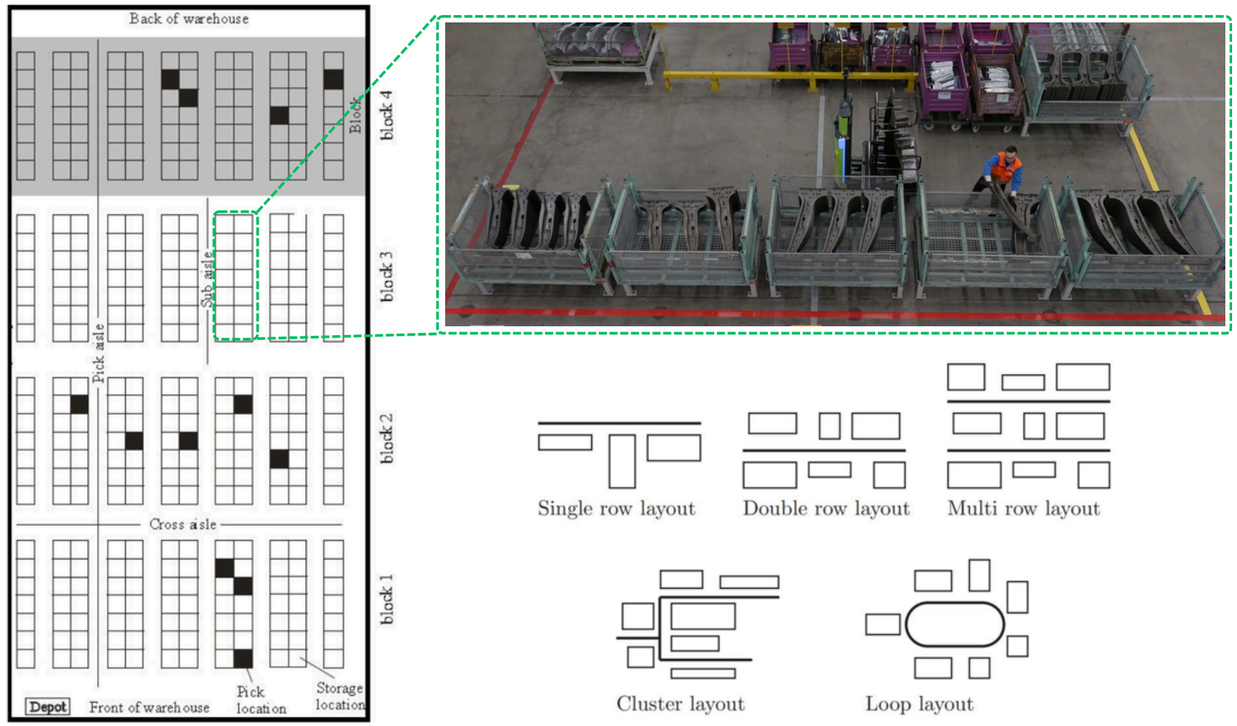
27 July 2026

1. Problem Statement

- **AMR for Warehouse Logistic :** Autonomous mobile robots (AMRs) have made a significant contribution to manufacturing specifically in warehousing improving performance and productivity.



Russell Keith, Hung Manh La; Review of Autonomous Mobile Robots for the Warehouse Environment; 2024,
https://www.press.bmwgroup.com/global/video/detail/PF0009726/autonomous-driverless-transport-vehicle-navigates-precisely-through-bmw-group-plant-regensburg-press-plant

1. Problem Statement

- **Identified Gensurv's issues**

Current Solutions

LMR : Latent Mobile Robot

↓

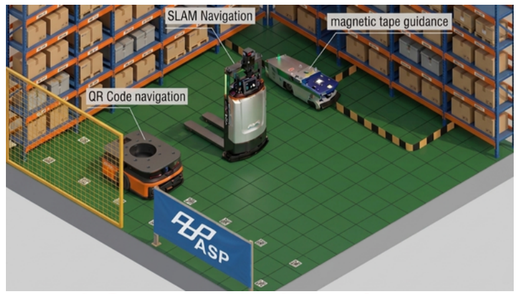
QR-navigated robot

PLY AGV Autonomous Tow truck

↓

2D LiDAR Navigation

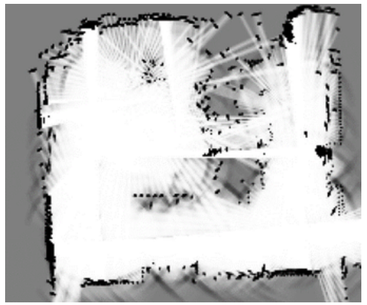
Limitations



Infrastructure dependence

Problem

High cost on changing warehouse infrastructure



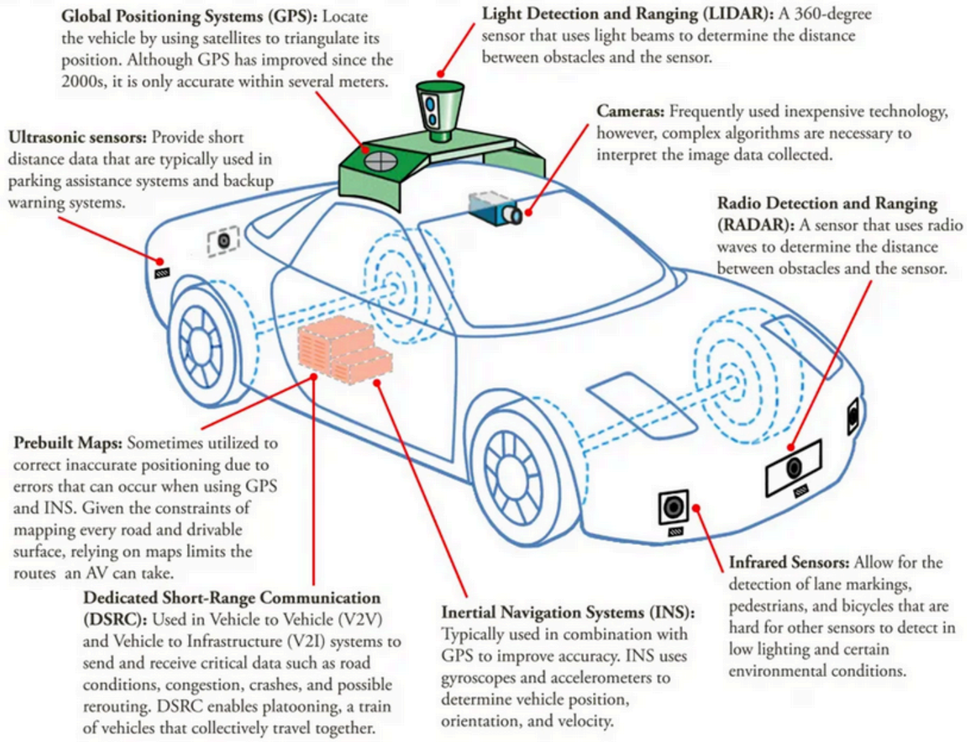
Low Robustness for different environment

https://gensurv.com/products-th/

1. Problem Statement

- **Why do we use camera-based SLAM ?**

Examples of Technologies Supporting Autonomous Vehicles



Source: Center for Sustainable Systems, University of Michigan (2024)

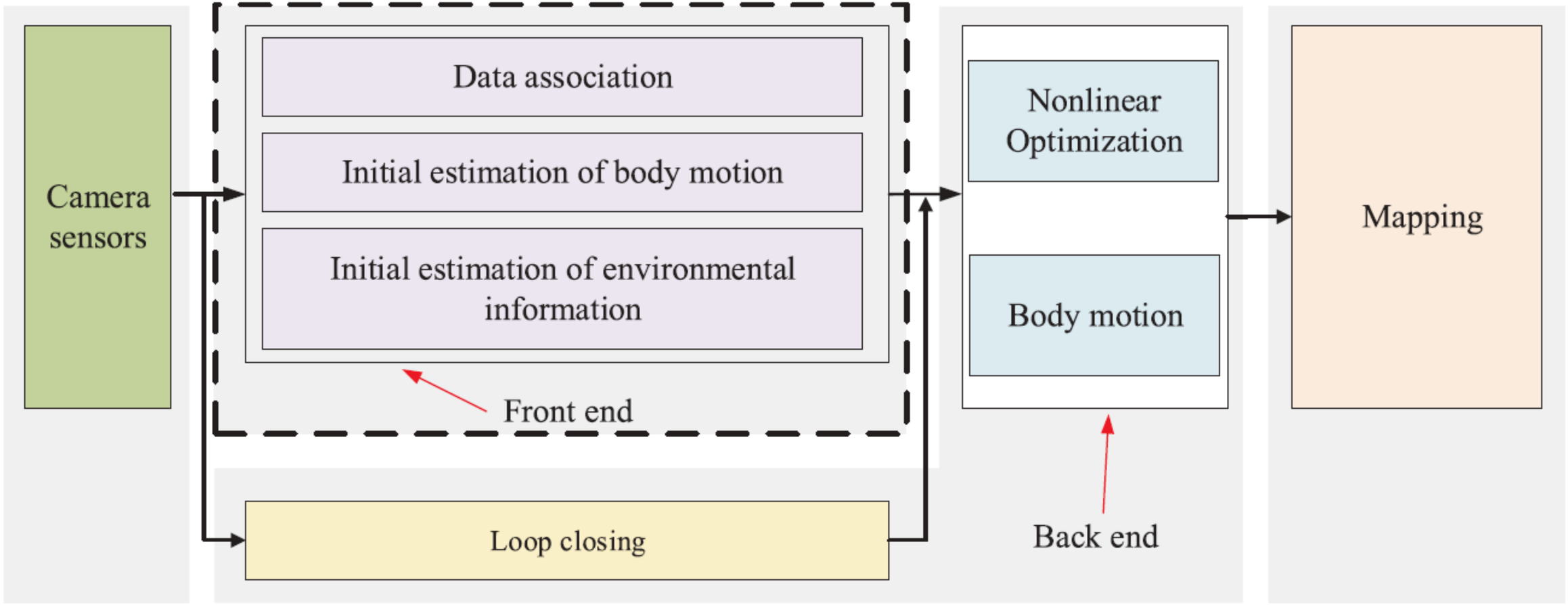
Modality	Affected by Illumination	Affected by weather	Color	Depth	Range	Accuracy	Size	Cost
Lidar	-	✓	-	✓	medium (< 200m)	high	large*	high*
Radar	-	-	-	✓	high	medium	small	medium
Ultrasonic	-	-	-	✓	short	low	small	low
Camera	✓	✓	✓	-	-	-	smallest	lowest
Stereo Camera	✓	✓	✓	✓	medium (< 100m)	low	medium	low
Flash Camera [77]	✓	✓	✓	✓	medium (< 100m)	low	medium	low
Event Camera [78]	limited	✓	-	-	-	-	smallest	low
Thermal Camera [79], [80]	-	✓	-	-	-	-	smallest	low

Sensor	Estimated Cost per unit (USD)
Lidar	500 - 75000
Camera	20 - 500 (required 8-12 cameras)

https://www.krungsri.com/en/research/research-intelligence/ai-autonomous-car-2024,
https://patentpc.com/blog/the-cost-of-self-driving-technology-how-much-do-av-components-really-cost-market-breakdown

1. Problem Statement

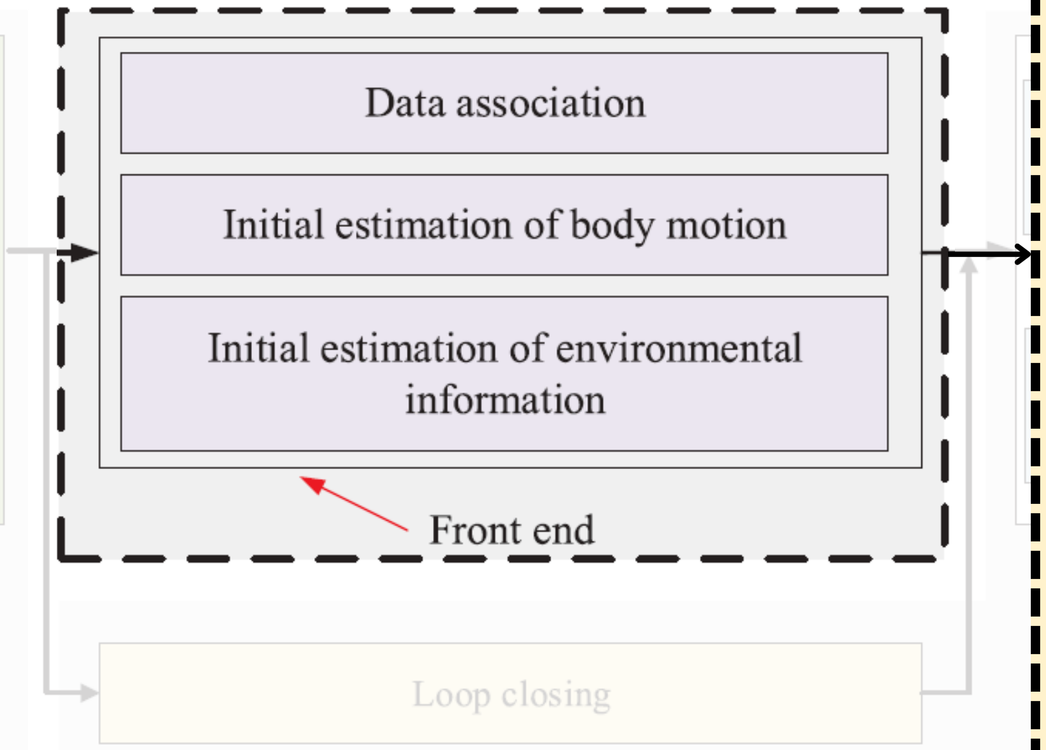
- **What is Visual SLAM ? :** Visual SLAM (Simultaneous Localization and Mapping) is the process where a robot or autonomous vehicle uses a camera to figure out its location in an unknown environment while simultaneously building a map of that environment.



Ke Wang , Guoliang Zhao, and Jianbo Lu ; A Deep Analysis of Visual SLAM Methods for Highly Automated and Autonomous Vehicles in Complex Urban Environment ; 2024

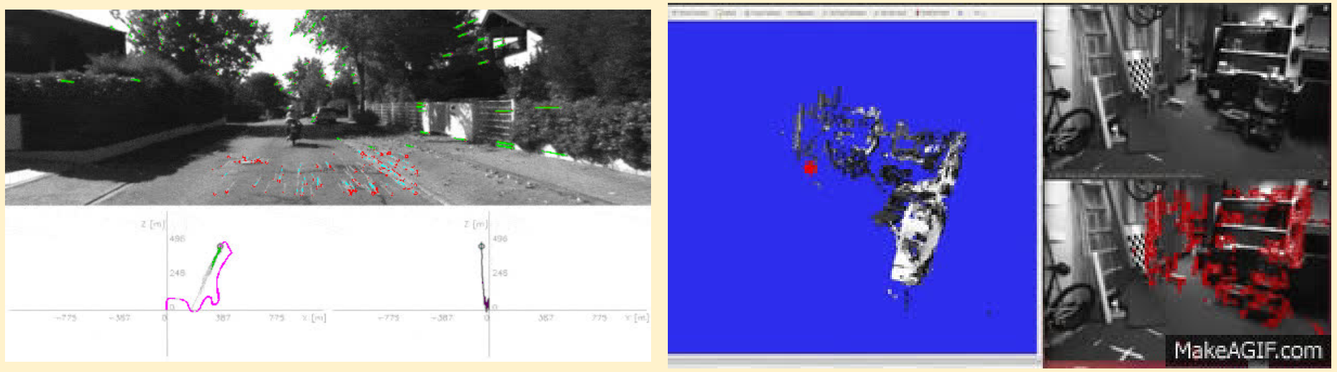
1. Problem Statement

- **What is Visual SLAM ? :**



Frontend Visual Odometry

- It estimates camera motion and feature orientation based on adjacent frames.
- VO methods are mainly divided into feature-based and direct/semi-direct approaches.
  - **Feature Method** : instead of looking at every pixel, the front-end extracts a sparse set of 2D point features (such as corners) and matches their descriptors across consecutive camera frames
  - **Direct Method** : Bypass feature extraction and intermediate abstractions entirely, computing localization and mapping directly from the raw sensory data



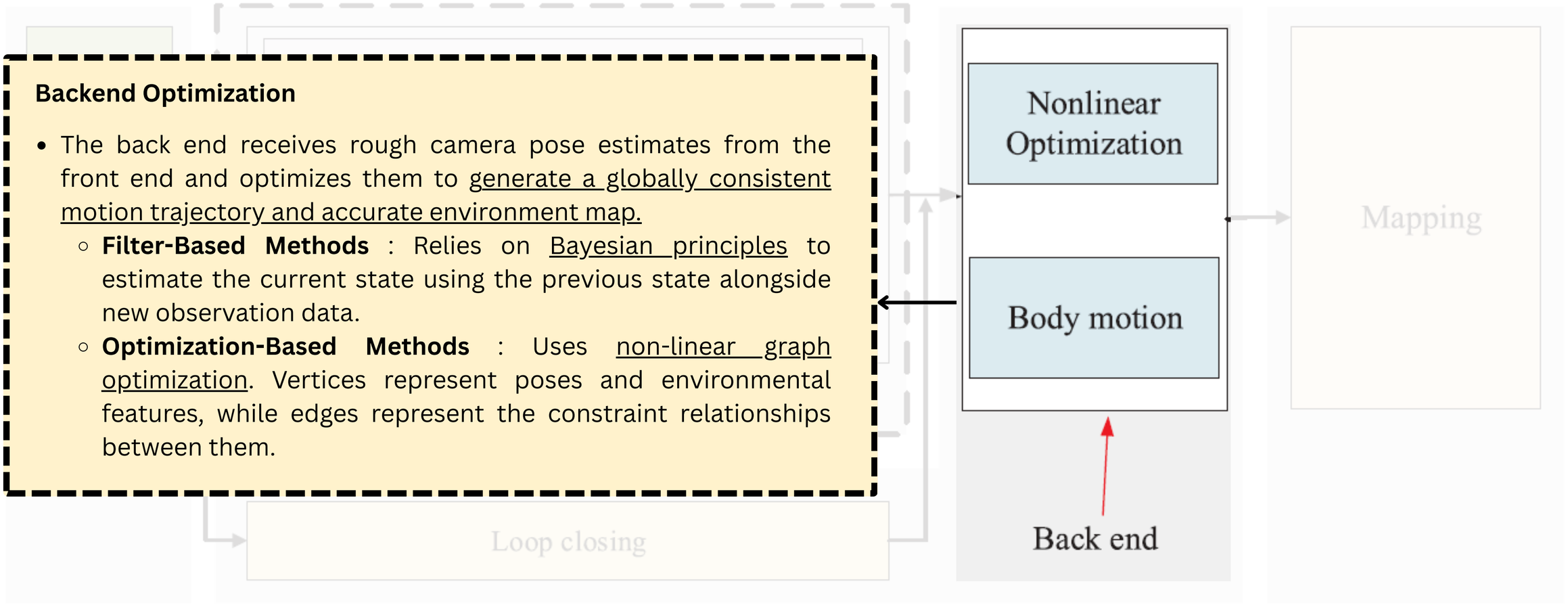
Ke Wang , Guoliang Zhao, and Jianbo Lu ; A Deep Analysis of Visual SLAM Methods for Highly Automated and Autonomous Vehicles in Complex Urban Environment ; 2024

1. Problem Statement

- **What is Visual SLAM ? :**

Backend Optimization

- The back end receives rough camera pose estimates from the front end and optimizes them to generate a globally consistent motion trajectory and accurate environment map.
  - **Filter-Based Methods** : Relies on Bayesian principles to estimate the current state using the previous state alongside new observation data.
  - **Optimization-Based Methods** : Uses non-linear graph optimization. Vertices represent poses and environmental features, while edges represent the constraint relationships between them.



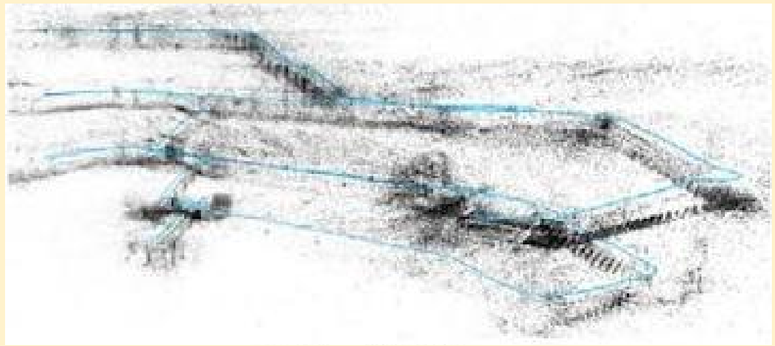
Ke Wang , Guoliang Zhao, and Jianbo Lu ; A Deep Analysis of Visual SLAM Methods for Highly Automated and Autonomous Vehicles in Complex Urban Environment ; 2024

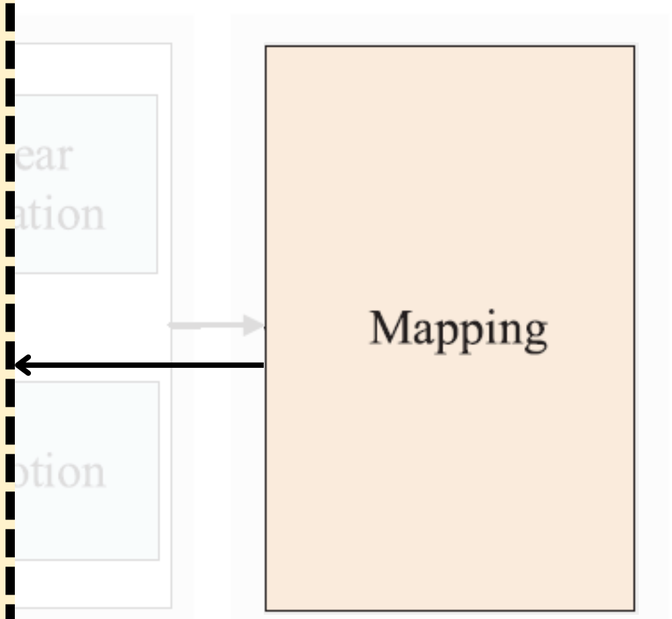
1. Problem Statement

- **What is Visual SLAM ? :**

Mapping

- Mapping is one of the two tasks of the visual SLAM system (i.e. localization and mapping), which plays an important role in navigation, obstacle avoidance and environmental reconstruction for autonomous driving.
  - **Metric Map** : describes the relative positional relationship between map elements
    - **Sparse Map** : Contain only a small amount of information in the scene which is suitable for positioning the ego.
    - **Dense Map** : Contain more information, which is beneficial for vehicles to perform navigation tasks according to the map.
  - **Topological Map** : Emphasize the connection relationship between map elements.





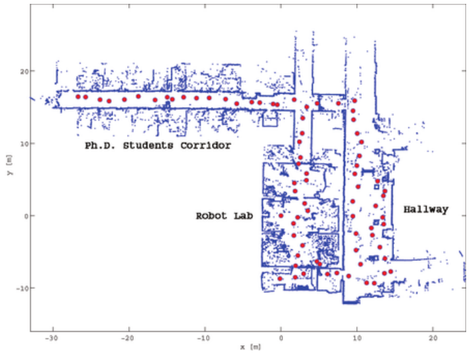
Ke Wang , Guoliang Zhao, and Jianbo Lu ; A Deep Analysis of Visual SLAM Methods for Highly Automated and Autonomous Vehicles in Complex Urban Environment ; 2024

1. Problem Statement

- **Indoor Environment Problem**

Sensor signal blocking

Close-Proximity Dynamic Objects

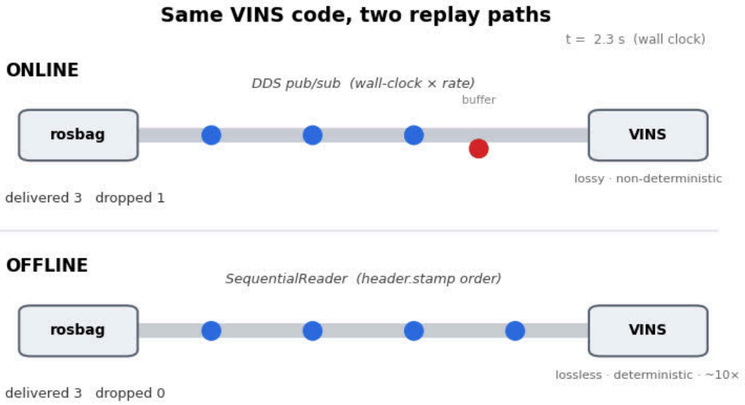


Repetitive Structures

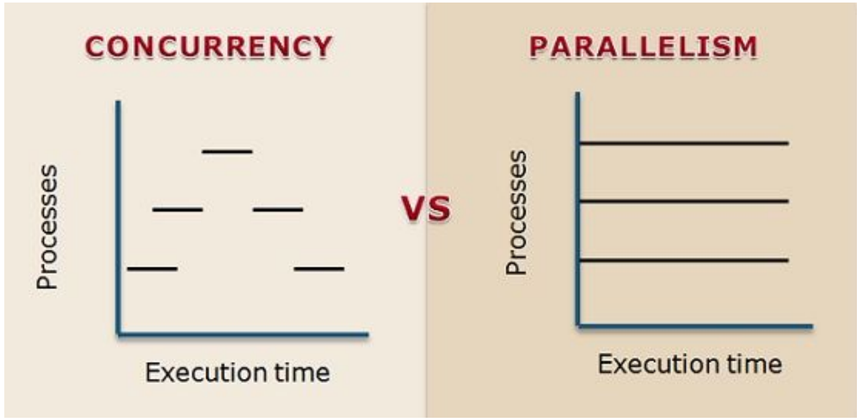
Balveer Singh; Puneet Kumar; Narinder Kaur; Advancements and Challenges in Visual-Only Simultaneous Localization and Mapping (V-SLAM):A Systematic Review; 2024

1. Problem Statement

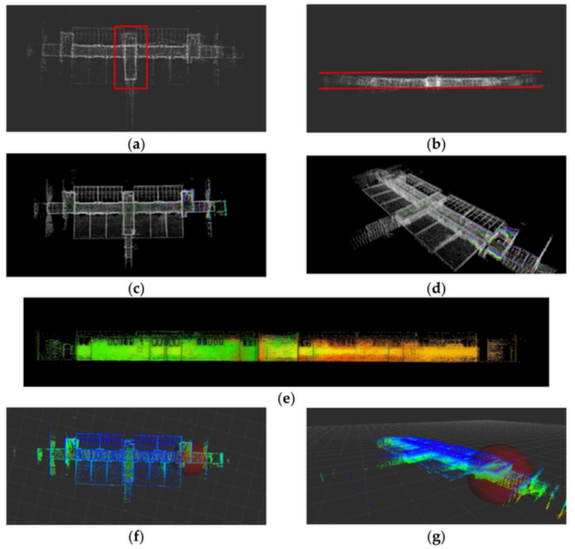
- **Implementation challenges**



Timeliness Challenge



The Concurrency Challenge



Context awareness Challenge

SOFIYA SEMENOVA; STEVEN KO; YU DAVID LIU; LUKASZ ZIAREK; KARTHIK DANTU; A Comprehensive Study of Systems Challenges in Visual Simultaneous Localization and Mapping Systems; 2024

2. Requirement and Scope

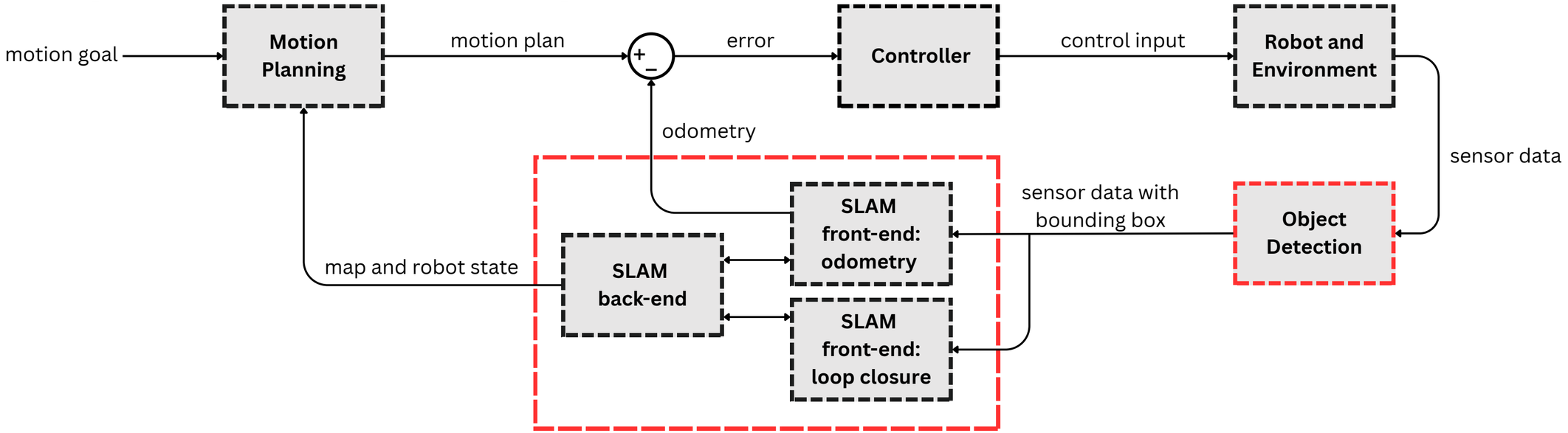
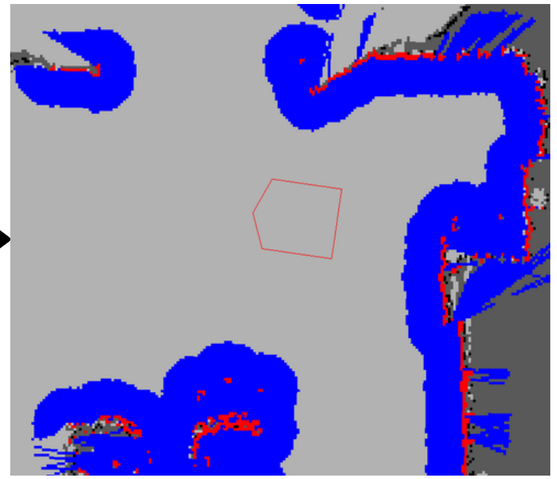


Image Preprocessing

- Stereo Calibration
- Light Enhancement
- Object Detection

Visual-Inertial Odometry

- Odometry (x,y,z,roll,pitch,yaw)
- Back-End Lifelong Mapping



2D costmap

2. Requirement and Scope

Requirement

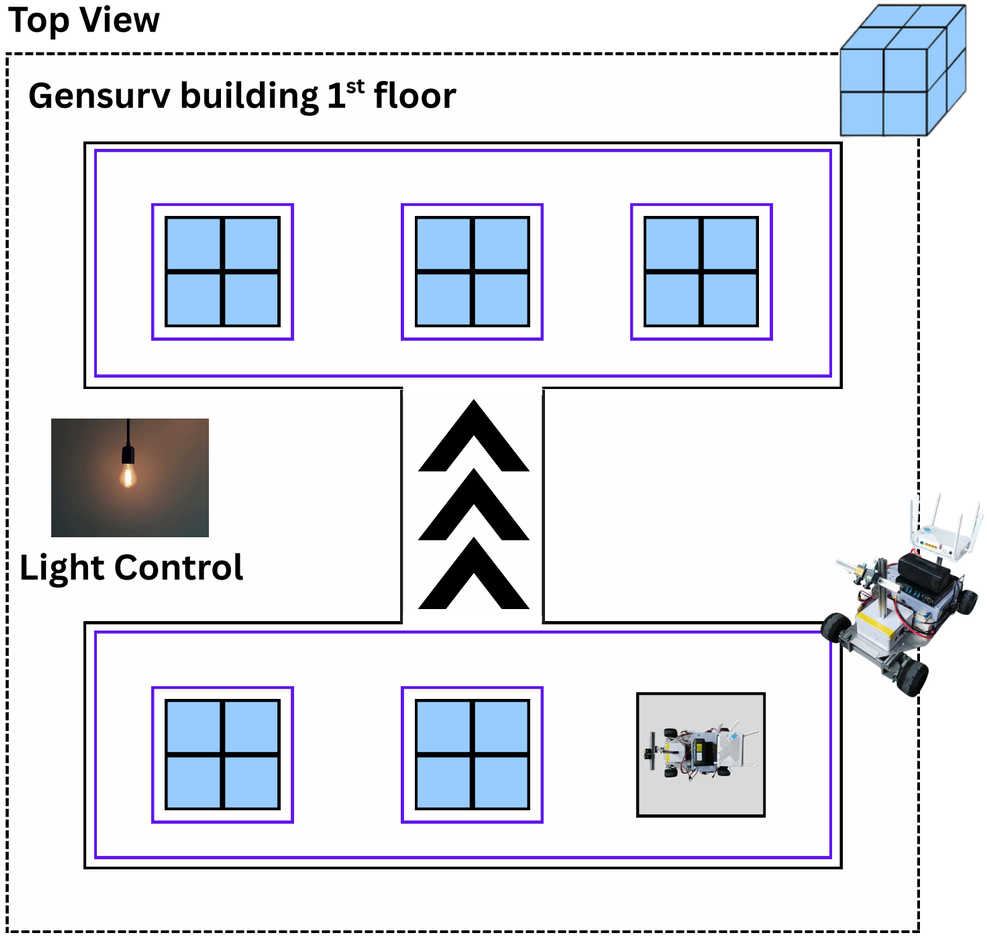
1. ทำการพัฒนาระบบ Visual-Inertial odometry โดยใช้ทั้ง Monocular-Inertial Odometry และ Stereo-Inertial Odometry
2. ทำการ Plugin Lifelong Mapping Algorithm สำหรับการอัปเดต Map อย่าง Real time
3. การรับรู้ Dynamic object จากการทำ Object detection
4. ทำการติดตั้ง Lane detection สำหรับ Feature Extraction ข้อมูลประเภทลวดลาย
5. ทำการทดสอบหลักผ่านหุ่นยนต์ Mobile robot ขนาดเล็กที่ประมวลผลด้วย NRU-51V และใช้ camera รุ่น AC-IMX390-H190 ในตึกสถาบันนวัตกรรมหุ่นยนต์และยานยนต์อัตโนมัติ (สนอ.) ชั้น 1
6. ทดสอบประสิทธิภาพการทำ Localization ผ่าน Dataset ในอินเตอร์เน็ต โดยมีความคลาดเคลื่อนน้อยกว่า 5 เซนติเมตร
7. Integrated system สามารถระบุตำแหน่งหุ่นยนต์ โดยมีความคลาดเคลื่อนน้อยกว่า 5 เซนติเมตร ในสภาพแวดล้อมที่มีความเข้มแสงแตกต่างกันได้
8. Integrated system สามารถสร้าง 2D cost map ที่มีความคลาดเคลื่อนน้อยกว่า 5 เซนติเมตร ในสภาพแวดล้อมที่มีความเข้มแสงแตกต่างกันได้

https://www.qmhinc.com/warehouse-loading-dock-ramps/?srsltid=AfmBOoqxYd7chZEsEL70iDazLzT-uJruQZmu-bQhUn_PdQjpCNHRp35P

2. Requirement and Scope

Scope

1. ทำการทดสอบหลักผ่านหุ่นยนต์ Mobile robot ขนาดเล็กที่ประมวลผลด้วยหน่วยประมวลผล NRU-51V และใช้กล้องรุ่น AC-IMX390-H190) เท่านั้น
2. ใน AMR navigation system นั้น ผู้จัดทำได้วางกรอบการทำงานไว้ในส่วนของ Visual SLAM, Object prediction และ Lane tracking controller (กรอบสีแดง) ดังรูปที่ 1 เท่านั้น
3. ทดสอบประสิทธิภาพของ Integrated system ของหุ่นยนต์ในสนามจำลอง Warehouse ที่มีการเปลี่ยนแปลงของพัสดุจำลอง, มีทางเดินที่มีความชัน 5 องศา, มีพัสดุ 5 กอง กองละ 4 กล่อง, มี Starting Point บริเวณขวาล่างที่เป็นสี่เหลี่ยมสีเทา ดังรูปที่ 2 เท่านั้น โดยตำแหน่งของคลังสินค้าจำลอง จะมีการเปลี่ยนแปลงระหว่างการทดสอบเพื่อจำลองสภาพแวดล้อมแบบ Dynamic
4. รถ mobile robot มีความเร็วในการเคลื่อนที่สูงสุดที่ 1.5 m/s



https://www.qmhinc.com/warehouse-loading-dock-ramps/?srsltid=AfmBOoqxYd7chZEsEL70iDazLzT-uJruQZmu-bQhUn_PdQjpCNHRp35P

3. Literature Review

Problem : Dynamics environment

- Survey paper of Structure from Motion

Solution : Semantic Understanding

- Survey paper of semantic visual SLAM
- Survey paper of object detection model

Problem : Memory & Computational Explosion

- Future direction of visual SLAM
- Visual SLAM comparison

Solution :

Visual Odometry	Mapping Back-End
• VINS-fusion	• RTAB-MAP
• ORB-SLAM3	

3. Literature Review

Paper: Semantic SLAM: A comprehensive survey of methods and applications

- Improved localization is achieved as semantic information helps in more robust data association and loop closure detection, reducing drift, and improving the accuracy of the SLAM solution.

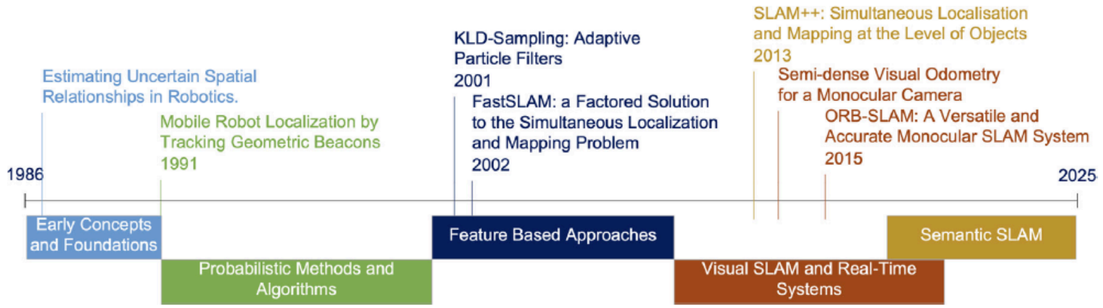
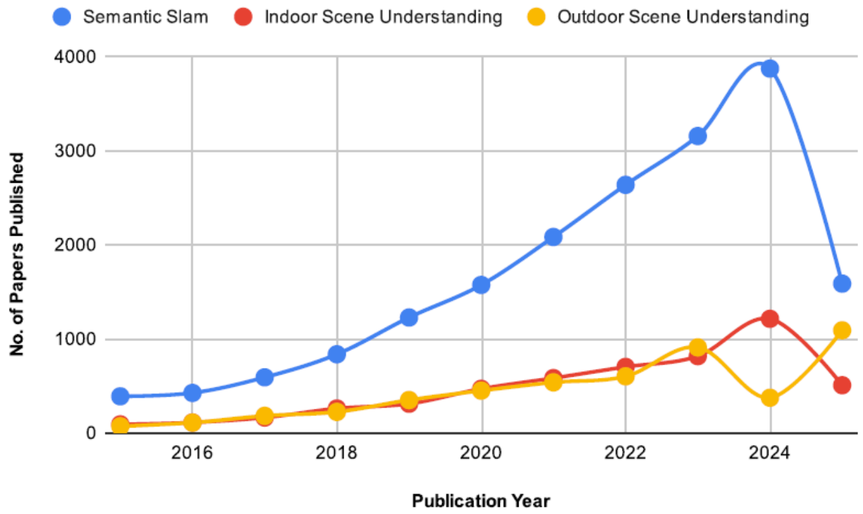


Fig. 2. History and evolution of SLAM over the years from 1986 to 2025.



Sequence	ORB-SLAM [1]		DynaSLAM (Monocular)	
	ATE [m]	% Traj	ATE [m]	% Traj
fr3/walking_halfsphere	**0.017**	87.16	0.021	**97.84**
fr3/walking_xyz	**0.012**	57.63	0.014	**87.37**
fr2/desk_with_person	**0.006**	95.30	0.008	**97.07**
fr3/sitting_xyz	**0.007**	91.44	0.013	**100.00**

Berta Bescos, Jos´e M. F´acil, Javier Civera and Jos´e Neira ;DynaSLAM: Tracking, Mapping and Inpainting in Dynamic Scenes ; 2018
Houssein Kanso, Abhilasha Singh, Etaf El Zarif ;Semantic SLAM: A comprehensive survey of methods and applications ; 2025

3. Literature Review

Paper: A Comprehensive Survey of Machine Learning Techniques and Models for Object Detection

- **Problem Statement:** The paper identifies the core problem as the difficulty of maintaining accurate and reliable detection under complex conditions. Specifically, systems struggle with managing occlusions (partially hidden objects), varying object scales, and the intense demand for real-time processing.

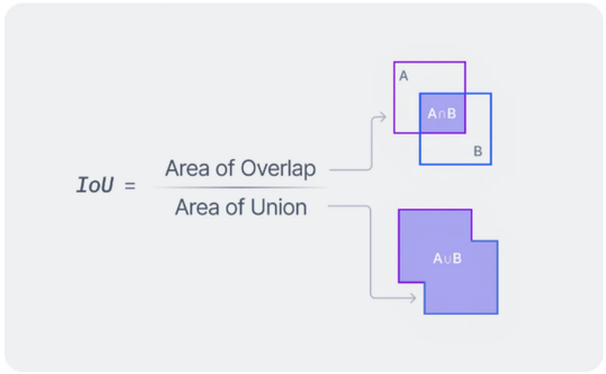
Model Architecture	Category	Key Strengths	Limitations	Ideal Use Cases
Faster R-CNN	Two-Stage Detector	Excels in classification and localization precision, achieving high mAP and IoU. Demonstrates lower overall detection error rates and high spatial precision.	Its multi-stage pipeline results in slower inference speeds.	High-accuracy scenarios where real-time processing is not a strict requirement.
YOLO Series	Single-Stage Detector	Reframes detection as a single regression problem, explicitly optimizing for speed. Balances speed and accuracy effectively, yielding competitive precision, recall, and a lower False Positive Per Image (FPPI) rate.	While highly optimized now, earlier versions struggled with small objects and dense	Real-time tasks where fast processing and low latency are critical.
SSD (Single Shot Detectors)	Single-Stage Detector	Eliminates the need for separate region proposal stages and optimizes multi-scale detection within a single, unified network.	Can sometimes lag behind YOLO in pure speed, though it remains highly efficient.	Applications requiring a very strong, reliable balance of computational efficiency and accuracy.
CenterNet / CornerNet	Anchor-Free Detector	Bypasses the complexity and inefficiency of predefined anchor boxes by predicting key points (like centers or corners) directly.		Scenarios requiring robust detection of objects with varying scales, dense object clusters, and non-standard shapes.
MobileNet / EfficientNet	Lightweight Model	Utilizes optimization techniques like depthwise separable convolutions to vastly reduce parameters and computations without sacrificing representational power.	Raw accuracy may be lower than heavy architectures like Faster R-CNN or large YOLO variants.	Edge devices with limited memory and processing power, such as mobile phones, IoT devices, and autonomous drones

Maria Trigka and Elias Dritsas ; A Comprehensive Survey of Machine Learning Techniques and Models for Object Detection ; 2025

3. Literature Review

Paper: A Comprehensive Survey of Machine Learning Techniques and Models for Object Detection

- **Evaluation Metrics:** The survey details several quantitative metrics crucial for assessing model effectiveness in classification and localization.

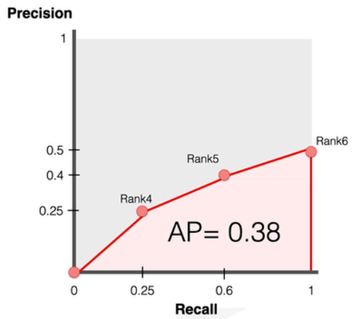


- **Intersection over Union (IoU) :** $IoU = \frac{|B_p \cap B_{gt}|}{|B_n \cup B_{at}|}$

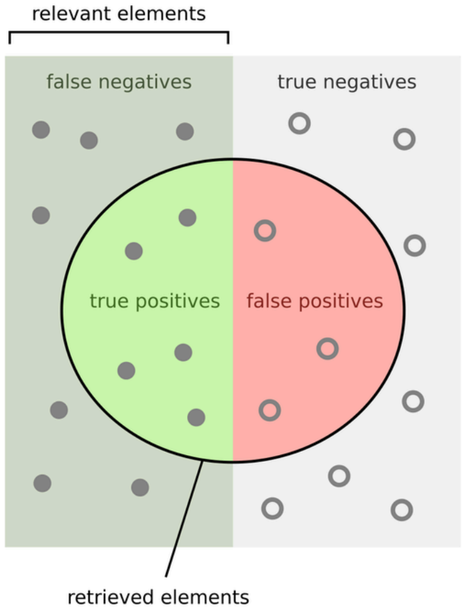
	Precision @k	Recall @k
1	0	0
2	0	0
3	0	0
4	0.25	0.25
5	0.4	0.6
6	0.5	1

AP = 1.15/3 = 0.38

Not relevant Relevant



- **Average Precision (AP) :** $mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i$



- **Precision :** $Precision = \frac{TP}{TP+FP}$
- **Recall :** $Recall = \frac{TP}{TP+FN}$
- **F1 Score :** $F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision+Recall}$

How many retrieved items are relevant?

Precision =

How many relevant items are retrieved?

Recall =

Maria Trigka and Elias Dritsas ; A Comprehensive Survey of Machine Learning Techniques and Models for Object Detection ; 2025

3. Literature Review

Article : Best Object Detection Models in 2026

- Based on the Roboflow article, the 2026 object detection landscape is defined by a split between high-performance supervised models and highly flexible zero-shot models.
- Result : coco dataset

Model	Category	Accuracy (mAP / AP)	Speed / Latency
YOLO26	Supervised	57.50%	~11.8ms (588 FPS)
RF-DETR (Medium)	Supervised	54.70%	~4.52 ms (~220 FPS)
YOLOv12 (X)	Supervised	55.20%	~11.79 ms (~85 FPS)
YOLOv12 (S)	Supervised	52.50%	~4.86 ms (~205 FPS)
RTMDet (Extra-Large)	Supervised	52.80%	300+ FPS
RTMDet (Tiny)	Supervised	40.50%	1,020+ FPS

https://blog.roboflow.com/best-object-detection-models/

3. Literature Review

Paper: A review of visual SLAM methods for autonomous driving vehicles

Problem Statement :

- The development of Visual-Inertial (VI) SLAM stems from the inherent limitations of using pure visual SLAM or Inertial Measurement Units (IMUs) independently. These are state of the art VI SLAM methods over the last decade.

Framework	Feature	Sensor	Pros	Cons	Year
OKVIS	BRISK	Monocular	Lightweight	No loop closing and map	2014
		Binocular	Higher accuracy	Complex configuration	
VI-ORBSLAM	ORB	Monocular	High accuracy with map reuse	Poor IMU initialization	2017
VI-DSO	High grad	Monocular	High accuracy and good robustness under weak texture	Not easy to maintain the global map	2018
VINS-Mono	Shi Tomasi	Monocular	Good real-time performance	Poor robustness during fast rotation	2018
VINS-Fusion	Shi Tomasi	Monocular	High accuracy with GPS integrated	Time consuming	2019
		Binocular	Low occupancy	Complex configuration	
		Fisheye	Wide field of view	High occupancy	
BASALT	FAST	Binocular	High accuracy with globally consistent keyframe-based BA	Time consuming	2020
Kimera	Shi Tomasi	Binocular	Good real-time performance with 3D metric-Semantic mesh	Poor robustness	2020
ORB-SLAM3	ORB	Binocular	long term navigation	Time consuming	2020
		Fisheye	Wide field of view	High occupancy	

Jun Cheng , Liyan Zhang , Qihong Chen ; A review of visual SLAM methods for autonomous driving vehicles ; 2024

3. Literature Review

Paper: A review of visual SLAM methods for autonomous driving vehicles

Result :

- The quantitative RMSE of the pose (translation and orientation) and Frame rate tested in the all sequences of EuRoC dataset.

Seq.	VI-ORBSLAM			VINS-MONO			ORB-SLAM3		
	Translation. (m)	Orientation (deg)	Frequency (fps)	Translation. (m)	Orientation (deg)	Frequency (fps)	Translation. (m)	Orientation (deg)	Frequency (fps)
MH_01	0.075	2.251	140	0.084	2.31	**275**	**0.042**	**2.155**	145
MH_02	0.084	**2.523**	136	0.105	2.753	**275**	**0.053**	3.043	137
MH_03	0.087	**1.353**	134	0.074	1.262	**274**	**0.043**	1.935	140
MH_04	0.217	1.306	122	0.122	1.246	**272**	**0.099**	**1.106**	133
MH_05	0.082	0.559	138	0.147	0.395	**266**	**0.071**	**0.529**	136
V1_01	**0.027**	**2.112**	123	0.047	2.733	**268**	0.043	2.422	140
V1_02	0.028	1.692	131	0.066	1.732	**269**	**0.016**	**1.673**	134
V1_03	x	x	119	0.18	4.598	**279**	**0.025**	**2.632**	122
V2_01	0.053	1.326	114	0.058	1.525	**263**	**0.021**	**1.126**	129
V2_02	0.041	2.58	119	0.09	2.613	**273**	**0.015**	**1.18**	131
V2_03	0.074	**4.012**	111	0.244	5.532	**263**	**0.037**	4.362	124
Ave.	0.077	**1.971**	126.1	0.111	2.427	**270.6**	**0.042**	2.015	133.7

Jun Cheng , Liyan Zhang , Qihong Chen ; A review of visual SLAM methods for autonomous driving vehicles ; 2024

3. Literature Review

Paper: A Comprehensive Study of Systems Challenges in Visual Simultaneous Localization and Mapping Systems

- **Visual SLAM system comparison :** This paper focuses on how timeliness, concurrency, and context awareness impact performance.

- **Definition :**
  - **Timeliness :** SLAM system must generate mapping and localization information fast enough to support real-time applications such as avoiding physical obstacles during navigation.
  - **Concurrency** : The challenge of coordinating multiple parallel threads that are executing fundamentally sequential tasks at the same time.
  - **Context Awareness** : The ability of a SLAM system to understand its physical surroundings and internal state, allowing it to dynamically adjust which tasks are prioritized.

System	Latency	Robustness to tracking loss	Accuracy (best case)	Accuracy (difficult sequences)	Accuracy (resource-constrained)
ORB-SLAM3	High	High	Best	Best	Worst
Kimera	Medium	Low	Worst	Average	Average
OpenVINS	Low	Low	Average	Worst	Best

SOFIYA SEMENOVA; STEVEN KO; YU DAVID LIU; LUKASZ ZIAREK; KARTHIK DANTU; A Comprehensive Study of Systems Challenges in Visual Simultaneous Localization and Mapping Systems; 2024

3. Literature Review

Paper: A Comprehensive Study of Systems Challenges in Visual Simultaneous Localization and Mapping Systems

- **Timely Feature Tracking (Latency)**
  - The results lists the average processing duration (in milliseconds) and throughput of individual pipeline modules (Tracking, Local Mapping, and Loop Closing) under real-time (20 fps) and resource-ample (0.5 fps) settings on both devices

System	Latency	Robust tracki
ORB-SLAM3	High	H
Kimera	Medium	L
OpenVINS	Low	L

OpenVINS

	Jetson, 20 fps					
	4 OpenCV threads			1 OpenCV thread		
Module	Input Frequency	Throughput	Duration	Input Frequency	Throughput	Duration
Tracking	20	–	14.51 ± 3.37	20	–	14.38 ± 3.74
Propagation	–	–	0.75 ± 0.23	–	–	0.74 ± 0.20
MSCKF Update	–	–	4.08 ± 4.98	–	–	4.01 ± 4.29
SLAM Update	–	–	18.00 ± 14.42	–	–	18.07 ± 13.95
Re-tri & Marg	–	16.12 ± 0.17	15.51 ± 0.56	–	16.30 ± 0.06	15.22 ± 0.58
TOTAL	20	16.12 ± 0.17	52.85 ± 16.04	20	16.30 ± 0.06	52.42 ± 15.58

ORB-SLAM3

	Jetson, 20 fps			Laptop, 20 fps		
Module	Input Freq.	Throughput	Duration	Input Freq.	Throughput	Duration
Tracking	20.00 ± 0.00	9.64 ± 0.54	92.95 ± 15.23	20.00 ± 0.00	17.92 ± 4.01	46.07 ± 17.32
Local Mapping	2.16 ± 0.09	2.11 ± 0.08	393.21 ± 183.12	3.31 ± 0.49	3.31 ± 0.50	226.51 ± 90.57
Loop Closing	2.16 ± 0.09	2.10 ± 0.08	18.76 ± 28.97	3.31 ± 0.49	3.30 ± 0.51	8.79 ± 10.42
GBA	0.00 ± 0.01	0.00 ± 0.01	2,421.77 ± 249.01	0.01 ± 0.00	0.01 ± 0.00	746.68 ± 84.35

Kimera

	Jetson, 20 fps			Laptop, 20 fps		
Module	Input Freq.	Throughput	Duration	Input Freq.	Throughput	Duration
VIO Frontend	20.00 ± 0.00	15.16 ± 0.11	65.12 ± 75.48	20.00 ± 0.00	20.00 ± 0.00	20.05 ± 27.45
VIO Backend	3.79 ± 0.03	3.78 ± 0.03	138.60 ± 38.67	5.00 ± 0.00	5.00 ± 0.01	25.99 ± 7.73
RPGO	3.26 ± 0.13	3.25 ± 0.12	307.52 ± 110.70	5.00 ± 0.00	5.00 ± 0.00	72.11 ± 26.70

SOFIYA SEMENOVA; STEVEN KO; YU DAVID LIU; LUKASZ ZIAREK; KARTHIK DANTU; A Comprehensive Study of Systems Challenges in Visual Simultaneous Localization and Mapping Systems; 2024

3. Literature Review

Paper: A Comprehensive Study of Systems Challenges in Visual Simultaneous Localization and Mapping Systems

- **Concurrency**
  - The EuRoC dataset provides an excellent sandbox for evaluating how SLAM systems handle environmental complexity because its indoor runs are strictly classified by difficulty. Comparing V101 (Vicon Room 1 01 - "Easy") and V103 (Vicon Room 1 03 - "Difficult") reveals how real-world conditions cascade into severe systems failures.

System	Latency	Robustness to tracking loss
ORB-SLAM3	High	High
Kimera	Medium	Low
OpenVINS	Low	Low

ORB SLAM3

Sequence	Experiment	Tracked Images	Created KeyFrames	Dropped KeyFrames	ATE (m)
V101	Laptop, 20 fps	2,824 ± 32	391 ± 1	1,871 ± 45	0.03 ± 0.00
	Jetson, 20 fps	1,160 ± 13	280 ± 1	500 ± 34	0.07 ± 0.06
	Laptop, 0.5 fps	2,911	560	0	0.03
	Jetson, 0.5 fps	2,911	544	0	0.03
V103	Laptop, 20 fps	2,145 ± 2	408 ± 1	1,335 ± 14	0.02 ± 0.00
	Jetson, 20 fps	1,121 ± 43	236 ± 11	494 ± 112	0.75 ± 1.33
	Laptop, 0.5 fps	2,149	1,014	0	0.02
	Jetson, 0.5 fps	2,149	1,009	2	0.02

Kimera

Sequence	Experiment	Tracked Images	Created KeyFrames	ATE VIO (m)	ATE Final (m)
V101	Laptop, 20 fps	2,910 ± 1	727 ± 0	0.07 ± 0.01	0.06 ± 0.03
	Jetson, 20 fps	2,460 ± 210	615 ± 52	0.05 ± 0.01	0.06 ± 0.02
	Laptop, 0.5 fps	2,910	727	0.07	0.04
	Jetson, 0.5 fps	2,910	727	0.08	0.06
V103	Laptop, 20 fps	2,146 ± 0	536 ± 0	0.16 ± 0.00	0.12 ± 0.03
	Jetson, 20 fps	1,678 ± 6	419 ± 2	0.19 ± 0.01	0.18 ± 0.01
	Laptop, 0.5 fps	2,143	536	0.16	0.09
	Jetson, 0.5 fps	2,147	536	0.20	0.11

OpenVINS

Sequence	Experiment	Tracked Images	MSCKF Features	SLAM Features	ATE (m)
V101	Laptop, 20 fps	2,912 ± 1	21.42 ± 31.37	44.00 ± 6.48	0.05 ± 0.00
	Jetson, 20 fps	2,080 ± 12	23.95 ± 26.43	39.05 ± 9.13	0.06 ± 0.01
	Laptop, 0.5 fps	2,912	18.24 ± 17.89	44.29 ± 5.96	0.05
	Jetson, 0.5 fps	2,912	18.00 ± 18.03	44.16 ± 6.59	0.05
V103	Laptop, 20 fps	2,149 ± 1	40.35 ± 71.17	25.91 ± 13.86	0.07 ± 0.01
	Jetson, 20 fps	1,792 ± 10	35.34 ± 49.38	18.95 ± 13.00	0.06 ± 0.01
	Laptop, 0.5 fps	2,149	26.31 ± 20.20	26.06 ± 13.81	0.06
	Jetson, 0.5 fps	2,149	29.85 ± 30.16	25.90 ± 13.84	0.06

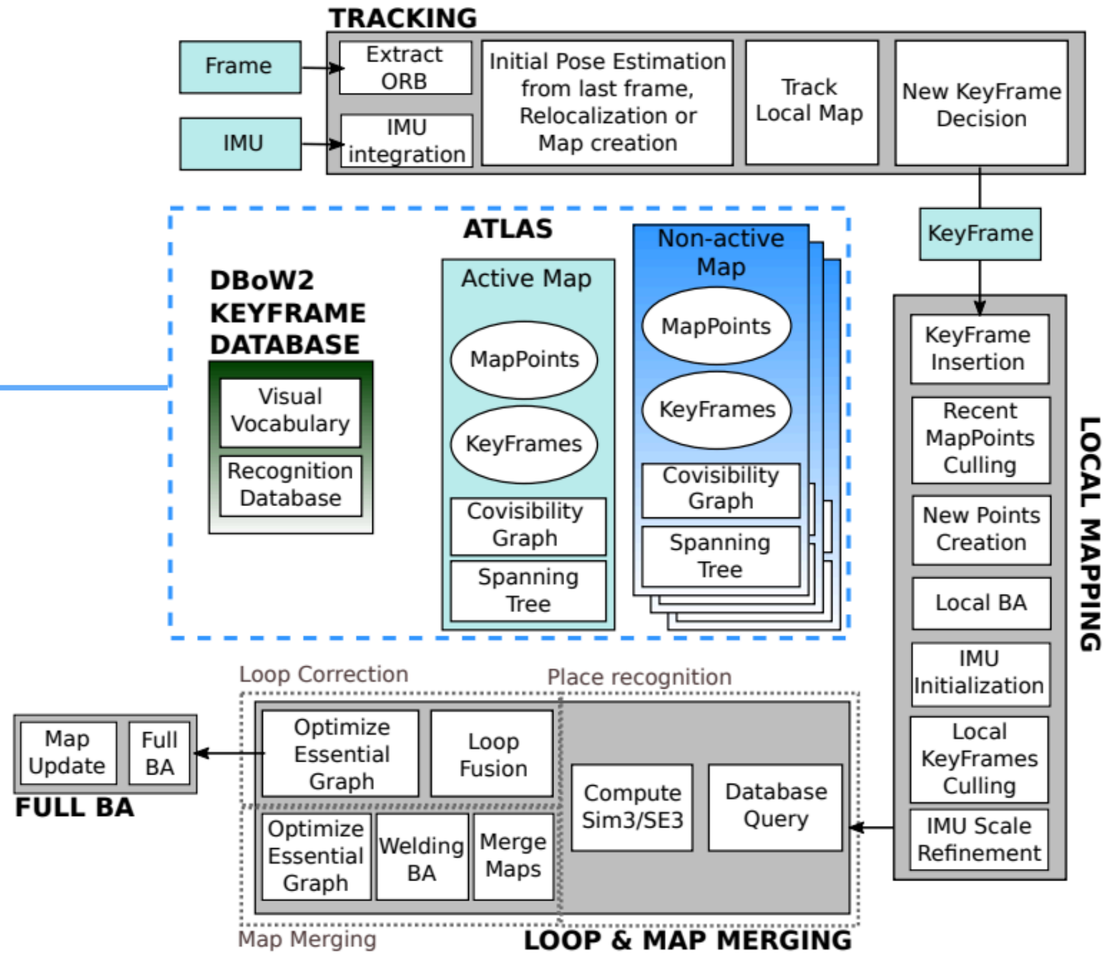
SOFIYA SEMENOVA; STEVEN KO; YU DAVID LIU; LUKASZ ZIAREK; KARTHIK DANTU; A Comprehensive Study of Systems Challenges in Visual Simultaneous Localization and Mapping Systems; 2024

3. Literature Review

Paper: ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial and Multi-Map SLAM

ORB-SLAM 3 : The first system able to perform visual, visual inertial and multi-map SLAM with monocular, stereo and RGB-D cameras, using pin-hole and fisheye lens models.

Atlas : A multi-map representation composed of a set of disconnected maps. There is an active map where the tracking thread localizes the incoming frames, and is continuously optimized and grown with new keyframes by the local mapping thread. the SLAM system can seamlessly switch its reference map depending on the current time of day, ensuring robust loop closures. (If an autonomous car maps a street at noon, its visual memory will match the street at midnight.)



Carlos Campos, Richard Elvira , Juan J. Gomez Rodrıguez; ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial and Multi-Map SLAM; 2021

3. Literature Review

Paper: ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial and Multi-Map SLAM

Result :

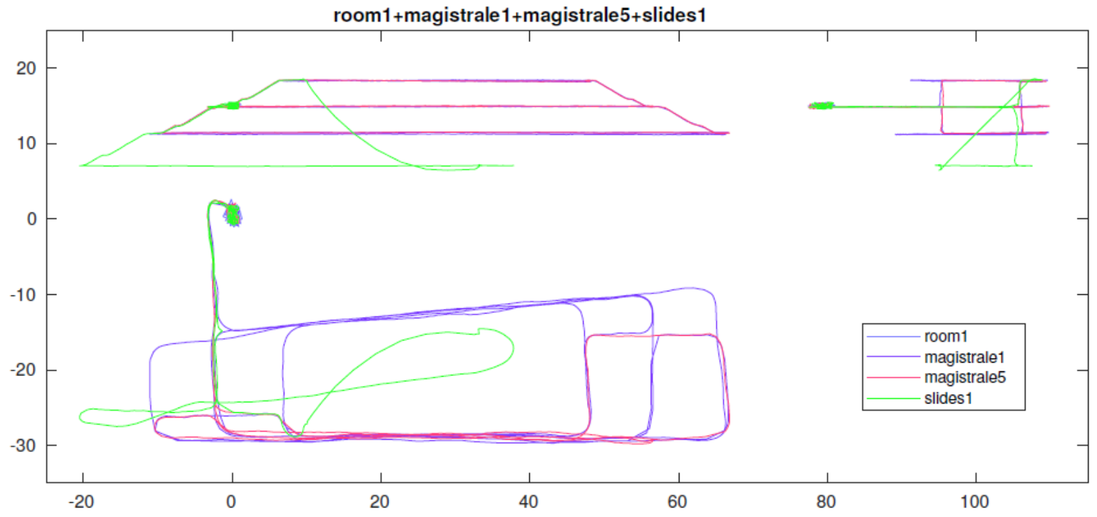


Figure 5: Multi-session stereo-inertial result with several sequences from TUM-VI dataset (front, side and top views).

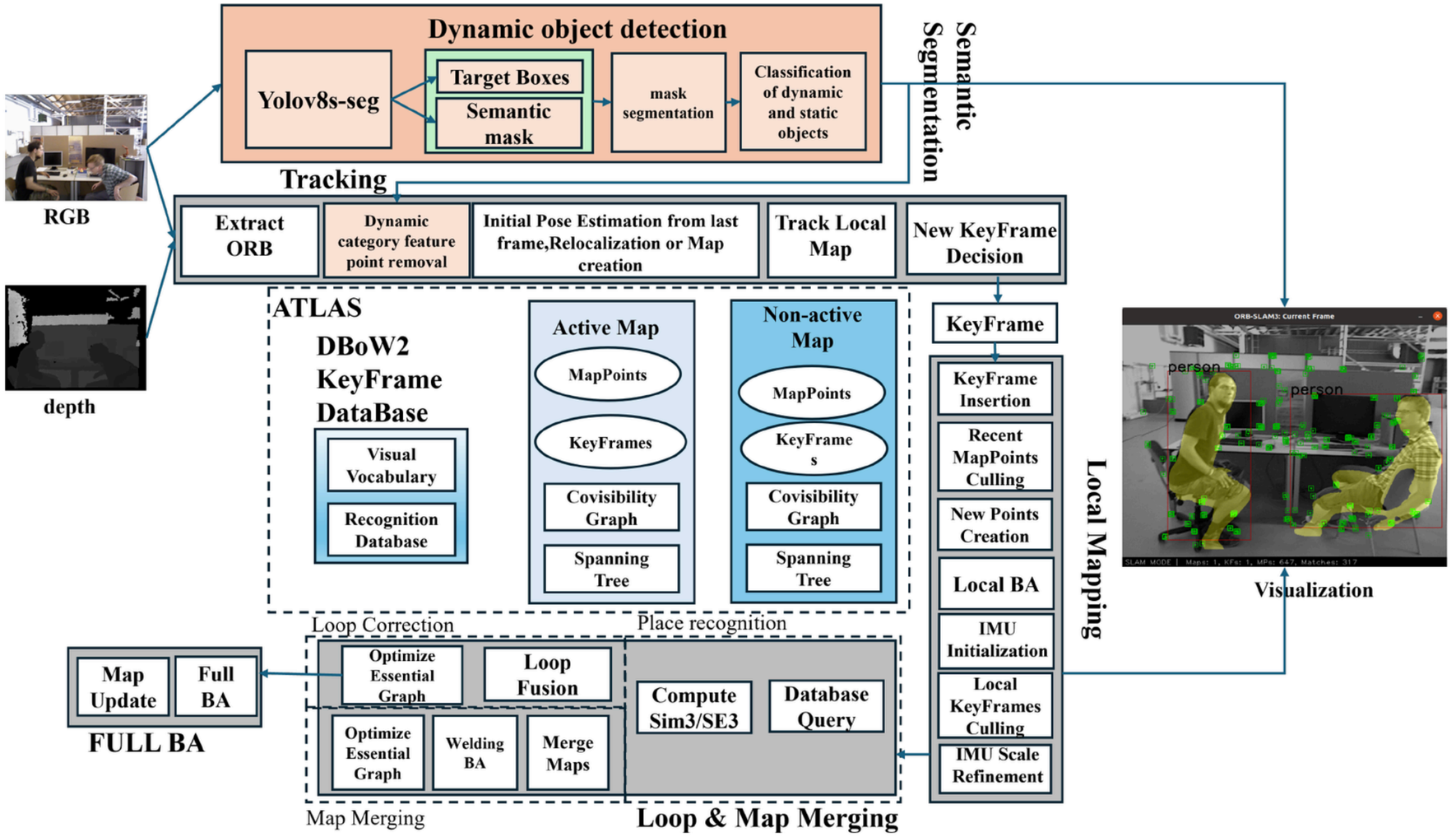
TUM-VI

	Seq.	Mono-Inertial		Stereo-Inertial				Length (m)	LC
		VINS-Mono	ORB-SLAM3	OKVIS	ROVIO	BASALT	ORB-SLAM3		
Indoor	corridor1	0.63	**0.04**	0.33	0.47	0.34	**0.03**	305	✓
	corridor2	0.95	**0.02**	0.47	0.75	0.42	**0.02**	322	✓
	corridor3	1.56	**0.31**	0.57	0.85	0.35	**0.02**	300	✓
	corridor4	0.25	**0.17**	0.26	**0.13**	0.21	0.21	114	
	corridor5	0.77	**0.03**	0.39	2.09	0.37	**0.01**	270	✓
Indoor	magistrale1	2.19	**0.56**	3.49	4.52	1.20	**0.24**	918	✓
	magistrale2	3.11	**0.52**	2.73	13.43	1.11	**0.52**	561	✓
	magistrale3	**0.40**	4.89	1.22	14.80	**0.74**	1.86	566	
	magistrale4	5.12	**0.13**	0.77	39.73	1.58	**0.16**	688	✓
	magistrale5	**0.85**	1.03	1.62	3.47	**0.60**	1.13	458	✓
	magistrale6	2.29	**1.30**	3.91	X	3.23	**0.97**	771	
Outdoor	outdoors1	74.96	**70.79**	X	101.95	255.04	**32.23**	2656	
	outdoors2	133.46	**14.98**	73.86	21.67	64.61	**10.42**	1601	
	outdoors3	**36.99**	39.63*	32.38	**26.10**	38.26	54.77	1531	
	outdoors4	**16.46**	25.26	19.51	X	17.53	**11.61**	928	
	outdoors5	130.63	**14.87**	13.12	54.32	**7.89**	8.95	1168	✓
	outdoors6	133.60	**16.84**	96.51	149.14	65.50	**10.70**	2045	
	outdoors7	21.90	**7.59**	13.61	49.01	**4.07**	4.58	1748	✓
	outdoors8	83.36	**27.88**	16.31	36.03	13.53	**11.02**	986	
Indoor	room1	0.07	**0.01**	0.06	0.16	0.09	**0.01**	146	✓
	room2	0.07	**0.02**	0.11	0.33	0.07	**0.01**	142	✓
	room3	0.11	**0.04**	0.07	0.15	0.13	**0.01**	135	✓
	room4	0.04	**0.01**	0.03	0.09	0.05	**0.01**	68	✓
	room5	0.20	**0.02**	0.07	0.12	0.13	**0.01**	131	✓
	room6	0.08	**0.01**	0.04	0.05	0.02	**0.01**	67	✓
dark tubular slide	slides1	**0.68**	0.97	0.86	13.73	**0.32**	0.41	289	
	slides2	**0.84**	1.06	2.15	0.81	**0.32**	0.49	299	
	slides3	**0.69**	**0.69**	2.58	4.68	0.89	**0.47**	383	

Carlos Campos, Richard Elvira , Juan J. Gomez Rodrıguez; ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial and Multi-Map SLAM; 2021

3. Literature Review

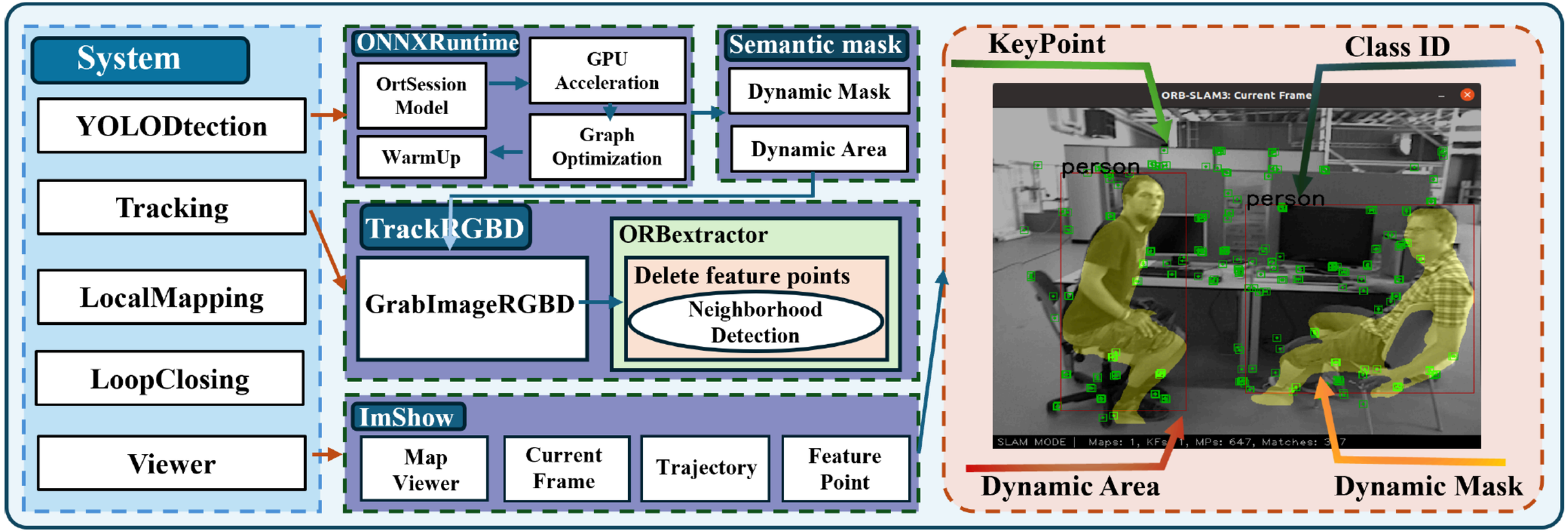
Paper: RY-SLAM: A robust YOLO-based real-time visual SLAM solution for dynamic environments



Xiaohong Chen, Siwen Li, Shuangyin Ren; RY-SLAM: A robust YOLO-based real-time visual SLAM solution for dynamic environments, 2025

3. Literature Review

Paper: RY-SLAM: A robust YOLO-based real-time visual SLAM solution for dynamic environments



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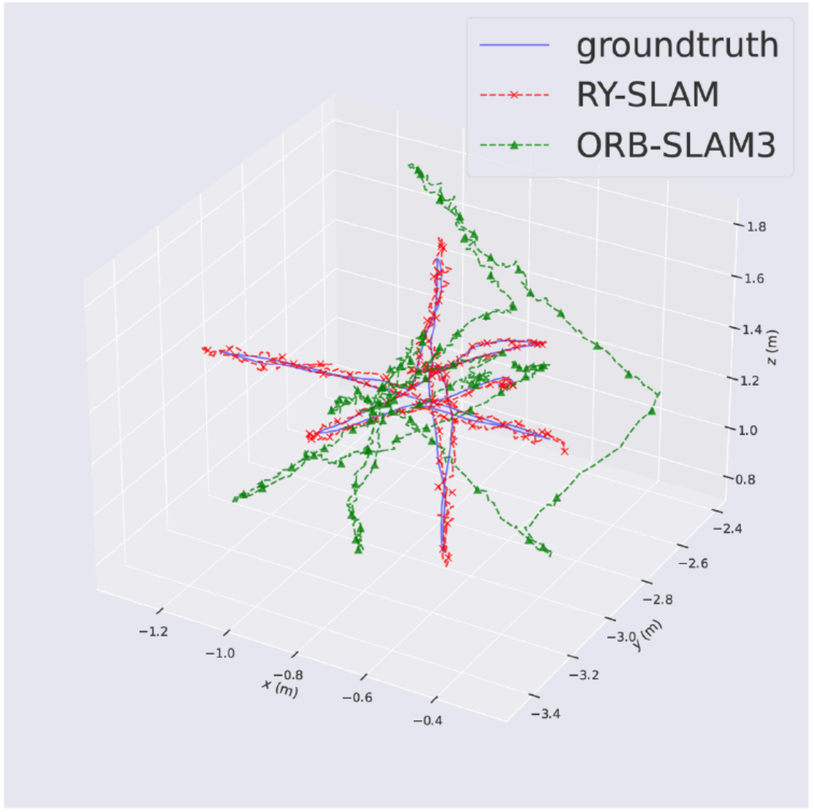
3. Literature Review

Paper: RY-SLAM: A robust YOLO-based real-time visual SLAM solution for dynamic environments



walking_xyz

Table II: Results Of Translation Relative Pose Error(RPE)

Sequences	RY-SLAM		ORB-SLAM3		DS-SLAM		Dyna-SLAM		Improvement(%)	
	RMSE	S.D.	RMSE	S.D.	RMSE	S.D.	RMSE	S.D.	RMSE	S.D.
walking_halfsphere	**0.0140**	**0.0076**	0.2166	0.1145	0.0297	0.0152	0.0284	0.0149	93.52%	93.39%
walking_rpy	**0.0201**	**0.0127**	0.5049	0.3077	0.1503	0.1168	0.0448	0.0262	96.02%	95.87%
walking_static	**0.0061**	**0.0036**	0.1165	0.1134	0.0102	0.0038	0.0089	0.0044	94.76%	96.83%
walking_xyz	**0.0127**	**0.0075**	0.2222	0.1136	0.0333	0.0229	0.0217	0.0119	94.27%	93.41%
sitting_halfsphere	0.0118	0.0065	**0.0081**	**0.0044**	0.0177	0.0084	0.0239	0.0120	-45.69%	-48.08%
sitting_xyz	0.0099	0.0052	**0.0086**	**0.0044**	0.0133	0.0069	0.0142	0.0073	-14.82%	-18.91%

Table III: Results Of Rotational Relative Pose Error(RPE)

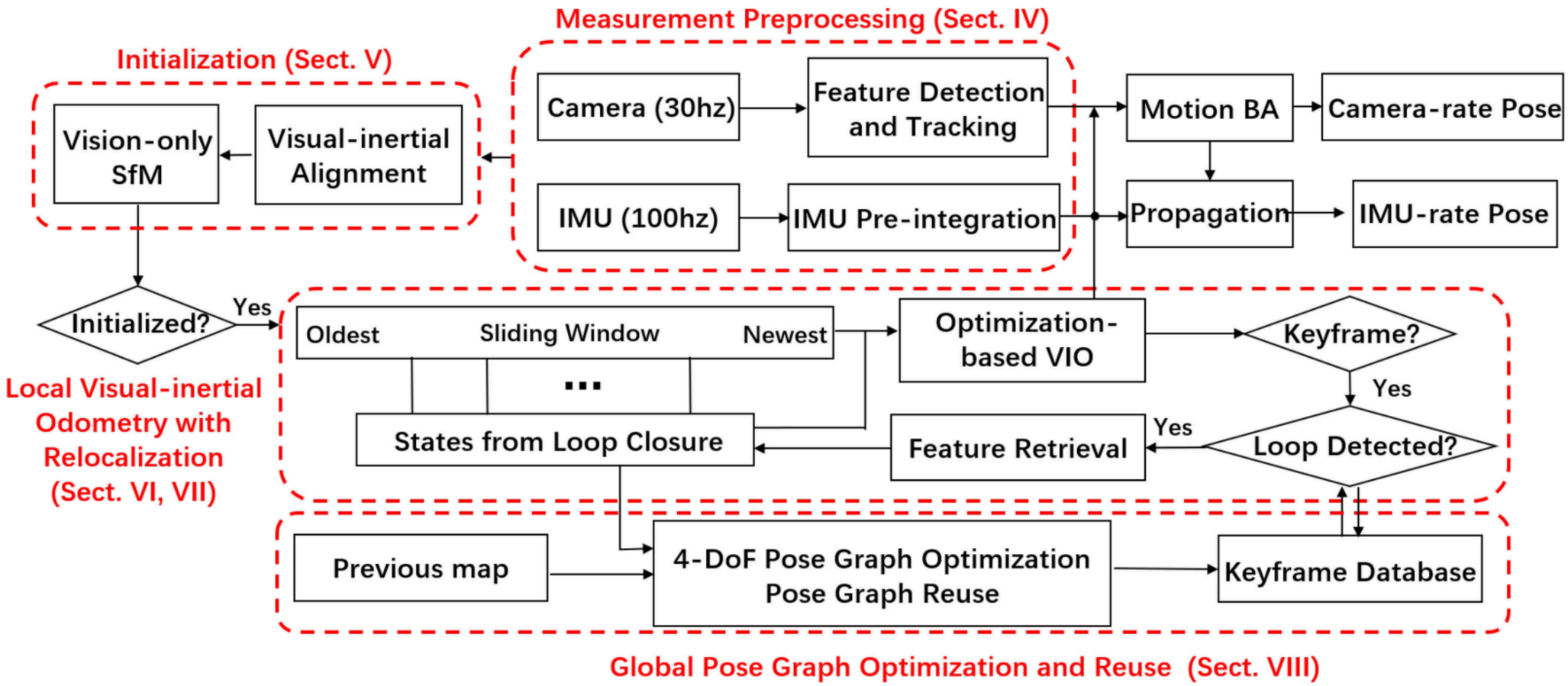
Sequences	RY-SLAM		ORB-SLAM3		DS-SLAM		Dyna-SLAM		Improvement(%)	
	RMSE	S.D.	RMSE	S.D.	RMSE	S.D.	RMSE	S.D.	RMSE	S.D.
walking_halfsphere	**0.4148**	**0.2141**	5.5333	3.3133	0.8142	0.4101	0.7842	0.4012	92.50%	93.54%
walking_rpy	**0.5049**	**0.3077**	6.6845	4.4043	3.0042	2.3065	0.9894	0.5701	92.45%	93.01%
walking_static	**0.1746**	**0.0907**	3.3217	2.2302	0.2690	0.1215	0.2612	0.1259	94.74%	95.93%
walking_xyz	**0.3905**	**0.2737**	5.5461	3.3482	0.8266	0.2826	0.6284	0.3848	92.96%	91.82%
sitting_halfsphere	0.3989	0.2096	**0.3747**	**0.1978**	0.6098	0.2850	0.7045	0.3488	-6.46%	-5.96%
sitting_xyz	0.3237	0.1663	**0.3180**	**0.1649**	0.5044	0.2623	0.5042	0.2651	-1.81%	-0.83%

Xiaohong Chen, Siwen Li, Shuangyin Ren; RY-SLAM: A robust YOLO-based real-time visual SLAM solution for dynamic environments, 2025

3. Literature Review

Paper: VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator

Vins fusion (Visual odometry) : VINS-Fusion is a tightly-coupled, optimization-based estimator : minimizing a sum of Mahalanobis residuals over a sliding window of recent keyframes.



Tong Qin , Peiliang Li , and Shaojie Shen; VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator ; 2018

3. Literature Review

Paper: VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator

Vins fusion (Visual-Inertial odometry) : The visual clues and the IMU clues often disagree slightly because sensors have noise. The system uses a nonlinear solver (like Ceres) to wiggle the estimated position, velocity, and rotation of every frame inside the window until all the clues match up as perfectly as mathematically possible.

Cost function : $$\min_{x}\{||r_p - H_p X||^2 + \sum_{K\in B} ||r_B(z_{b_k b_{k+1}}, X)||^2_{P_{b_k b_{k+1}}} + \sum_{(l,j)\in C} ||r_C(z_{c_l^j},)||^2_{P_{c_l^j}}\}$$

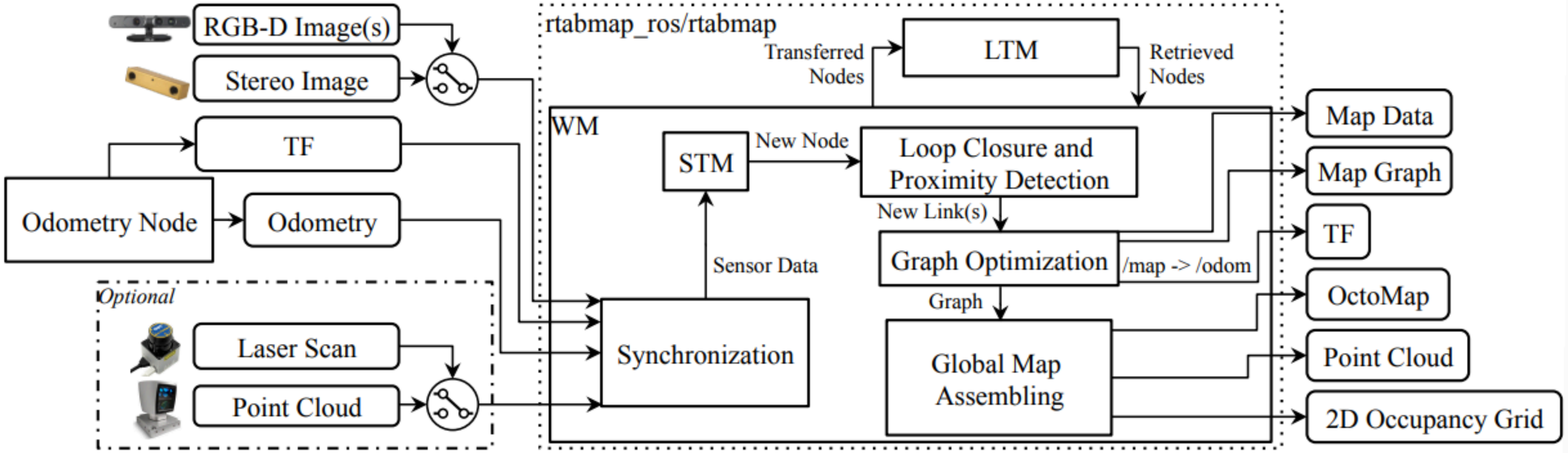
- $\min_{\mathcal{X}}$: "Find the best state ($\mathcal{X}$)—meaning position, velocity, orientation, and feature depths—that makes the total error as small as possible."
- **Part 1: The Prior Error ($\|r_p - \ldots\|^2$)**: The penalty for disagreeing with what the oldest, forgotten frames told us before they left the window.
- **Part 2: The IMU Error ($\sum \|r_{\mathcal{B}}\|^2$)**: The penalty for disagreeing with the physics recorded by the IMU between the frames.
- **Part 3: The Visual Error ($\sum \rho(\|r_{\mathcal{C}}\|^2)$)**: The penalty for the 3D corner features not lining up perfectly where the camera pixels say they should be. (The ρ is a Huber norm, which is a mathematical shield that ignores crazy outliers so one bad pixel doesn't ruin the whole estimate).

Tong Qin , Peiliang Li , and Shaojie Shen; VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator ; 2018

3. Literature Review

Paper: RTAB-Map as an Open-Source Lidar and Visual SLAM Library for Large-Scale and Long-Term Online Operation

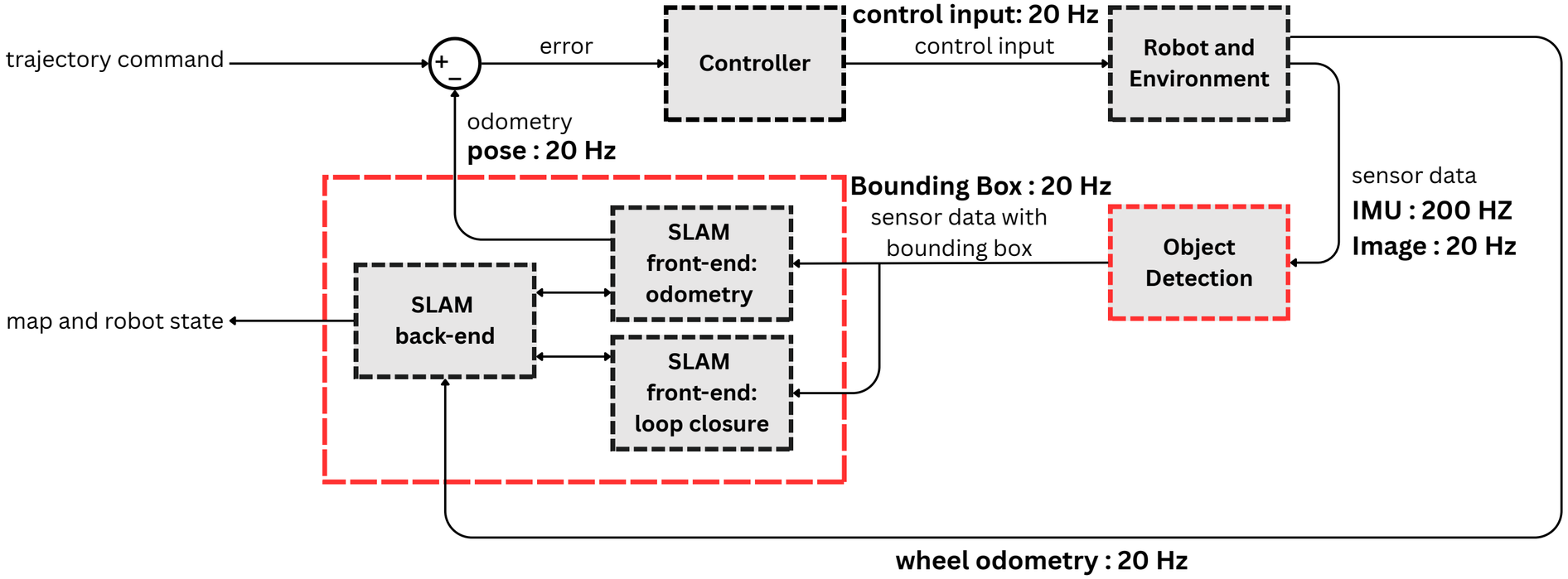
RTAB-Map : RTAB-Map is a graph-based SLAM approach that has been integrated in ROS as the rtabmap ros7 package since 2013. The odometry is an external input to RTABMap, which means that SLAM can also be done using any kind of odometry to use what is appropriate for a given application and robot.



Mathieu Labb, Fran¸cois Michaud ; RTAB-Map as an Open-Source Lidar and Visual SLAM Library for Large-Scale and Long-Term Online Operation ; 2024

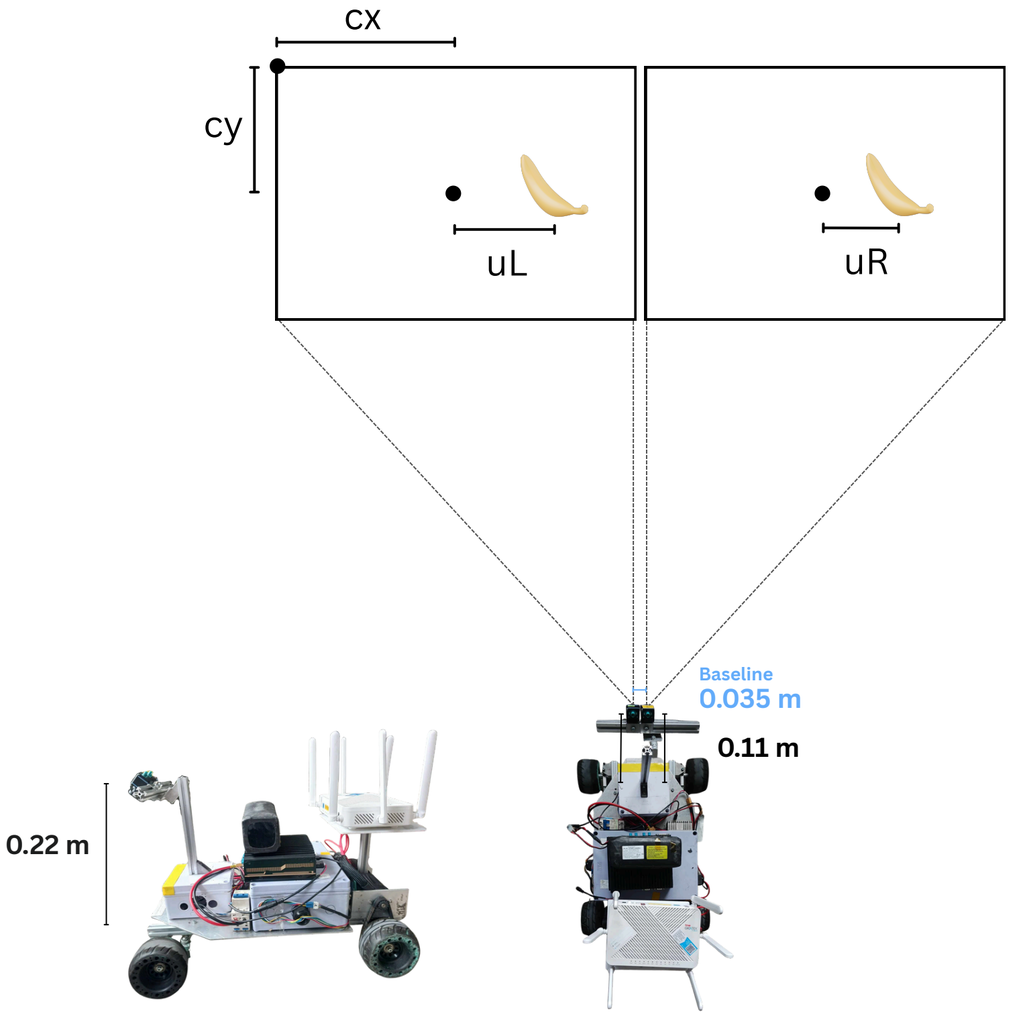
4. Methodology

- **System Architecture**



4. Methodology

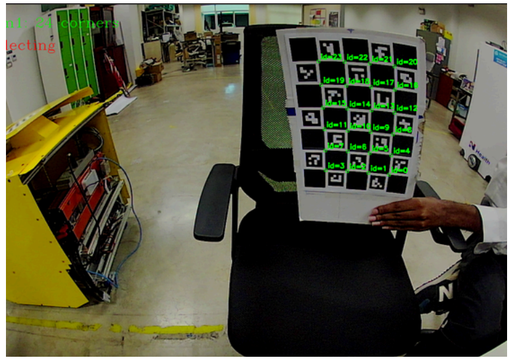
- **System Architecture : camera**

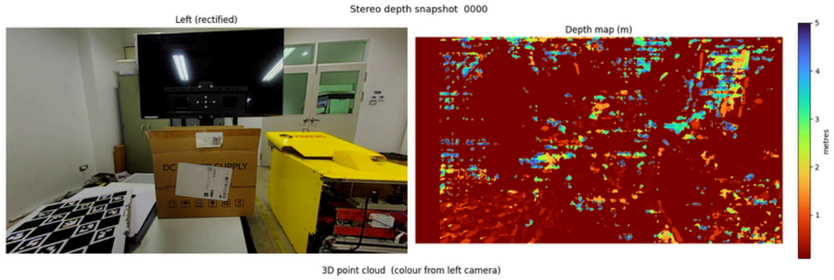


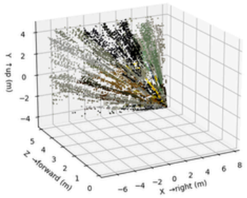
- **Instrinsic :** $depth = \frac{focal \times baseline}{disparity}$; $disparity = uL - uR$

Parameter	Source
model	camera datasheet
resolution	camera datasheet
Focal length (fx=fy) (from simulation config)	fx = $\frac{W}{2tan\,(FOV \div 2)} = \frac{960}{2tan45^{\circ}} = 480$
(cx,cy)	image center
k1,k2,p1,p2	Stereo calibration
Baseline	The distance between 2 cameras

- **Extrinsic : To be continue**

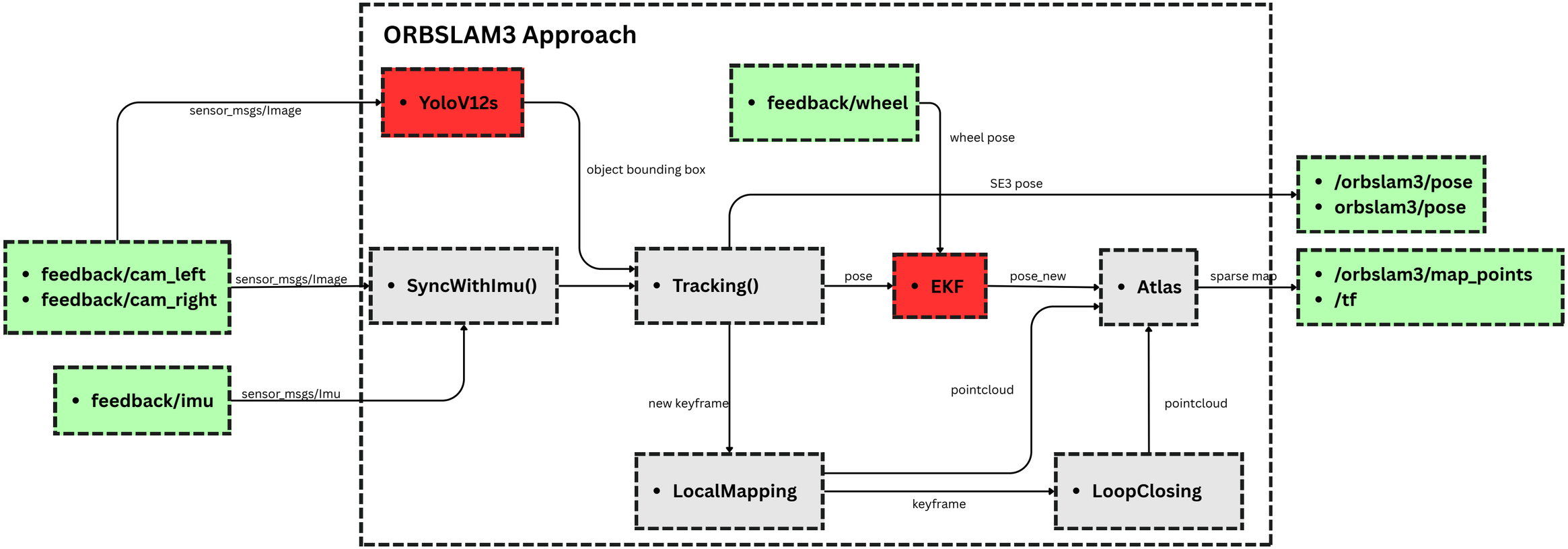






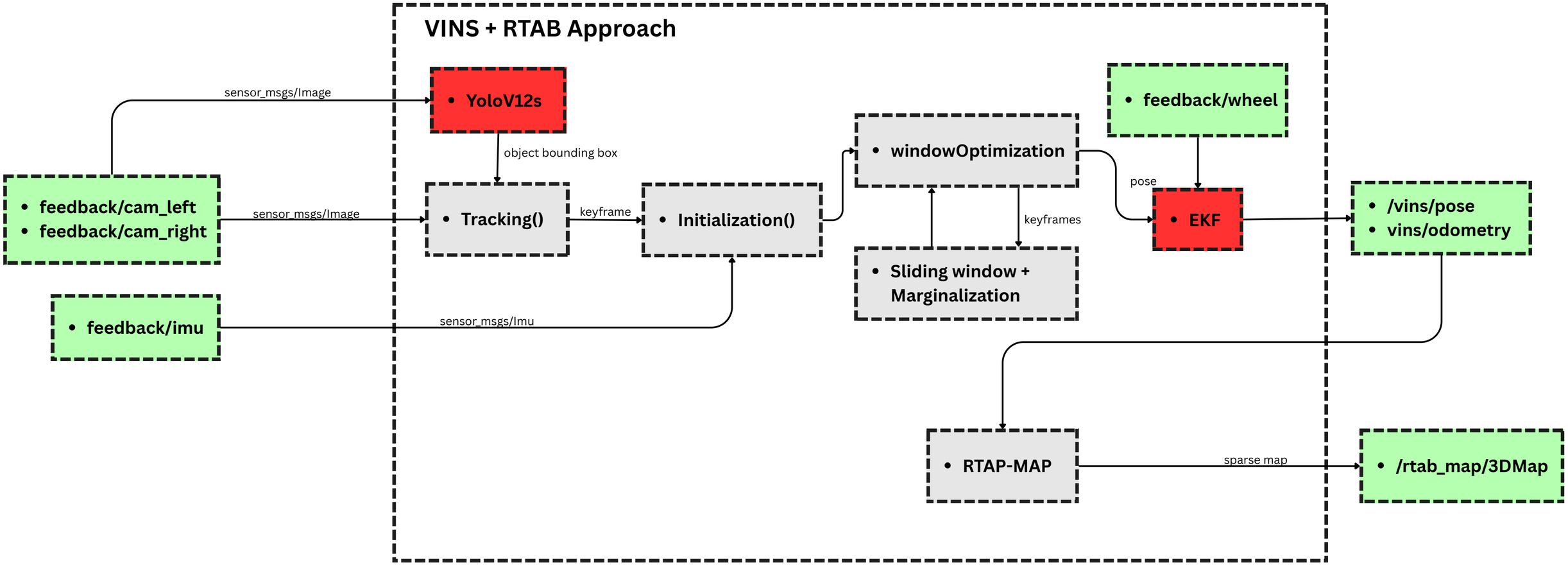
4. Methodology

- **System Architecture : ORBSLAM3 Approach**

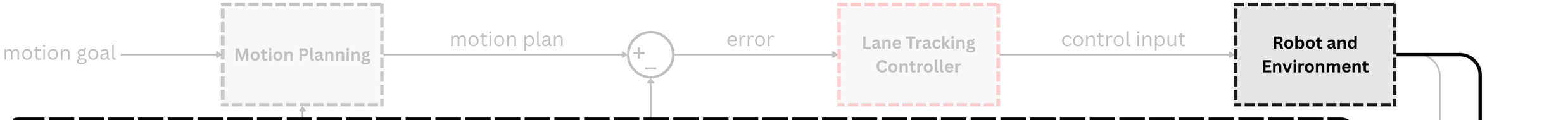


4. Methodology

- System Architecture : VINS + RTAB Approach



5. Experimental Design



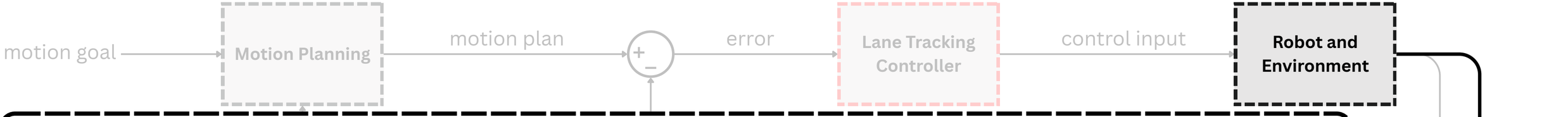
- **Robot**
  - **NRU-51V series**

- Processor: NVIDIA Jetson Orin NX or Xavier™ NX System-on-Module (SOM)
- Camera Support: 4× GMSL2 automotive cameras via FAKRA Z connectors with built-in frame synchronization
- Expansion & Storage: 2× mini-PCIe sockets (WiFi/GNSS/NVMe/CAN) and 1× M.2 3042/3052 B key socket (for 4G/5G mobile communication)

  - **AC-IMX390-H190 2 ตัว**

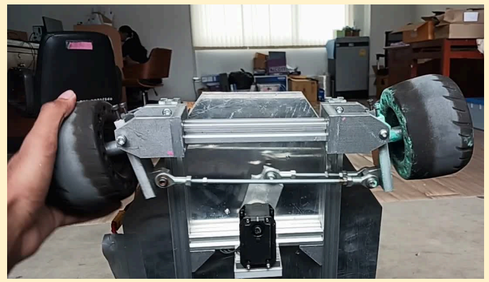
Specification	Details
Sensor & ISP	Sony IMX390 CMOS sensor and GEO SEMI GW5200 ISP
Resolution & FPS	1920x1080 (2MP) @ 30 FPS
Field of View (HFOV)	186°
Video Output	Uncompressed YUV422 (UYVY)
Dynamic Range	120dB High Dynamic Range (HDR)
Special Features	LED Flicker Mitigation (LFM), 5-axis active alignment
Environmental	IP67 and IP69K waterproof (lens side)
Operating Temperature	-40°C to 85°C
Connector	Male FAKRA connector

5. Experimental Design

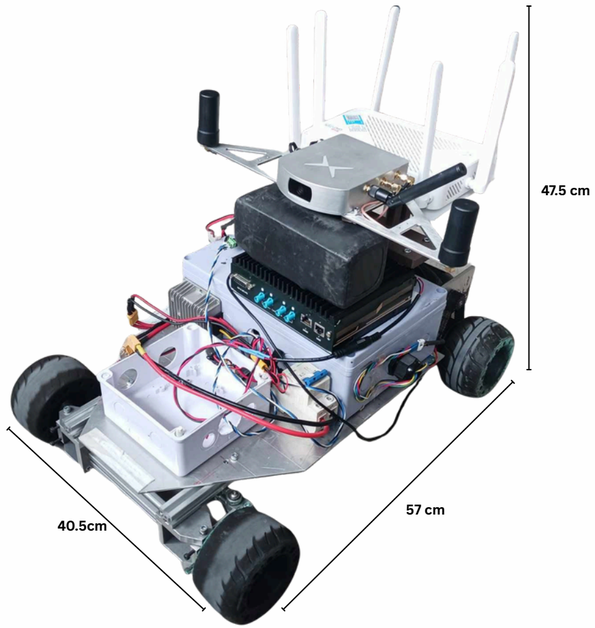


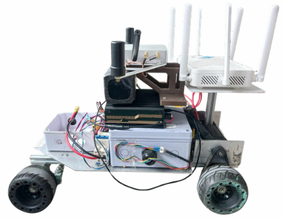
- **Robot**

DYNAMIXEL RX-64

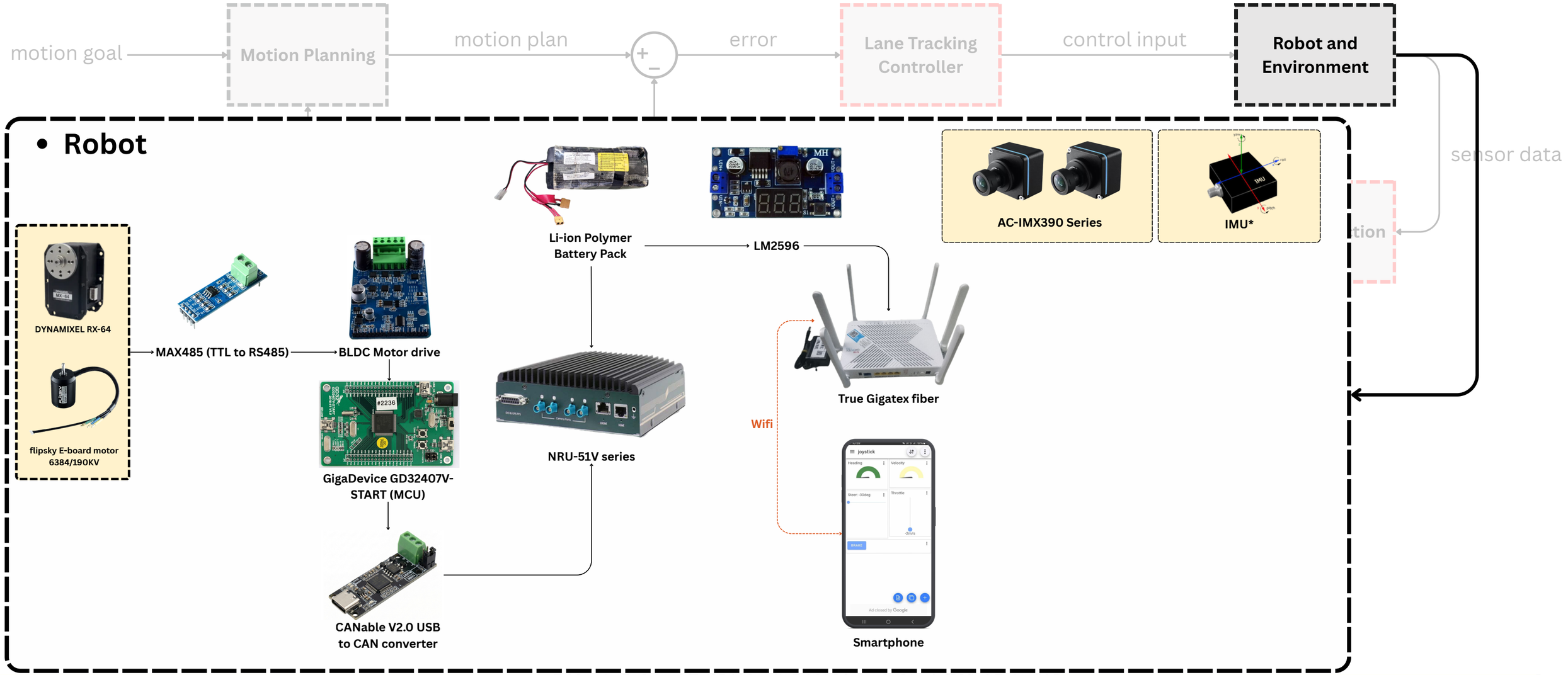


flipsky E-board motor 6384/190KV





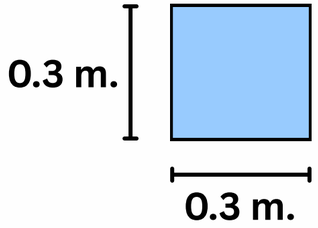
5. Experimental Design



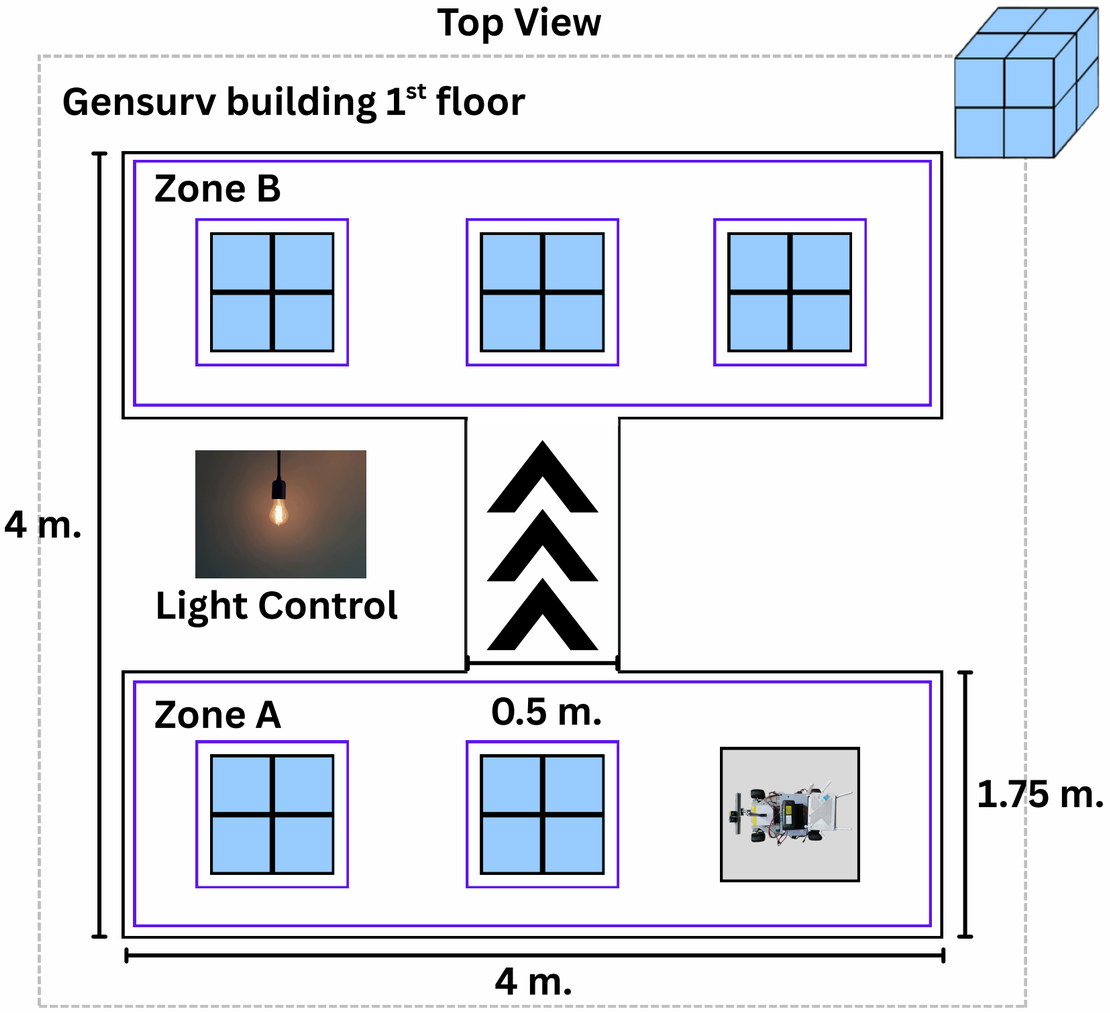
5. Experimental Design

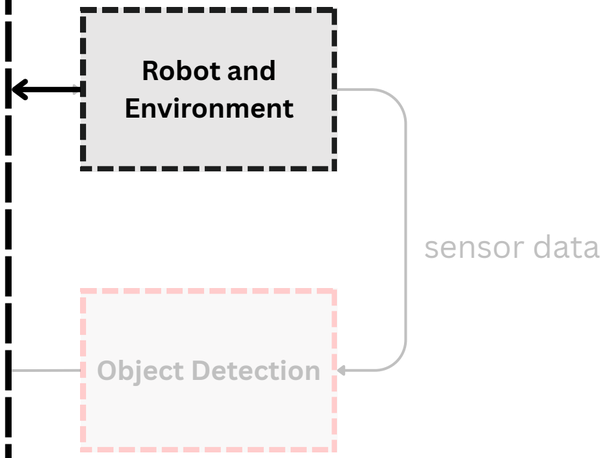
- **Environment (Indoor) : ตึกสถาบันนวัตกรรมหุ่นยนต์และยานยนต์อัตโนมัติ (สนอ.) ชั้น 1**

Motion capture camera

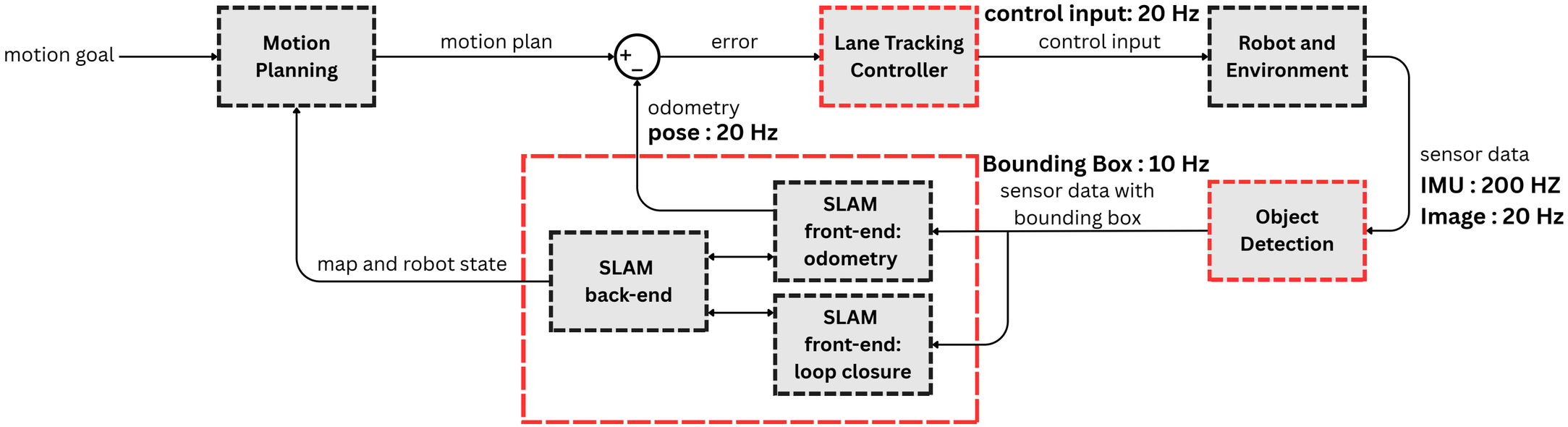


Top View





5. Experimental Design



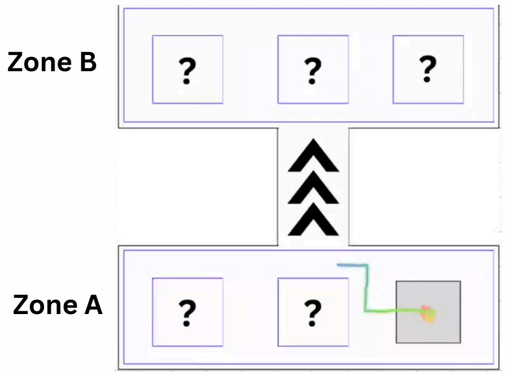
Experiments

- **Object Detection**
  - Dynamic Object Detection Accuracy
- **Visual SLAM**
  - Localization Accuracy
  - Costmap Accuracy
- **Integrated System**

5. Experimental Design

- **Object Detection Experiment (Offline test)**

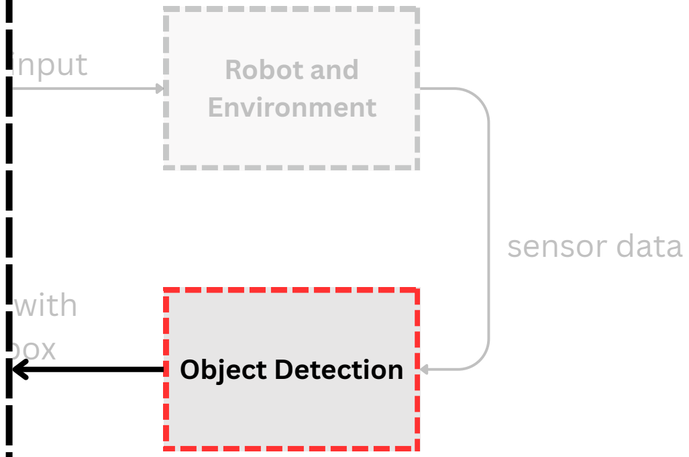
Procedure



Collecting dataset

Labeling images

Model training & evaluation



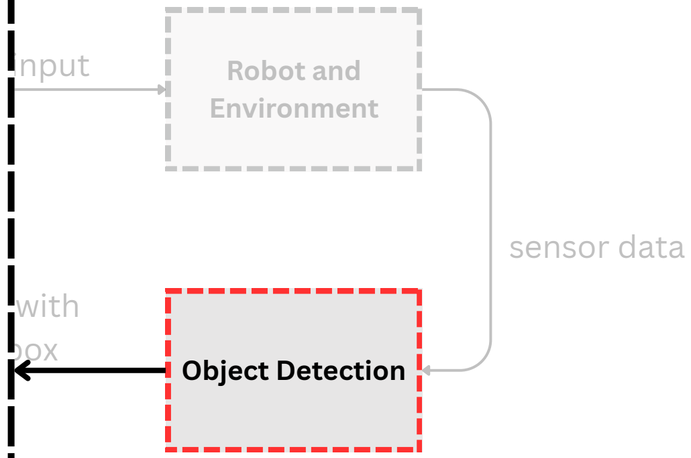
- **Light condition :** 8.00 am, 12.00 pm, 6.00 pm
- **Robot Speed** : 1.5 m/s
- **Duration** : 5 minutes per light condition
- **?** : Sets of parcel (In range 1-4 parcels). Whenever the robot transitions to a new zone, the number of parcels per set and the set's location are randomized.
- **Robot path** : The vehicle navigates back and forth between Zone A and Zone B. Within these zones, its trajectory must follow the pattern shown in the top-left figure.

5. Experimental Design

- **Object Detection Experiment**

Model Configuration

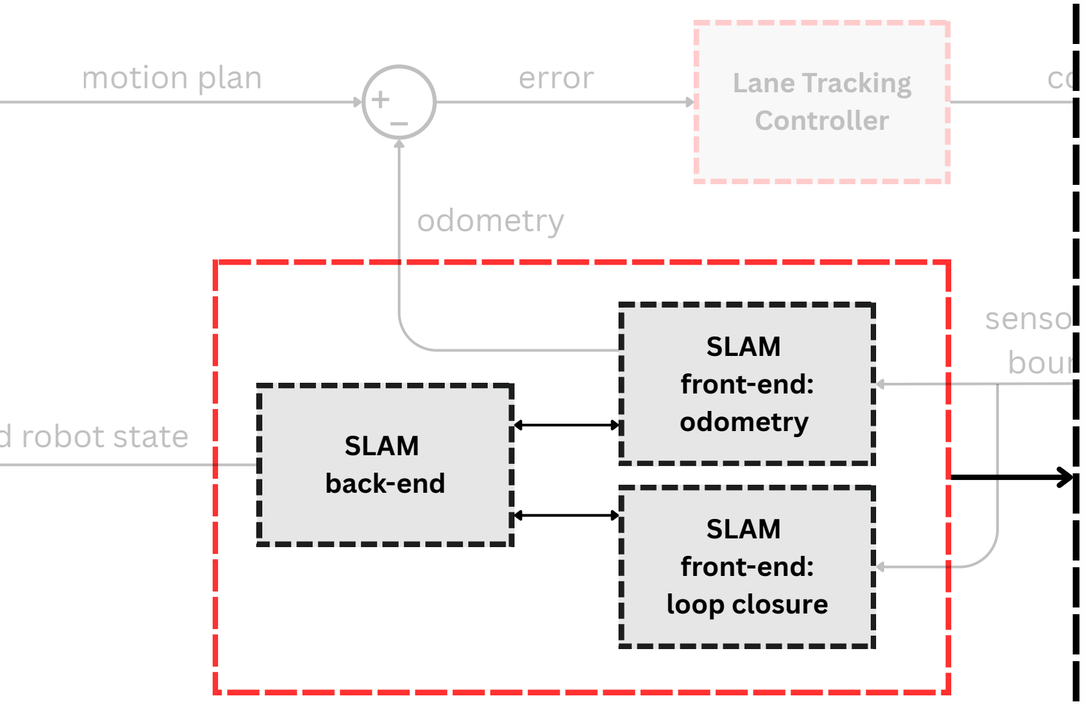
Variant	Parameters	Default Input Resolution	FLOPs (Complexity)
YOLOv12 Nano (N)	2.6 M	640 × 640	6.5 Billion
YOLOv12 Small (S)	9.3 M	640 × 640	21.4 Billion
YOLOv12 Medium (M)	20.2 M	640 × 640	67.5 Billion



Evaluation Metrics

- **Precision :** $Precision = \frac{TP}{TP+FP}$
- **Recall :** $Recall = \frac{TP}{TP+FN}$
- **F1 Score :** $F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision+Recall}$
- **Intersection over Union (IoU) :** $IoU = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|}$
- **Average Precision (AP) :** $mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i$

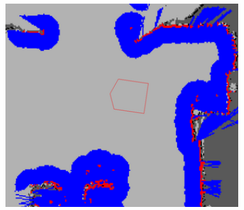
5. Experimental Design



- **VIsual SLAM Experiment : Costmap Accuracy (offline test)**

Procedure

- static environment



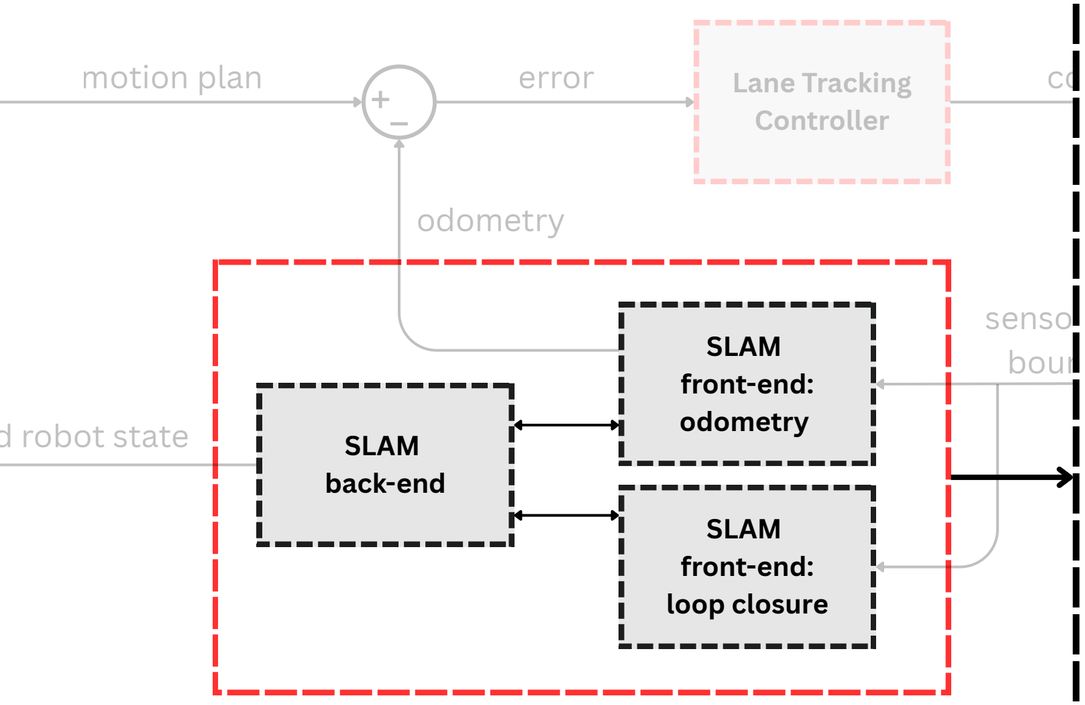
vs **GT**

- 2D map comparison

Evaluation Metrics

Metric	In plain words
Voxel IoU @0.3 m	overlap of occupied 30 cm boxes (very strict — see caveat 2)
F-score @0.5 m	single balanced score of the two above (**headline**)

5. Experimental Design



- **VIsual SLAM Experiment : Localization Accuracy (offline test)**

Procedure

8.00 am.
12.00 pm.
6.00 pm.

Light condition

- static environment
- dynamic environment

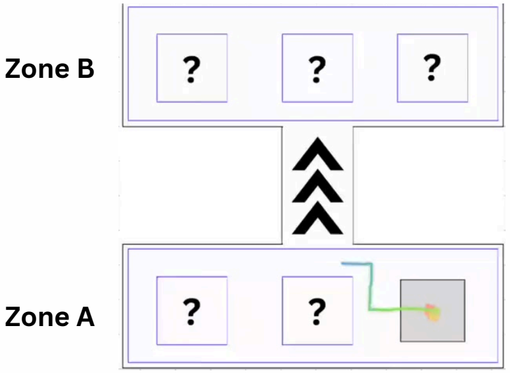
Collecting ground truth using tracking camera

Evaluation Metrics

- Absolute Trajectory Error (ATE) : $\mathrm{ATE}_{\mathrm{RMSE}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} ||\mathrm{trans}(E_i)||^2}$; $E_i = Q_i^{-1} S P_i$
- Relative Pose Error (RPE) : $\mathrm{RPE}_{\mathrm{RMSE}} = \sqrt{\frac{1}{m}\sum_{i=1}^{m} ||\mathrm{trans}(E_i)||^2}$; $E_i = (Q_i^{-1} Q_{i+\Delta})^{-1} (P_i^{-1} P_{i+\Delta})$
- Latency (milisecond, reference on camera frame rate)

5. Experimental Design

- **Integrated System Experiment (online test)**



- **Light condition :** 8.00 am, 12.00 pm, 6.00 pm
- **Robot Speed** : 1.5 m/s
- **Duration** : 5 minutes per light condition
- **?** : Sets of parcel (In range 1-4 parcels). Whenever the robot transitions to a new zone, the number of parcels per set and the set's location are randomized.
- **Robot path** : The vehicle navigates back and forth between Zone A and Zone B. Within these zones, its trajectory must follow the pattern shown in the top-left figure.

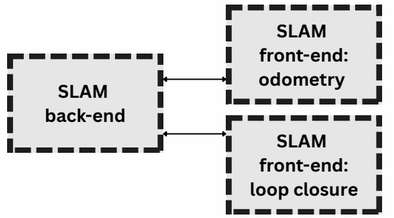
Object Detection

Evaluation Metrics

- **Precision :** $Precision = \frac{TP}{TP+FP}$
- **Recall :** $Recall = \frac{TP}{TP+FN}$
- **F1 Score :** $F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision+Recall}$
- **Intersection over Union (IoU) :** $IoU = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|}$
- **Average Precision (AP) :** $mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i$

Evaluation Metrics

- Absolute Trajectory Error (ATE) : $\mathrm{ATE}_{\mathrm{RMSE}} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} ||\mathrm{trans}(E_i)||^2}$ $E_i = Q_i^{-1} S P_i$
- Relative Pose Error (RPE) : $\mathrm{RPE}_{\mathrm{RMSE}} = \sqrt{\frac{1}{m} \sum_{i=1}^{m} ||\mathrm{trans}(E_i)||^2}$ $E_i = (Q_i^{-1} Q_{i+\Delta})^{-1} (P_i^{-1} P_{i+\Delta})$
- Latency (milisecond, reference on camera frame rate)



5. Gantt chart

ลำดับ	แผนดำเนินงาน	2569						
		มิ.ย.	ก.ค.	ส.ค.	ก.ย.	ต.ค.	พ.ย.	ธ.ค.
1	ศึกษาองค์ความรู้พื้นฐานเกี่ยวกับ Visual SLAM และ Object Prediction							
2	เตรียม Proposal Report และ สอบ Proposal							
3	พัฒนา Visual slam ตาม Pipeline ที่ได้สืบค้น							
4	จัดเตรียมสนามทดสอบ							
5	ทำการ Hardware setup ตัว Mobile robot ขนาดเล็ก และทดสอบระบบที่ได้จัดเตรียมไว้							
6	ทำรายงานและสอบ Final Presentation							

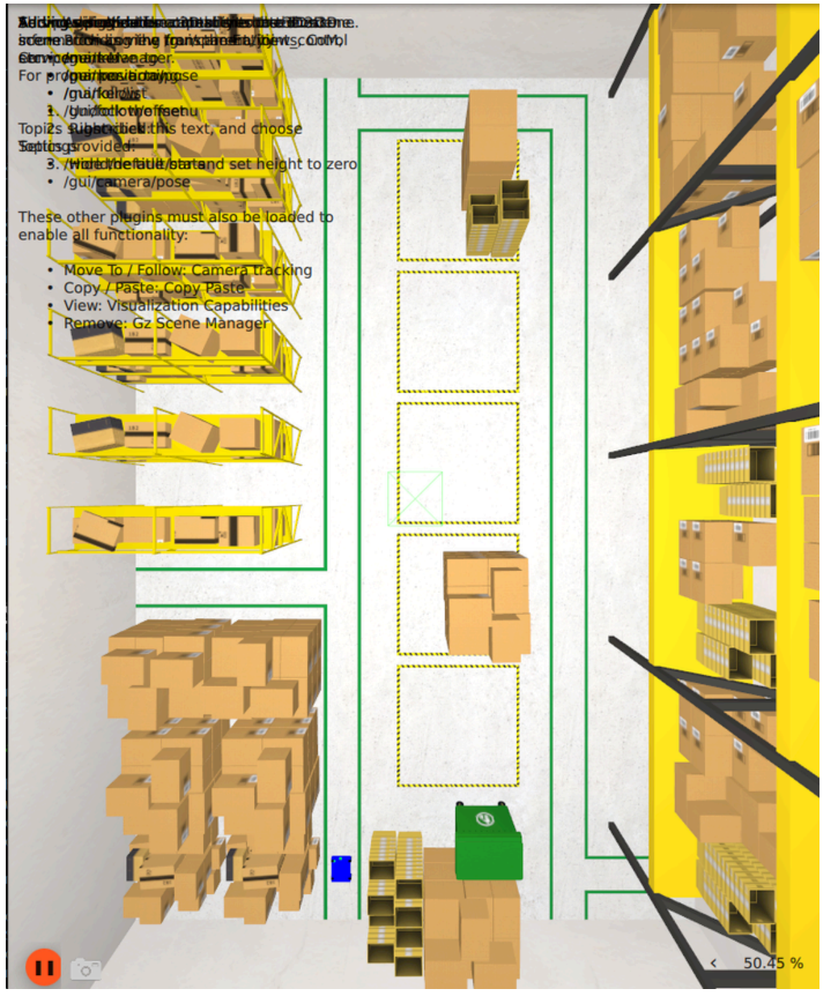
Thank you

Contact Us

Institute of Field Robotics
King Mongkut's University of Technology Thonburi

A Cradle of Future Leaders in Robotics

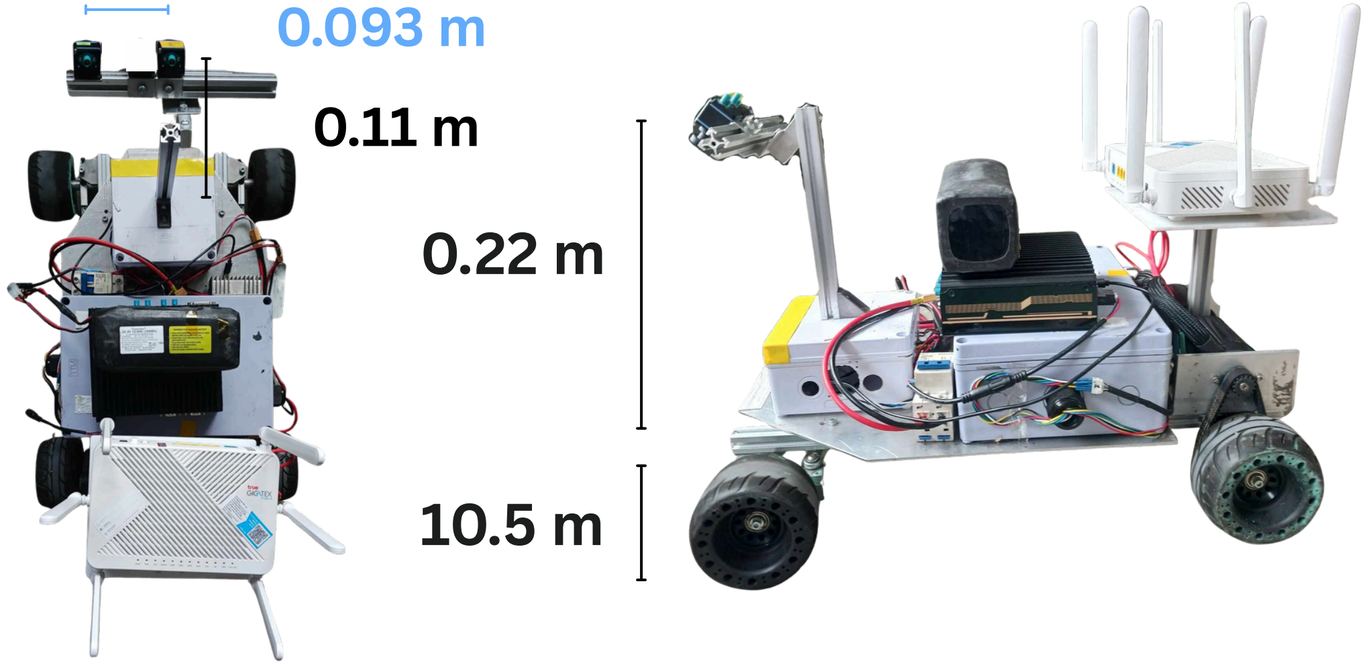
Update
26/8/69

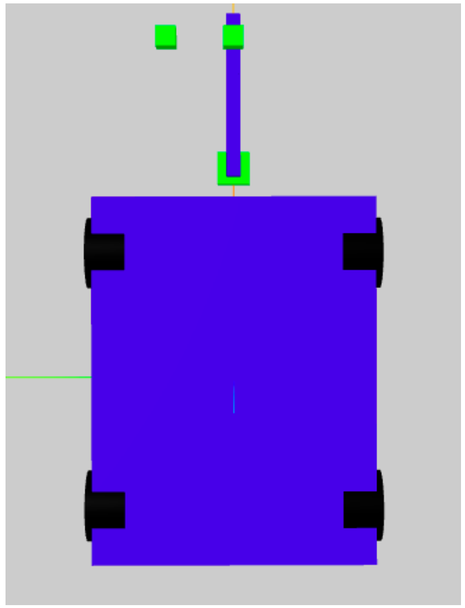


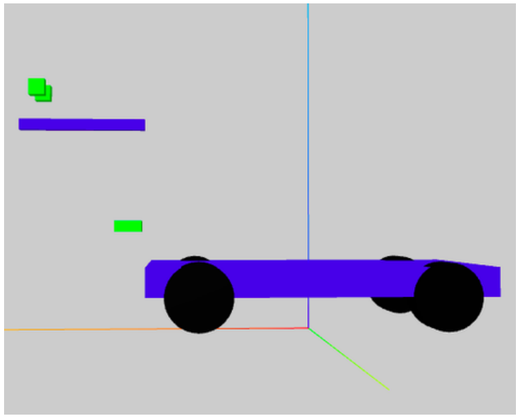
Static **Dynamic**

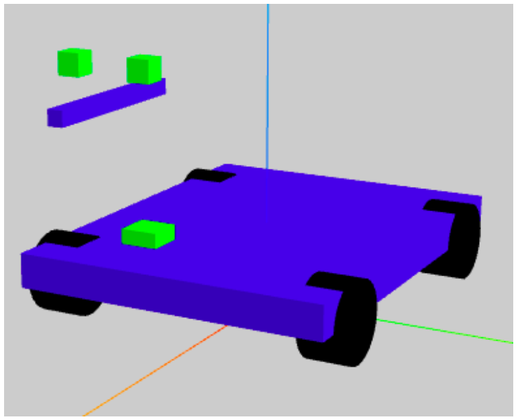
4. Methodology

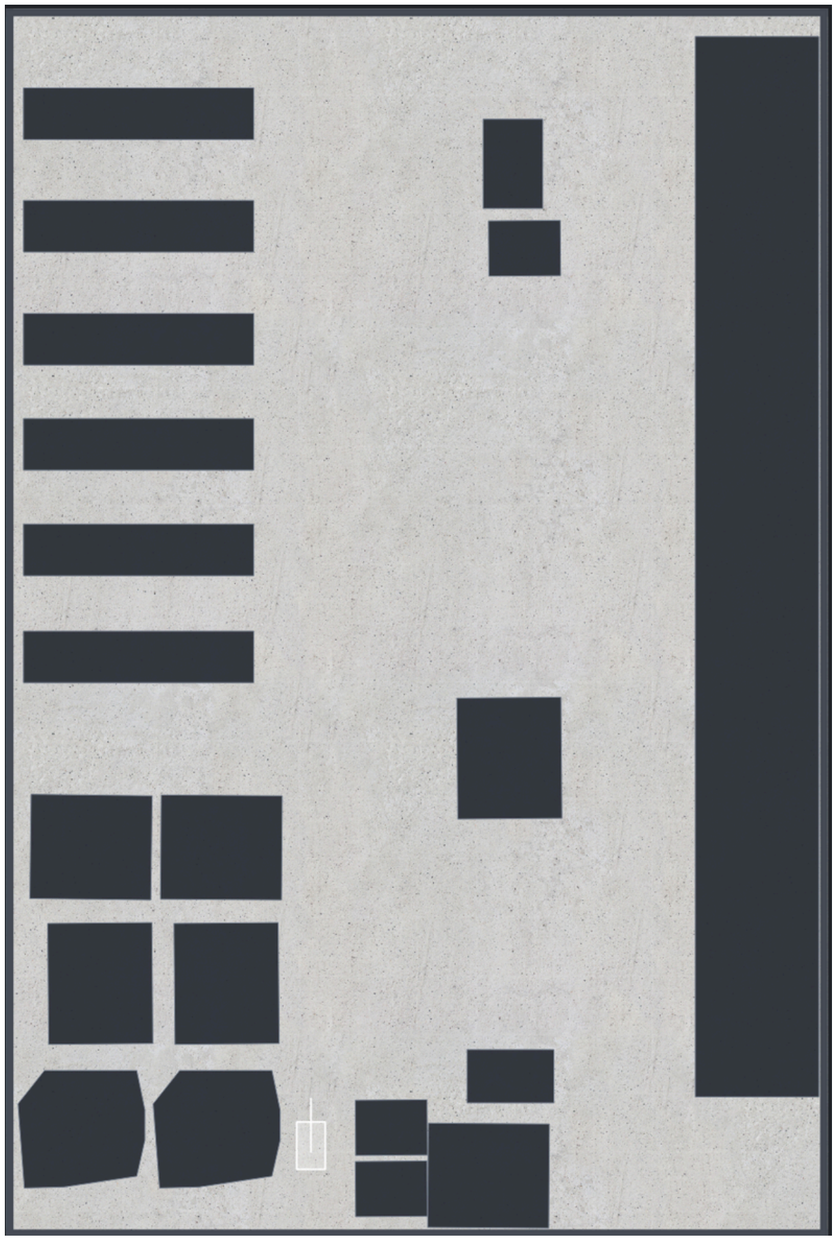
- System Architecture : camera







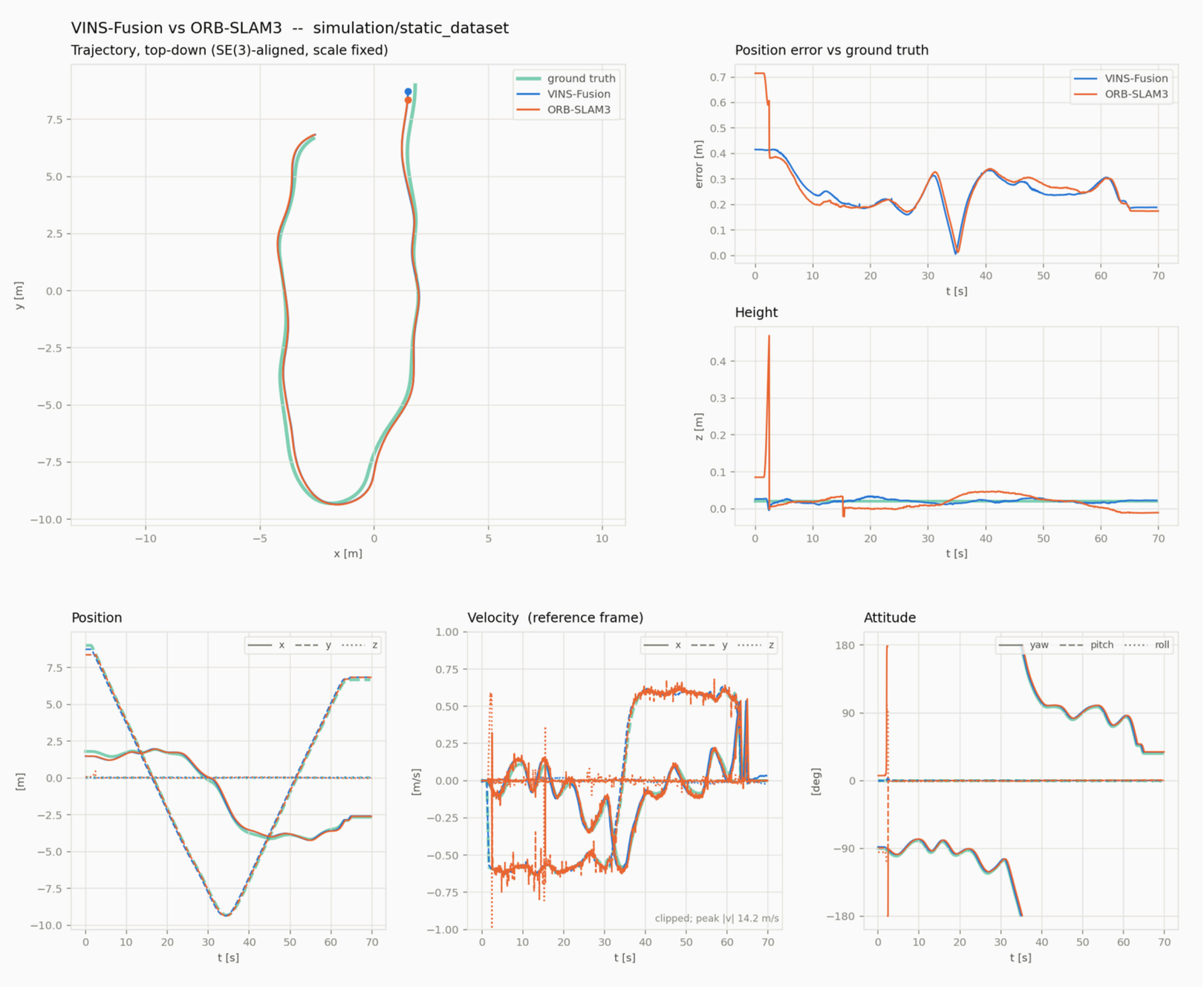




21 m

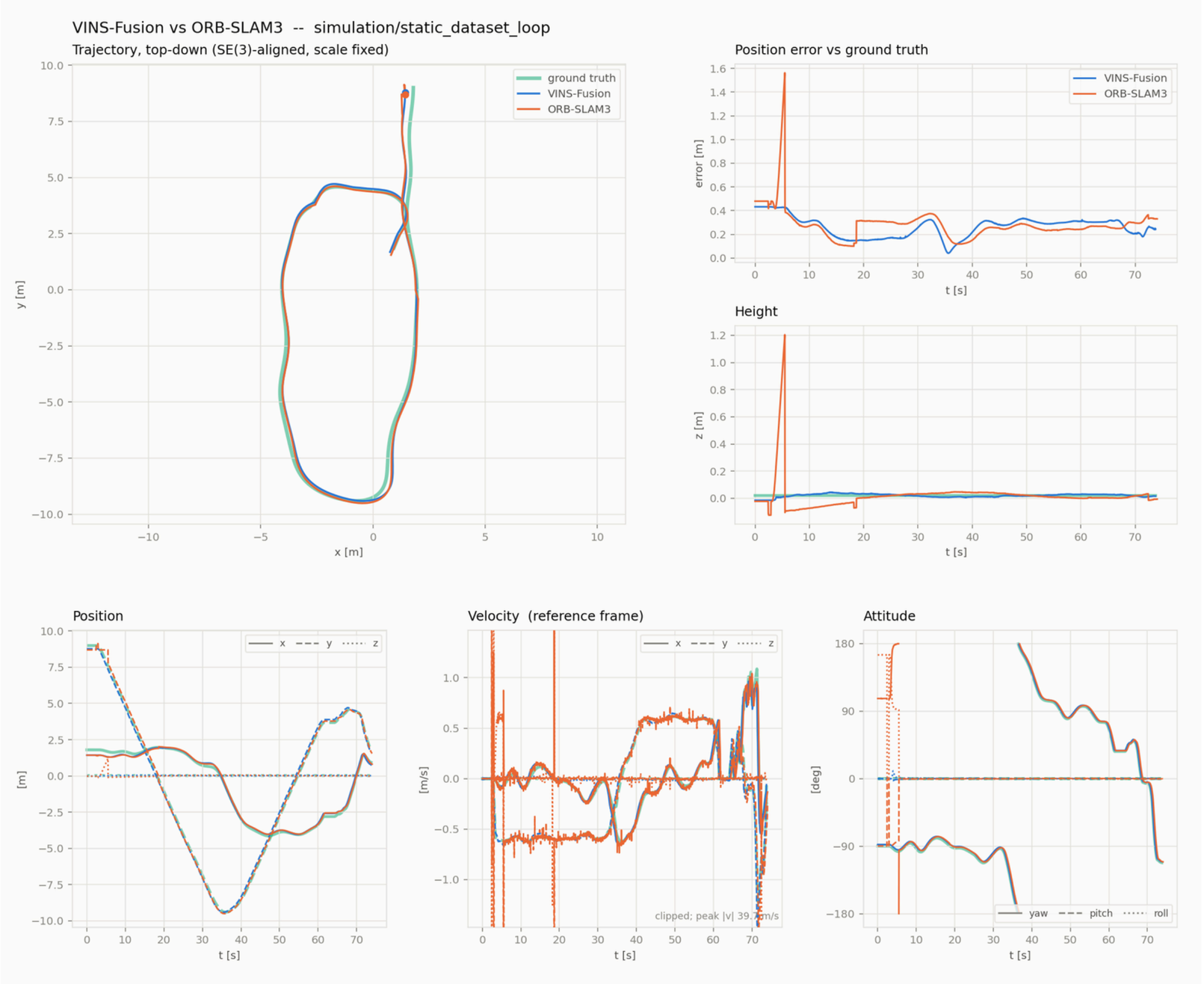
14 m

Simulation



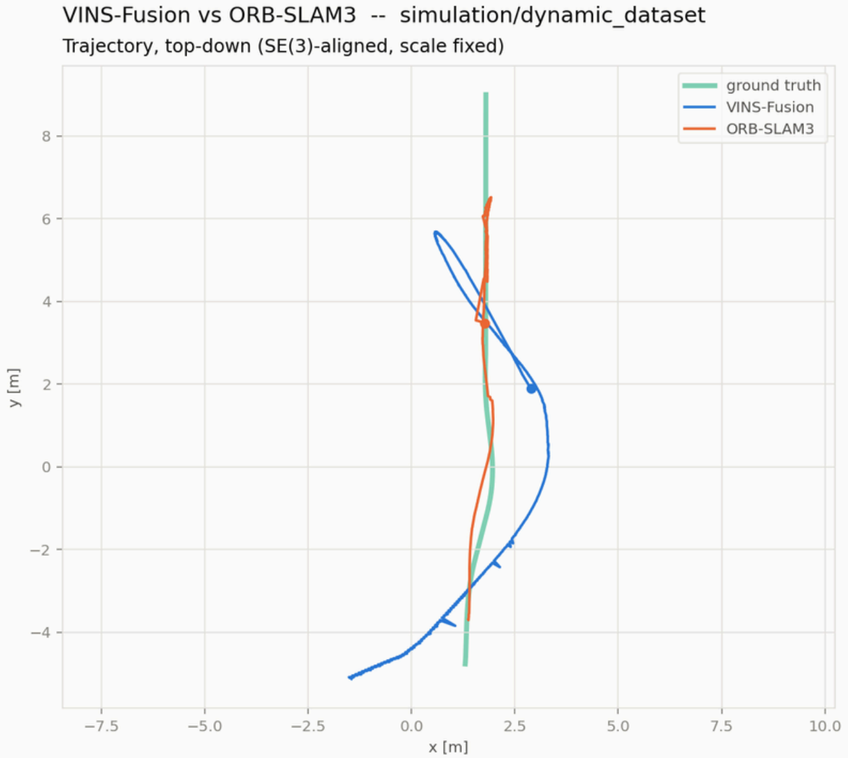
Static warehouse (no loop)

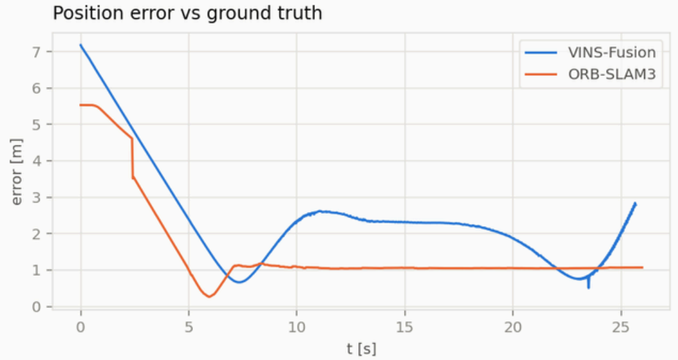
Simulation

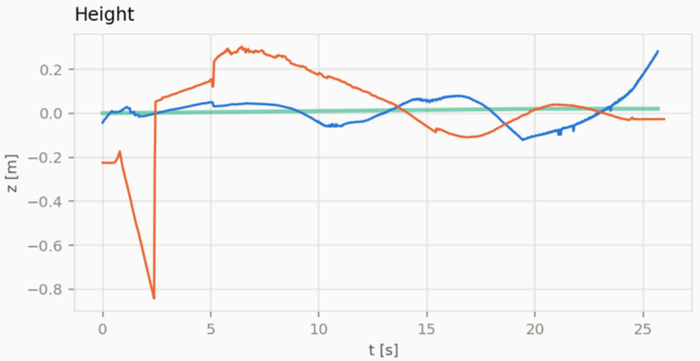


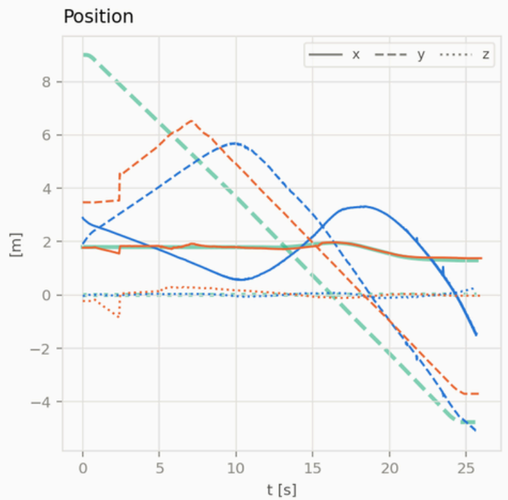
Static warehouse (loop)

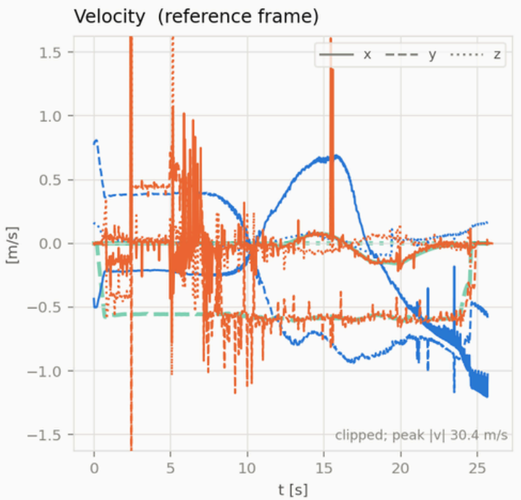
Simulation

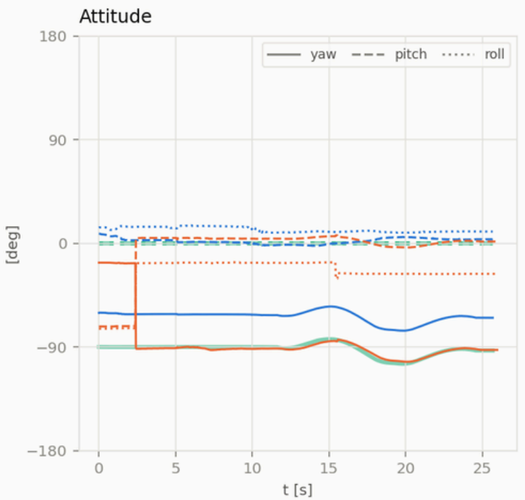








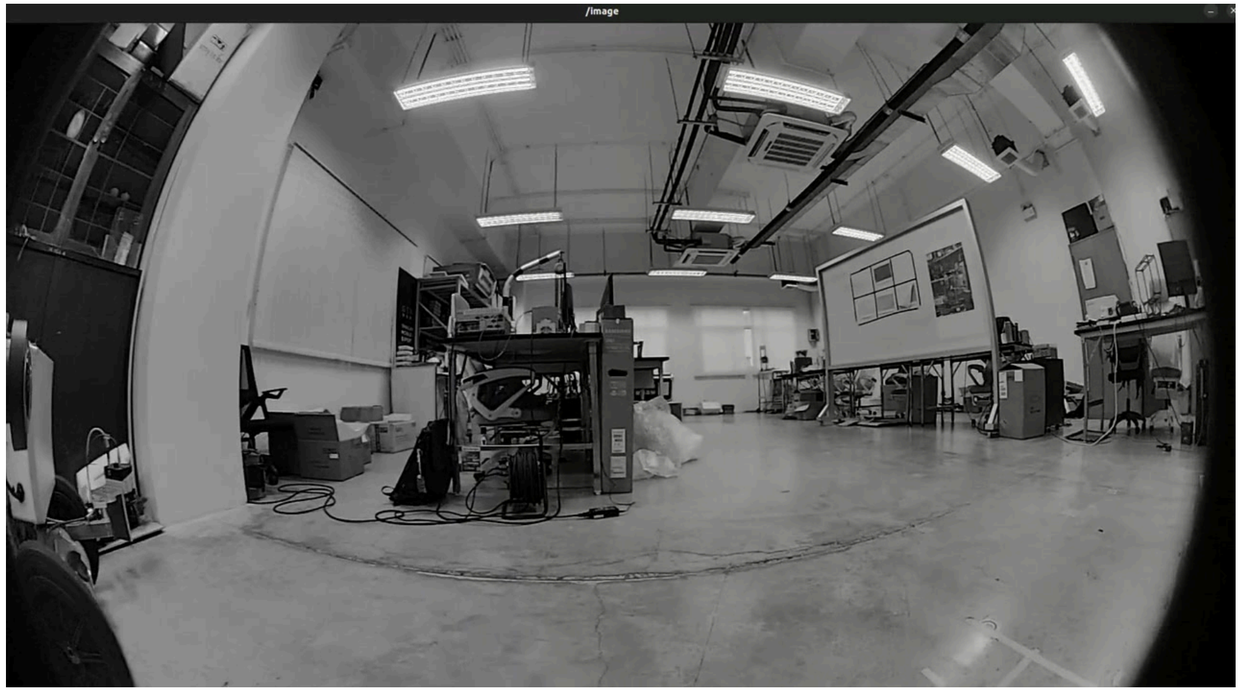




dynamic warehouse (straight line)

dataset	run	VINS ATE rms	ORB ATE rms
static_dataset	70 s, 37 m	0.262 m	0.251 m
static_dataset_loop	74 s, 42 m	0.279 m	0.205 m
dynamic_dataset	26 s, 14 m	160 m (Diverged)	1.14 m

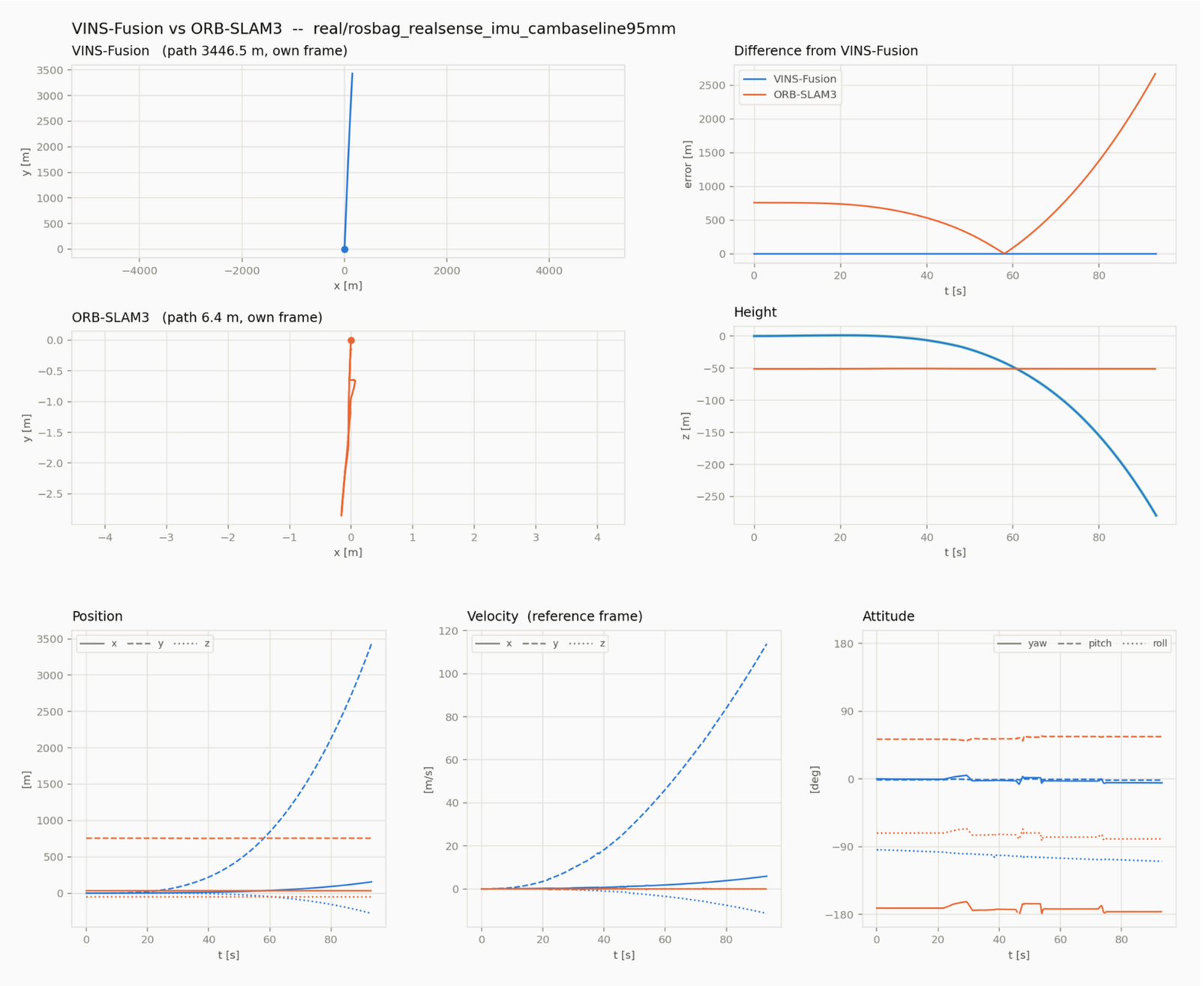
Real world



Straight line path

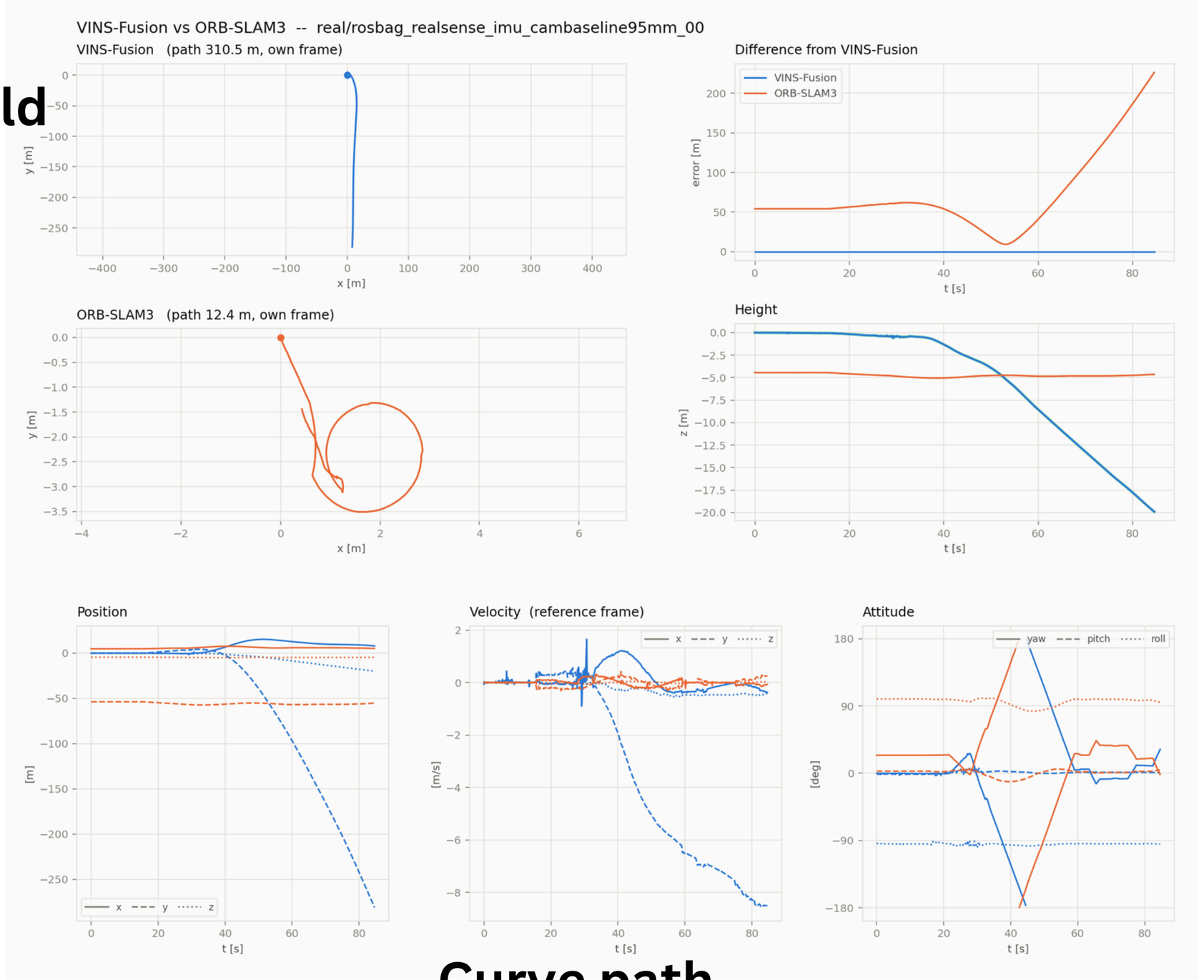
Curve path

Real world



Straight line path

Real world

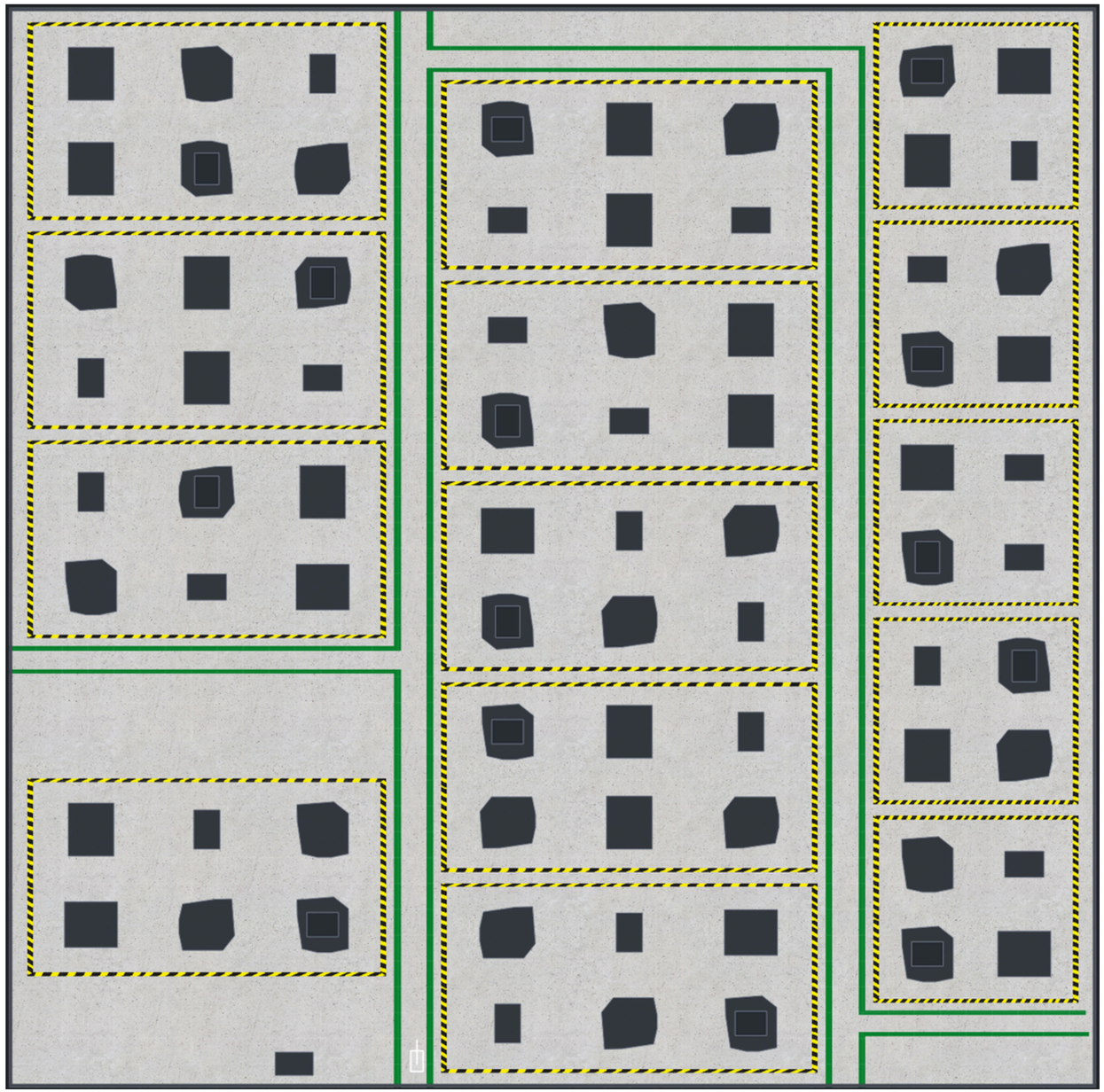


Curve path

Next plan

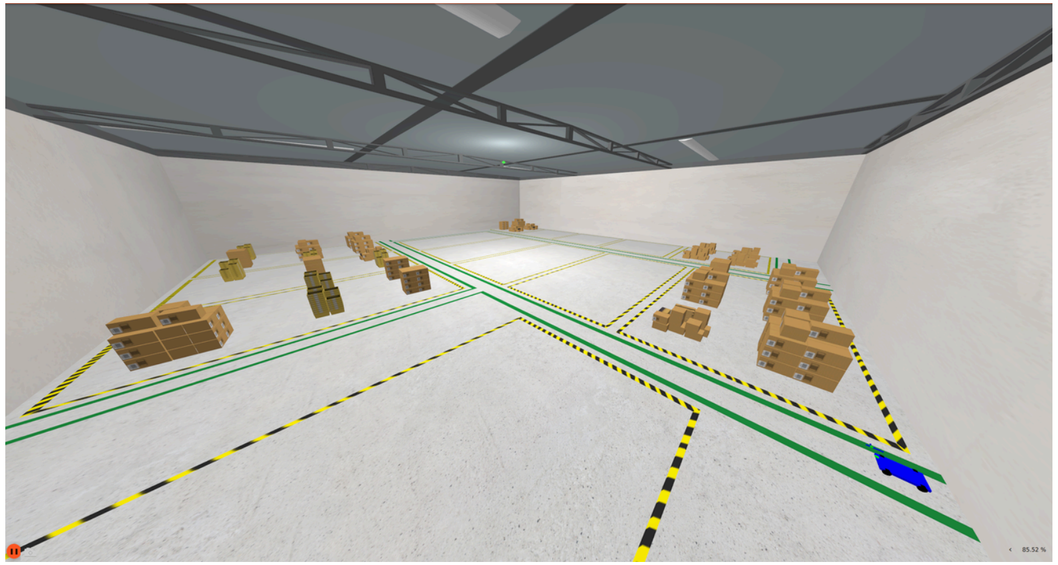
- sim
  - ลอง yolo
  - ลอง rtab
  - สร้าง simulation map ฉบับสมบูรณ์
- real
  - ทำกล้องให้หายสั่น

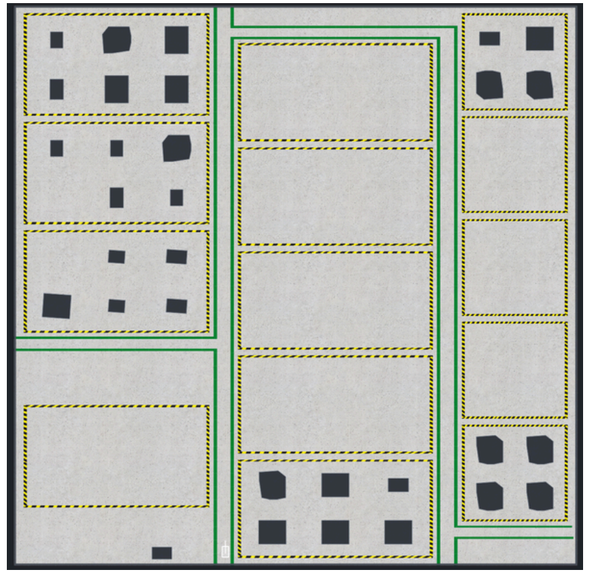
Update
9/9/69



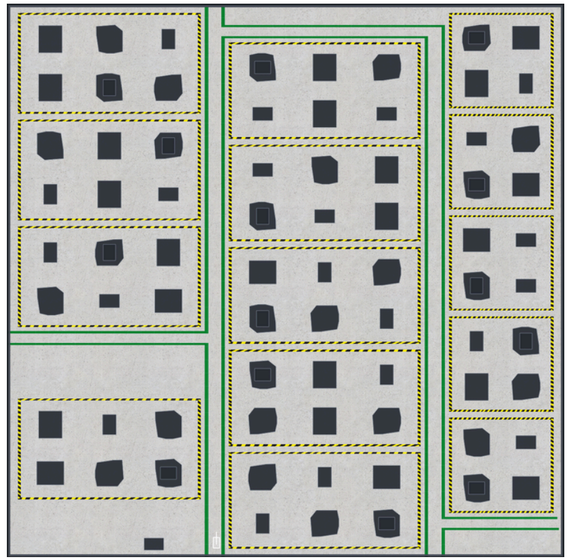
21 m

21 m

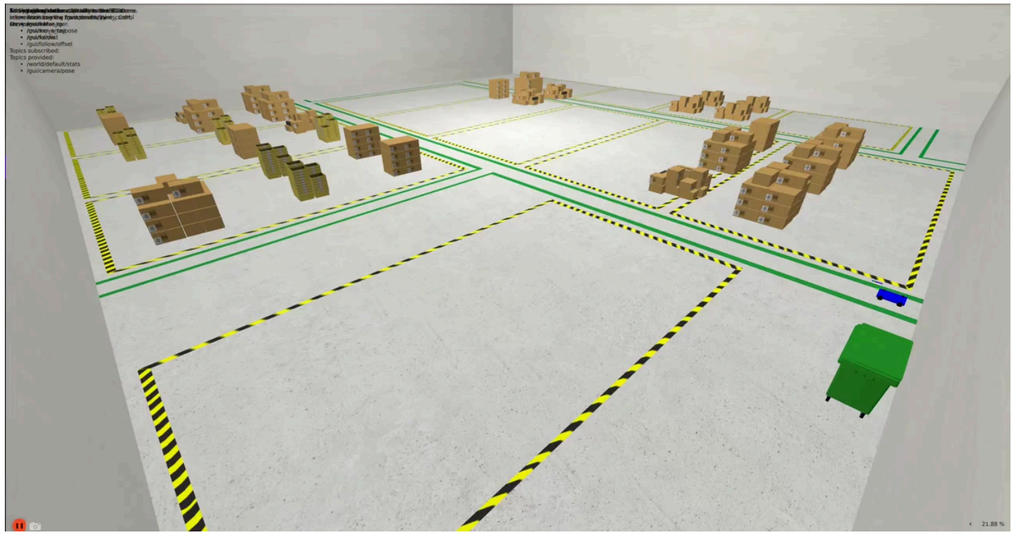


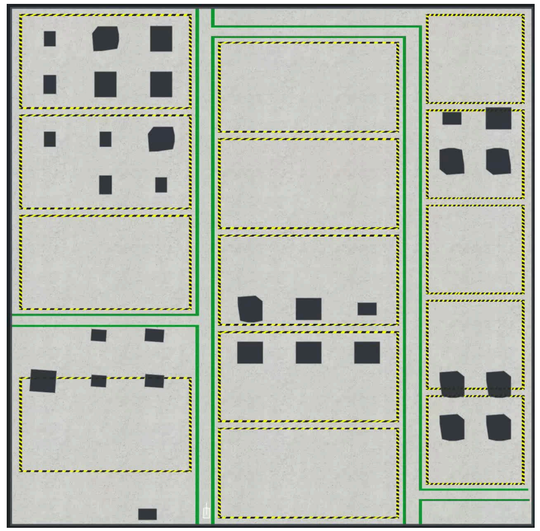


Static

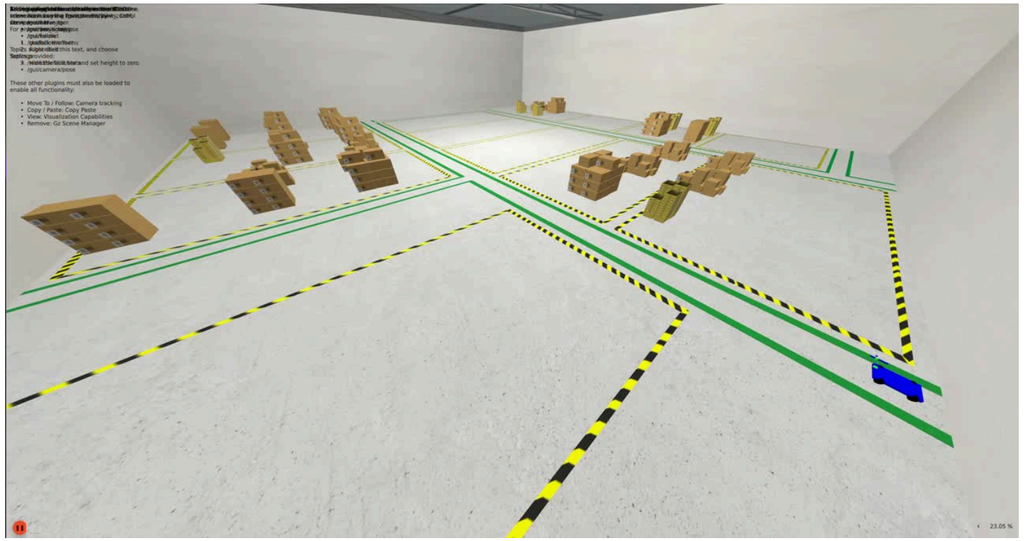


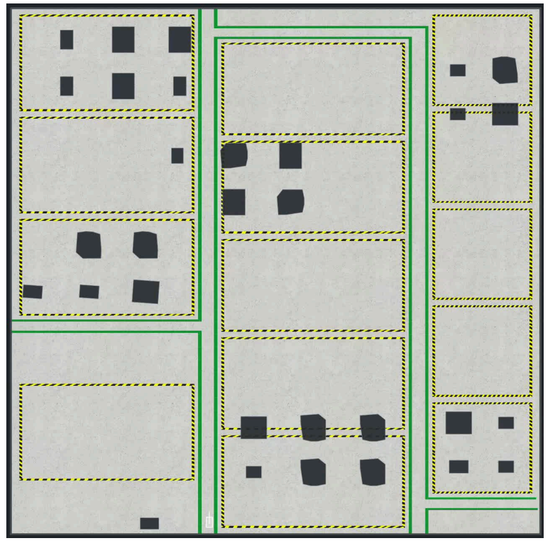
Static_01



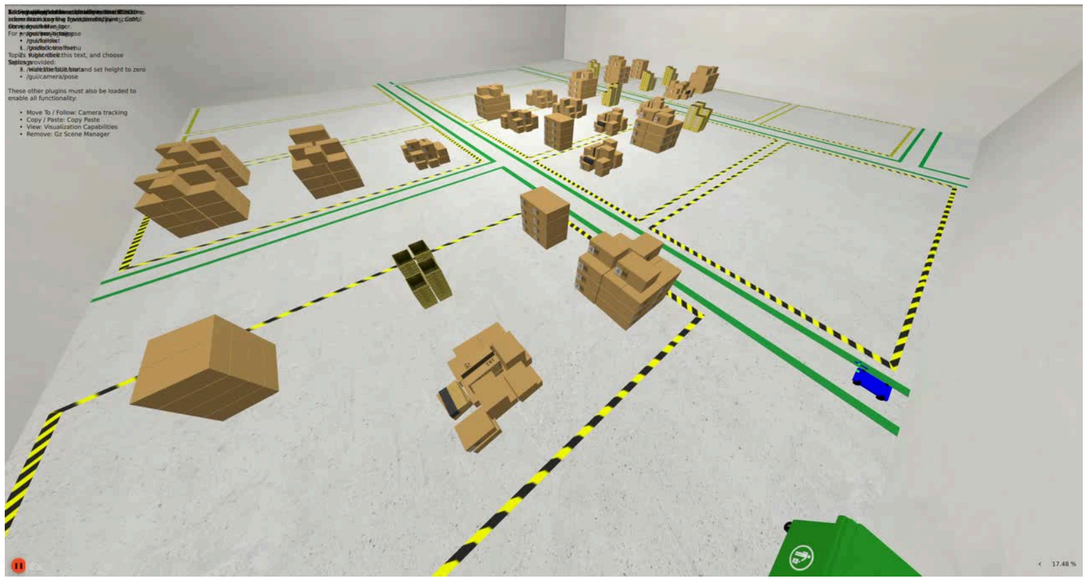


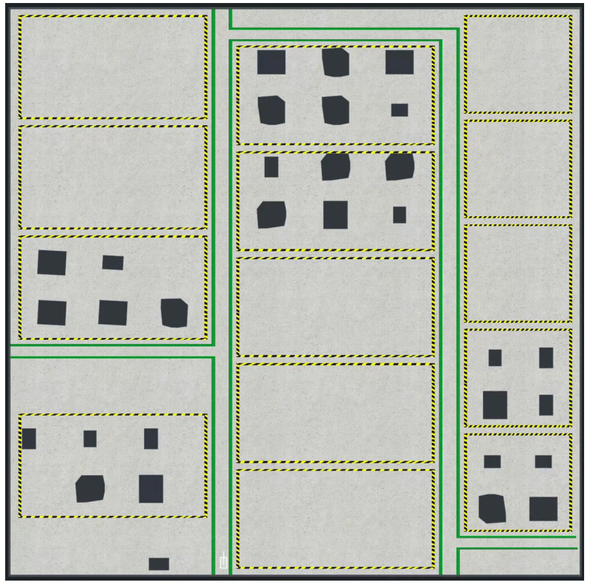
Dynamic



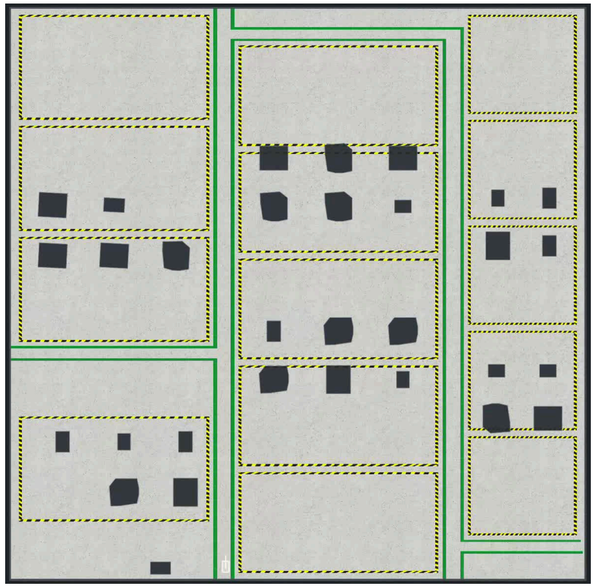


Dynamic_00





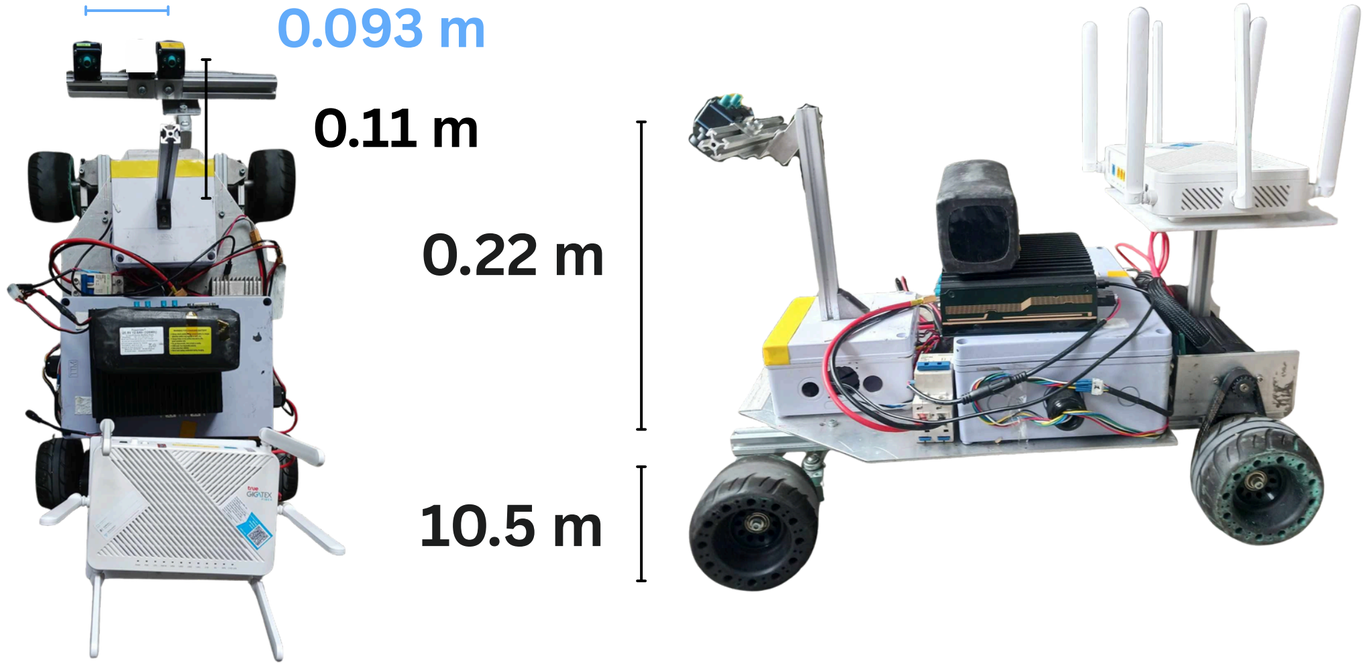
Dynamic_01

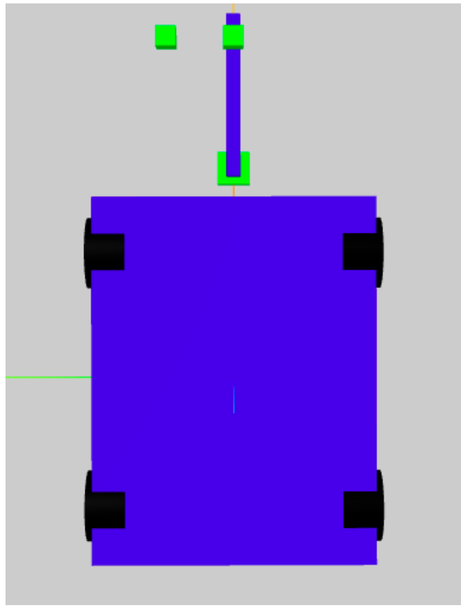


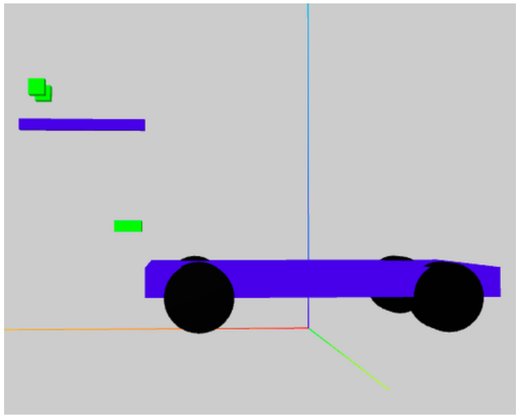
Dynamic_02

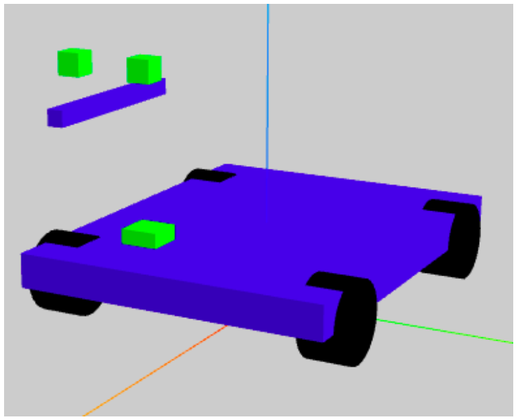
4. Methodology

- System Architecture : camera

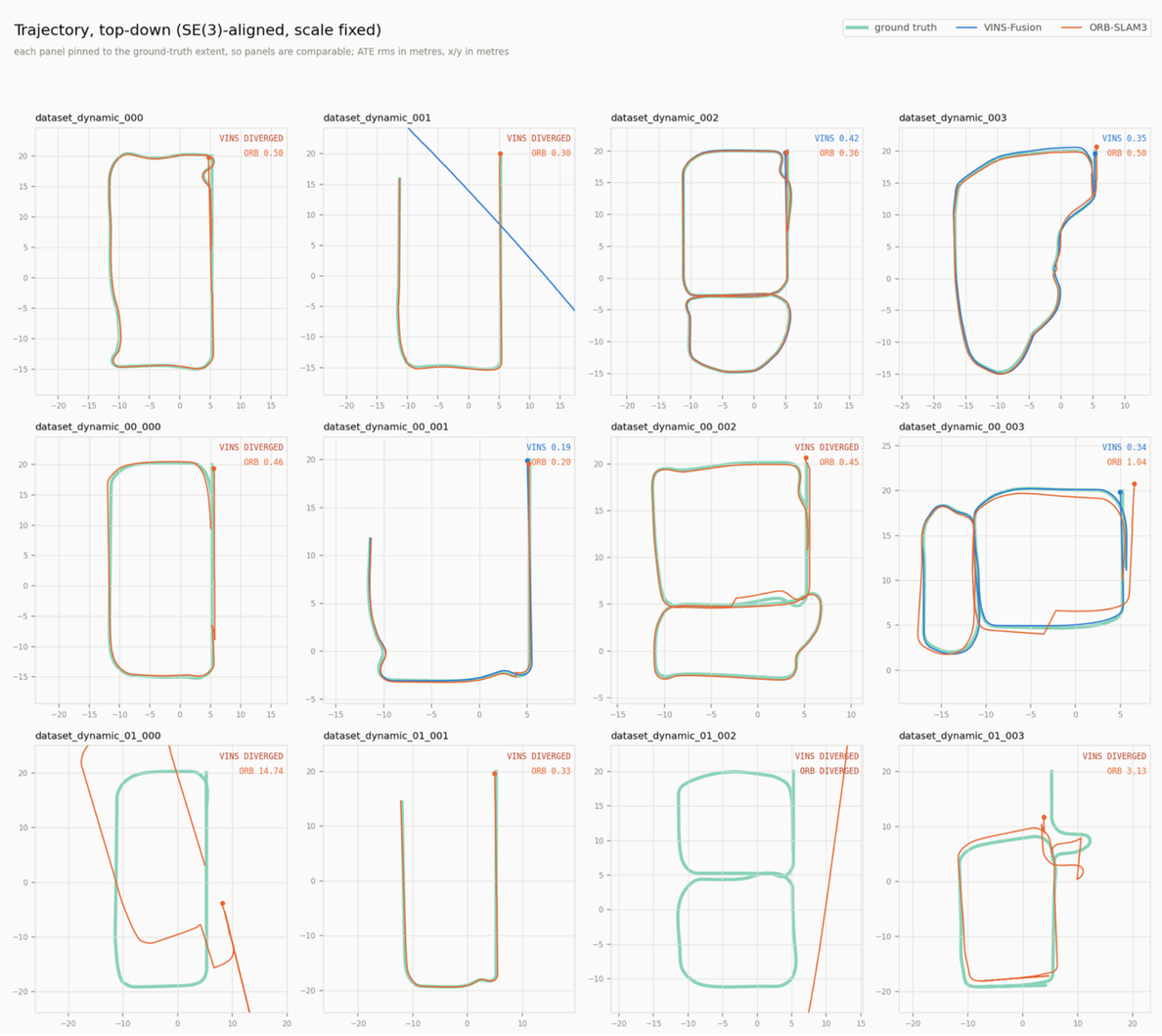


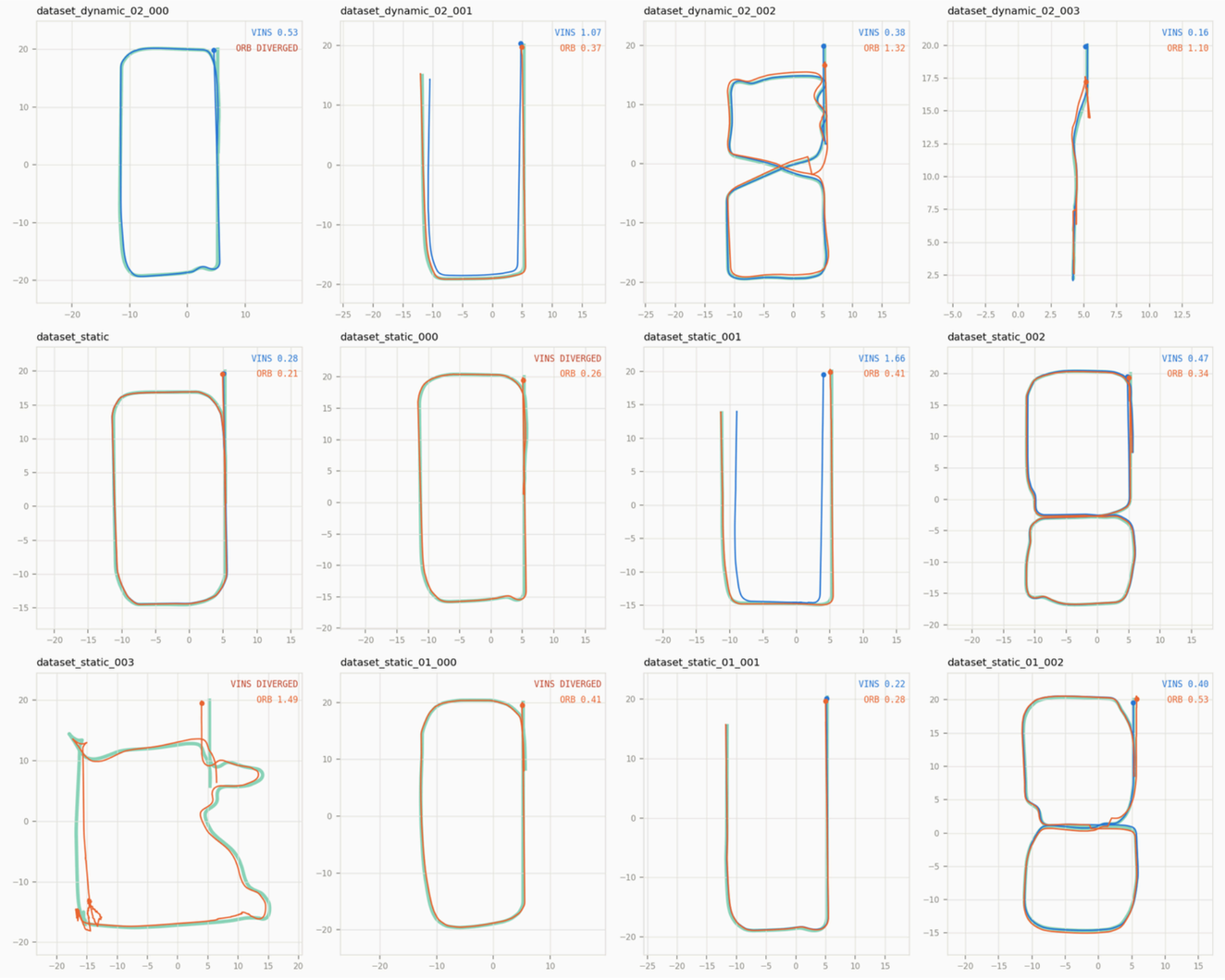






Simulation


Trajectory, top-down (SE(3)-aligned, scale fixed)
each panel pinned to the ground-truth extent, so panels are comparable; ATE rms in metres, x/y in metres
ground truth
VINS-Fusion
ORB-SLAM3
dataset_dynamic_000
VINS DIVERGED
ORB 0.50
dataset_dynamic_001
VINS DIVERGED
ORB 0.30
dataset_dynamic_002
VINS 0.42
ORB 0.36
dataset_dynamic_003
VINS 0.35
ORB 0.50
dataset_dynamic_00_000
VINS DIVERGED
ORB 0.46
dataset_dynamic_00_001
VINS 0.19
ORB 0.20
dataset_dynamic_00_002
VINS DIVERGED
ORB 0.45
dataset_dynamic_00_003
VINS 0.34
ORB 1.04
dataset_dynamic_01_000
VINS DIVERGED
ORB 14.74
dataset_dynamic_01_001
VINS DIVERGED
ORB 0.33
dataset_dynamic_01_002
VINS DIVERGED
ORB DIVERGED
dataset_dynamic_01_003
VINS DIVERGED
ORB 3.13


dataset_dynamic_02_000
VINS 0.53
ORB DIVERGED
dataset_dynamic_02_001
VINS 1.07
ORB 0.37
dataset_dynamic_02_002
VINS 0.38
ORB 1.32
dataset_dynamic_02_003
VINS 0.16
ORB 1.10
dataset_static
VINS 0.28
ORB 0.21
dataset_static_000
VINS DIVERGED
ORB 0.26
dataset_static_001
VINS 1.66
ORB 0.41
dataset_static_002
VINS 0.47
ORB 0.34
dataset_static_003
VINS DIVERGED
ORB 1.49
dataset_static_01_000
VINS DIVERGED
ORB 0.41
dataset_static_01_001
VINS 0.22
ORB 0.28
dataset_static_01_002
VINS 0.40
ORB 0.53

Simulation

VINS-Fusion vs ORB-SLAM3 -- all datasets

trans = position ATE rms in metres against ground truth, SE(3)-aligned, scale fixed. rot = orientation error rms in degrees (geodesic). DIVERGED = peak position error > 50 m.

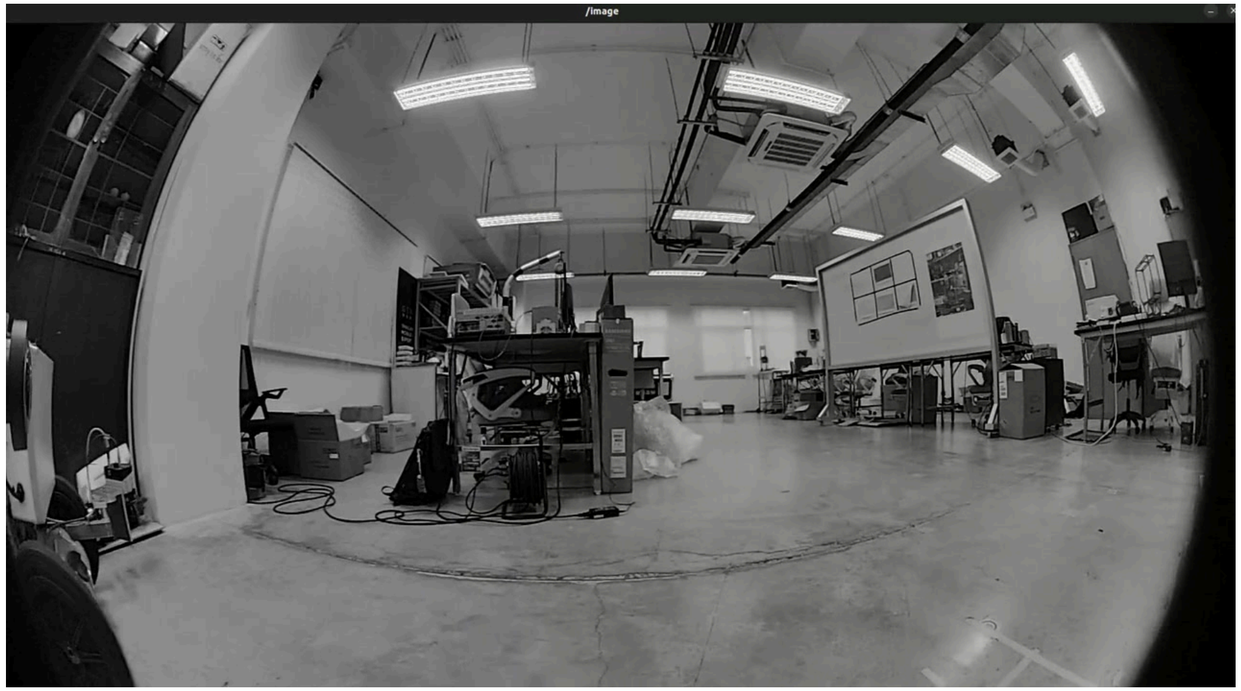
	VINS-Fusion				ORB-SLAM3						
dataset	trans	rot	clean	jmp	trans	rot	clean	jmp	secs	path m	GT m
REAL											
rosbag_realsense_imu_cambaseline9…	reference	--	--	1026	954.241	176.02	954.241	0	93	3446.5	--
rosbag_realsense_imu_cambaseline9…	reference	--	--	2	81.590	169.80	81.590	0	85	310.5	--
SIMULATION											
dataset_dynamic_000	DIVERGED	86.49	1.274	952	0.505	18.75	0.481	1	80	4520.2	116.4
dataset_dynamic_001	DIVERGED	166.79	8.639	604	0.299	19.47	0.180	2	66	625.0	80.8
dataset_dynamic_002	0.425	2.00	0.425	0	0.364	15.69	0.359	2	94	134.2	133.4
dataset_dynamic_003	0.351	1.87	0.351	0	0.498	17.81	0.164	4	71	102.6	101.7
dataset_dynamic_00_000	DIVERGED	140.97	5.251	1272	0.461	16.32	0.452	1	87	25185.7	110.7
dataset_dynamic_00_001	0.193	1.72	0.193	0	0.198	24.61	0.129	2	39	52.8	52.5
dataset_dynamic_00_002	DIVERGED	8.57	0.620	1265	0.452	16.77	0.331	3	81	5020.2	115.8
dataset_dynamic_00_003	0.339	1.54	0.339	0	1.037	17.58	0.333	2	73	106.5	105.6
dataset_dynamic_01_000	DIVERGED	125.93	1.687	1793	14.740	23.02	0.221	6	91	8879.4	132.7
dataset_dynamic_01_001	DIVERGED	12.50	21.427	785	0.333	18.74	0.322	2	61	1348.0	86.6
dataset_dynamic_01_002	DIVERGED	73.49	0.490	1647	DIVERGED	107.28	1.473	559	103	1716.6	127.4
dataset_dynamic_01_003	DIVERGED	122.91	1.714	2846	3.128	14.61	0.205	3	95	50302.8	131.4
dataset_dynamic_02_000	0.533	2.72	0.533	0	DIVERGED	49.93	0.020	1294	85	123.8	123.2
dataset_dynamic_02_001	1.074	7.08	1.074	0	0.371	16.61	0.285	3	61	84.0	86.6
dataset_dynamic_02_002	0.378	0.94	0.378	0	1.322	15.12	0.335	2	106	137.3	135.2
dataset_dynamic_02_003	0.155	1.97	0.155	0	1.101	45.91	0.170	6	38	24.8	23.6
dataset_static	--	--	--	--	0.213	17.12	0.213	0	85	107.6	106.9
dataset_static_000	DIVERGED	67.59	1.433	1070	0.257	17.13	0.257	0	89	3119.8	117.3
dataset_static_001	1.658	4.57	1.658	0	0.410	29.19	0.211	3	58	74.2	77.5
dataset_static_002	0.472	1.09	0.472	0	0.343	15.92	0.343	0	102	140.2	139.7
dataset_static_003	DIVERGED	96.01	11.233	113	1.487	14.50	0.443	18	125	505.6	153.1
dataset_static_01_000	DIVERGED	150.10	1.577	836	0.410	31.45	0.275	5	101	17782.4	116.4
dataset_static_01_001	0.215	1.30	0.215	0	0.279	19.33	0.243	1	64	87.3	87.6
dataset_static_01_002	0.395	0.87	0.395	0	0.531	21.16	0.438	3	102	132.4	131.6
dataset_static_01_003	0.346	1.70	0.346	0	0.398	14.87	0.215	2	102	133.0	131.5

trans: green < 1 m amber >= 1 m red = the run broke down grey = no ground truth, so the number grades nothing

'--' under a system means it was not run on that dataset. 'clean' is the longest jump-free segment (translational); 'jmp' counts steps implying >10 m/s. 'rot' is not colour-graded.

	static (n=8)	dynamic (n=8)
VINS diverged	3/8 (37.5%)	6/8 (75%)
VINS trans, non-diverged (median)	0.382 m	0.371 m
VINS rot, non-diverged (median)	1.42°	1.50°
ORB diverged	0/8 (0%)	1/8 (12.5%)
ORB trans, non-diverged (median)	0.384 m	0.393 m
ORB rot, non-diverged (median)	3.20°	2.14°

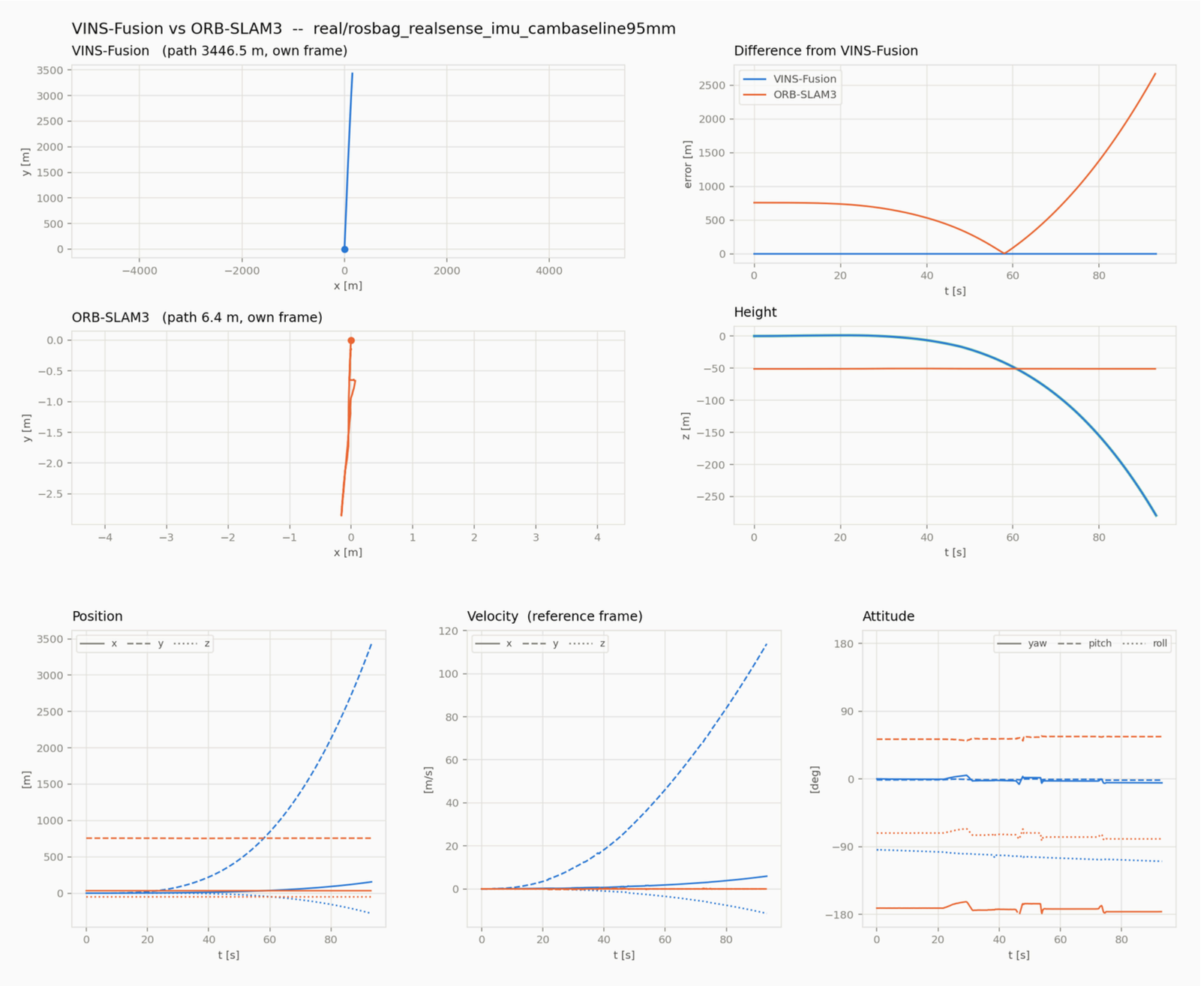
Real world



Straight line path

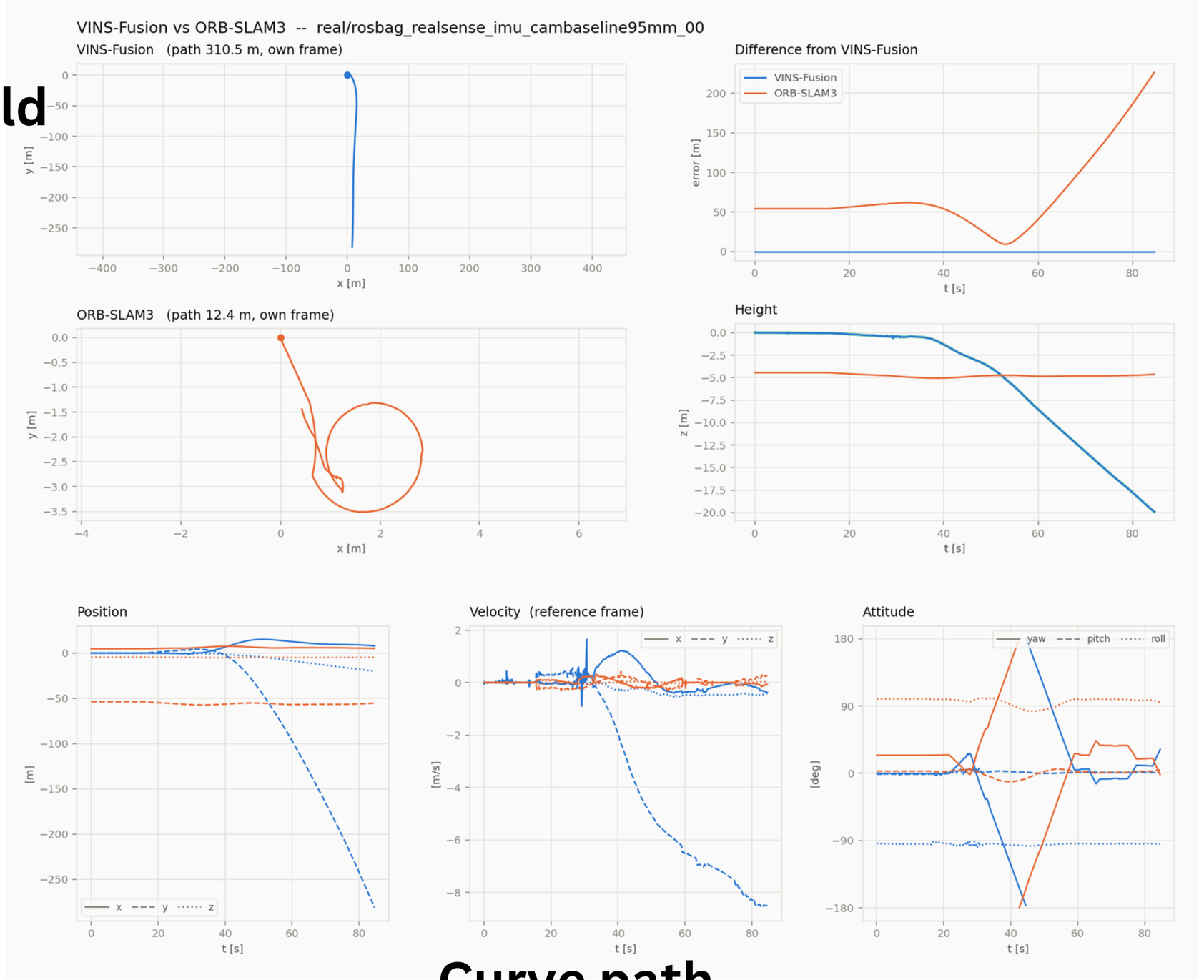
Curve path

Real world

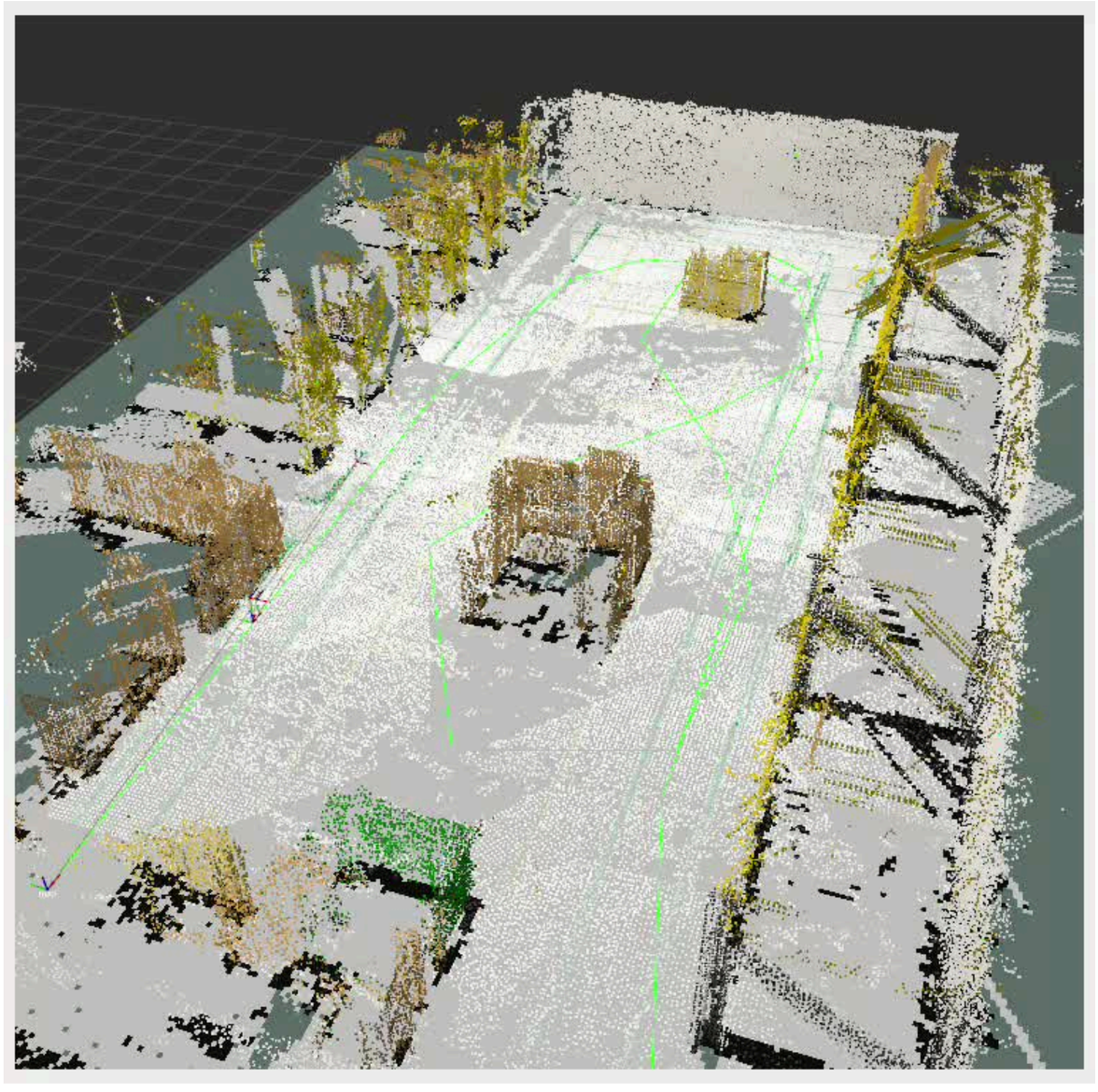


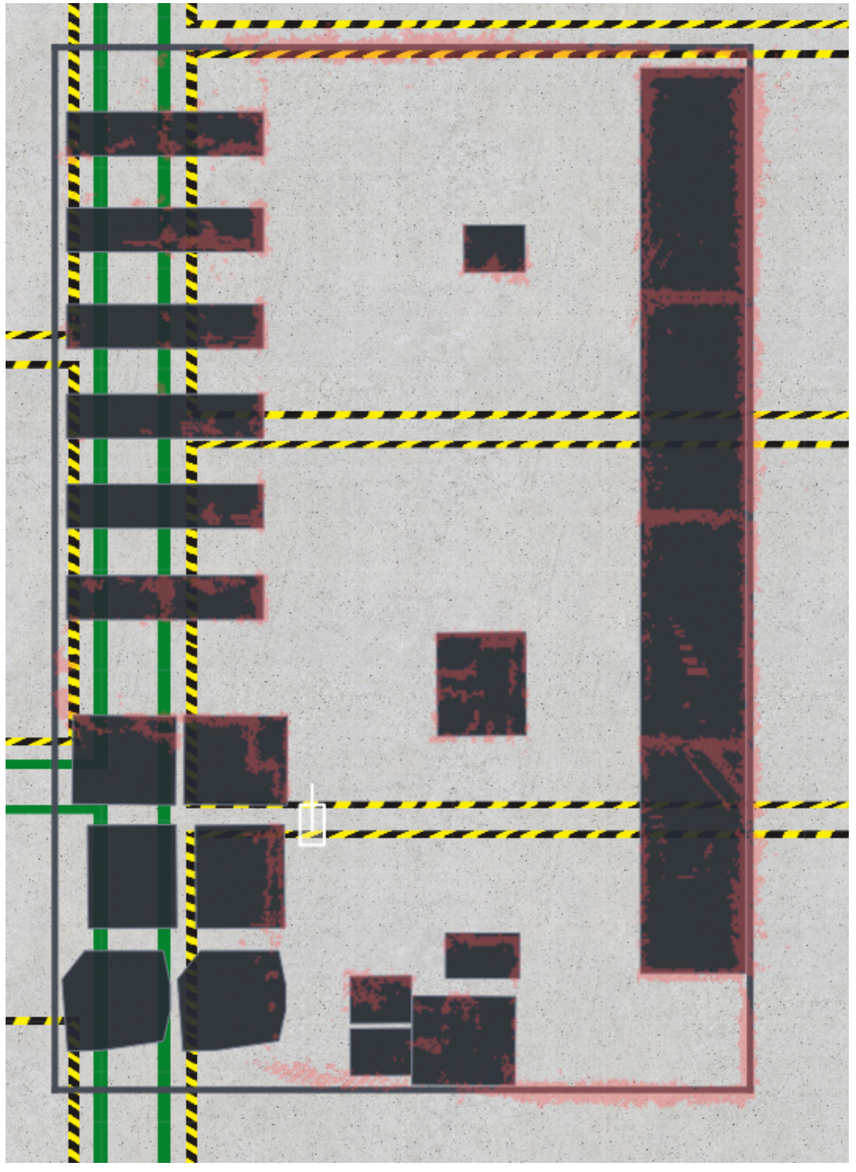
Straight line path

Real world

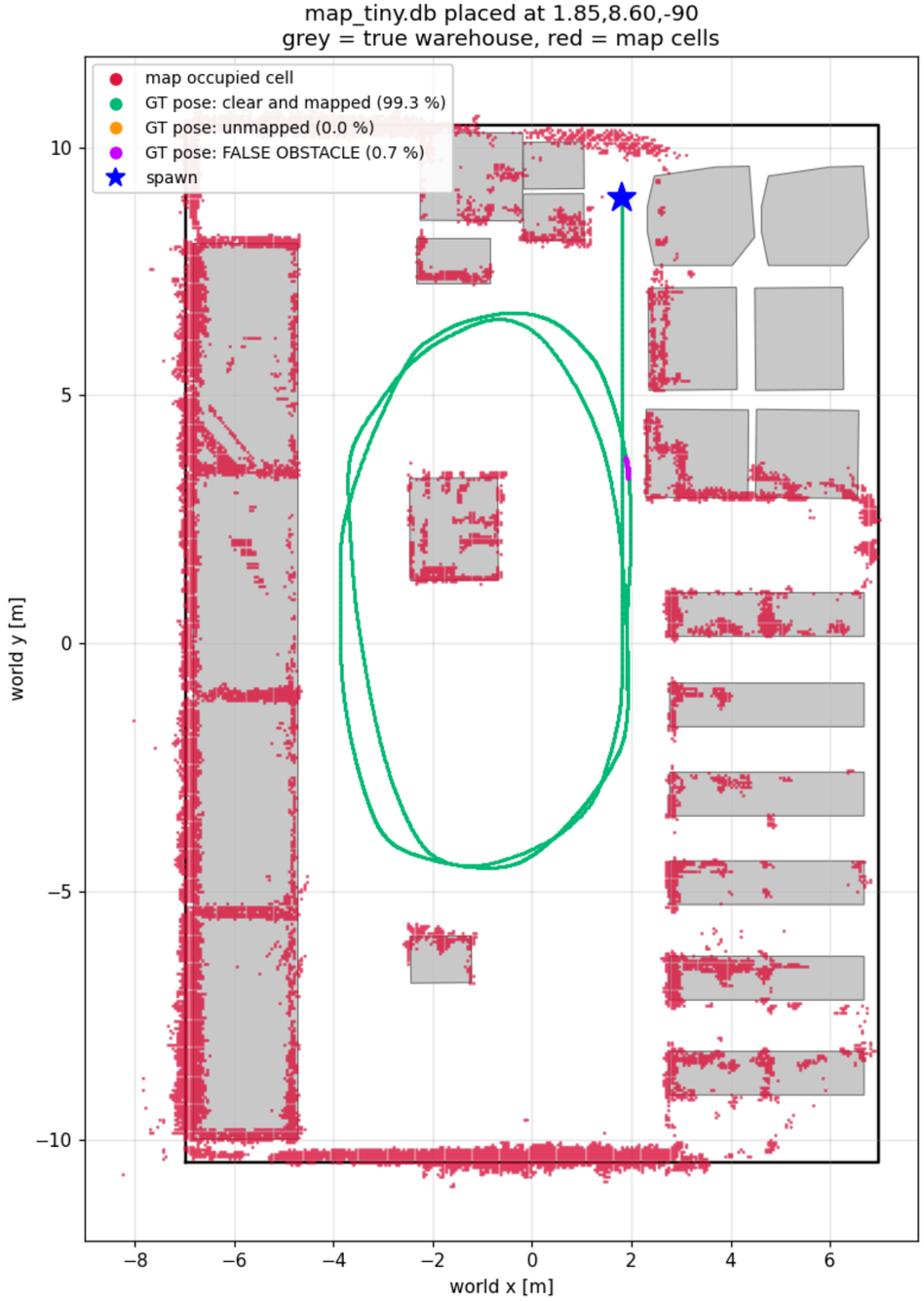


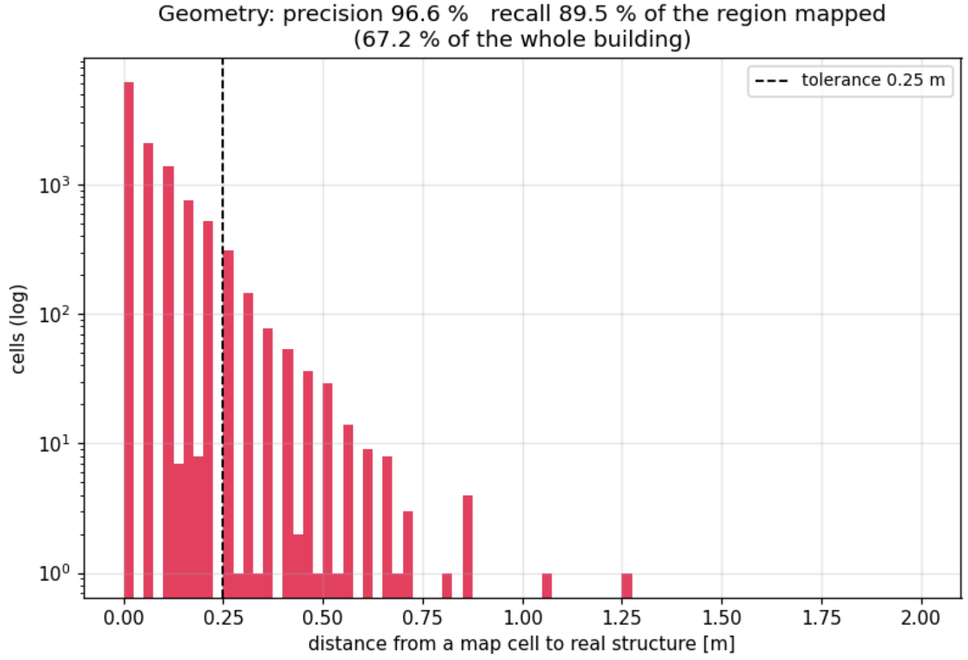
Curve path

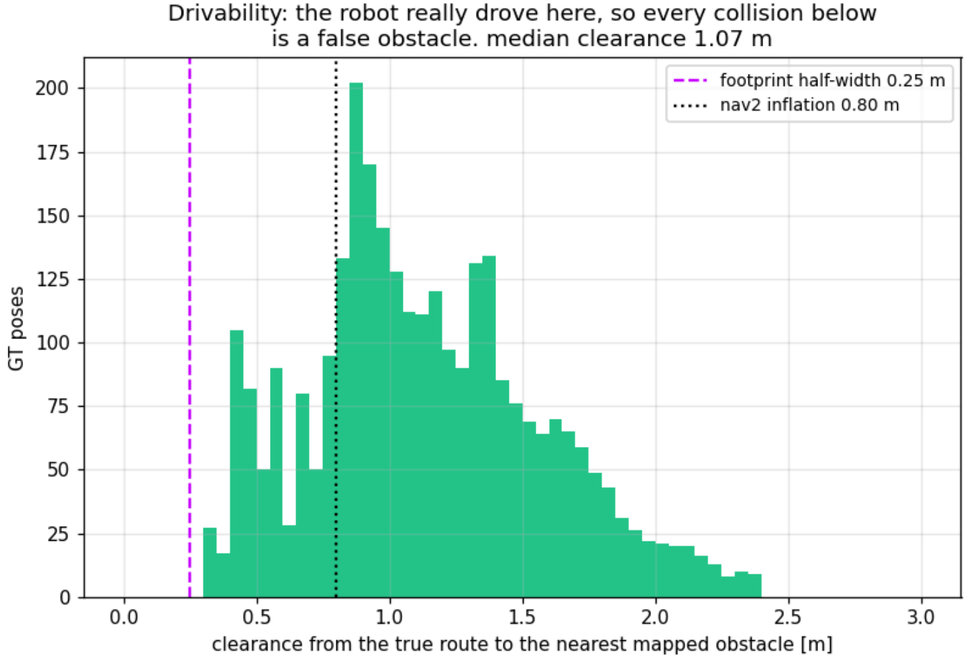




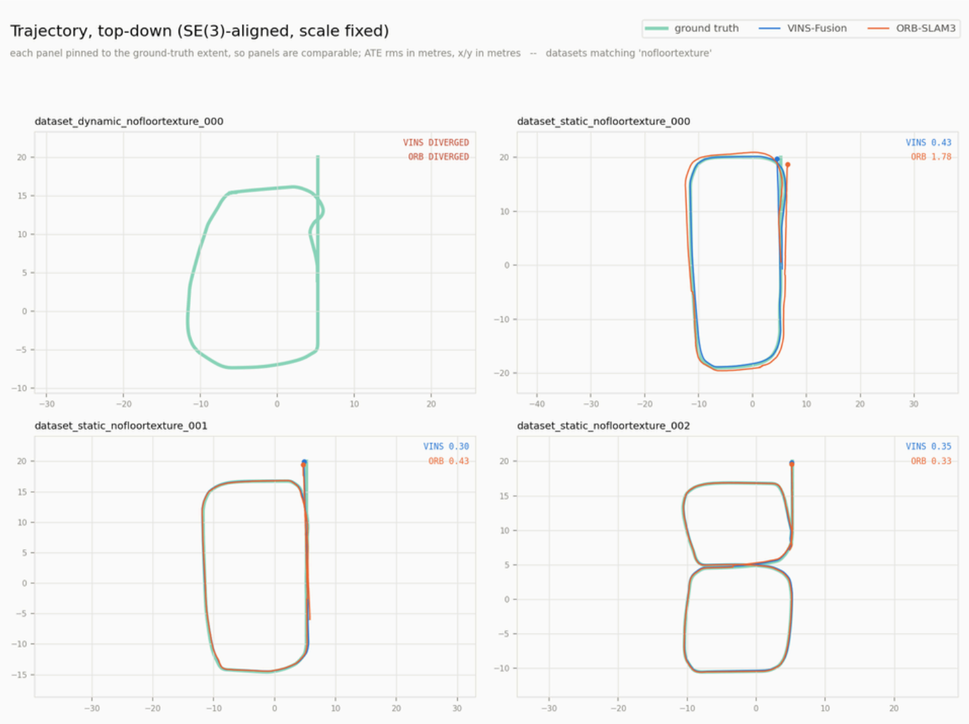
RTAB-Map database vs ground truth -- map_tiny.db vs gt_tiny.csv

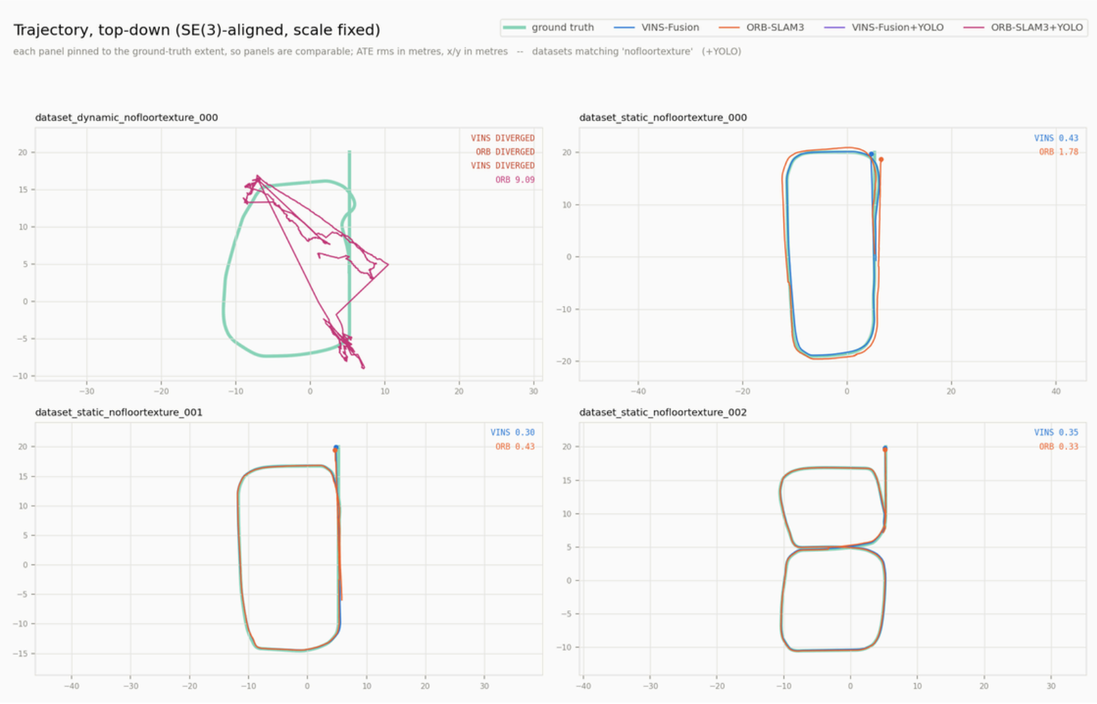






nofloortexture





nofloortexture

VINS-Fusion vs ORB-SLAM3 -- datasets matching 'nofloortexture'

trans = position ATE rms in metres against ground truth, SE(3)-aligned, scale fixed. rot = orientation error rms in degrees (geodesic). DIVERGED = peak position error > 50 m.

	VINS-Fusion				ORB-SLAM3						
dataset	trans	rot	clean	jmp	trans	rot	clean	jmp	secs	path m	GT m
SIMULATION											
dataset_dynamic_nofloortexture_000	DIVERGED	13.75	3.701	1312	DIVERGED	119.38	4.957	995	79	3302.6	89.0
dataset_static_nofloortexture_000	0.425	1.82	0.425	0	1.778	18.84	0.421	3	86	122.6	122.9
dataset_static_nofloortexture_001	0.300	2.01	0.300	0	0.431	17.51	0.380	1	82	113.6	113.1
dataset_static_nofloortexture_002	0.347	0.77	0.347	0	0.330	17.96	0.330	0	89	115.2	114.7

trans: green < 1 m amber >= 1 m red = the run broke down grey = no ground truth, so the number grades nothing

'--' under a system means it was not run on that dataset. 'clean' is the longest jump-free segment (translational); 'jmp' counts steps implying >10 m/s. 'rot' is not colour-graded.

VINS-Fusion vs ORB-SLAM3 vs VINS-Fusion+YOLO vs ORB-SLAM3+YOLO -- datasets matching 'nofloortexture'

trans = position ATE rms in metres against ground truth, SE(3)-aligned, scale fixed. rot = orientation error rms in degrees (geodesic). DIVERGED = peak position error > 50 m.

	VINS-Fusion				ORB-SLAM3				VINS-Fusion+YOLO				ORB-SLAM3+YOLO						
dataset	trans	rot	clean	jmp	trans	rot	clean	jmp	trans	rot	clean	jmp	trans	rot	clean	jmp	secs	path m	GT m
SIMULATION																			
dataset_dynamic_nofloortexture_000	DIVERGED	13.75	3.701	1312	DIVERGED	119.38	4.957	995	DIVERGED	5.48	7.697	1434	IMPLAUSIBLE	109.45	0.818	95	79	3302.6	89.0
dataset_static_nofloortexture_000	0.425	1.82	0.425	0	1.778	18.84	0.421	3	--	--	--	--	--	--	--	--	86	122.6	122.9
dataset_static_nofloortexture_001	0.300	2.01	0.300	0	0.431	17.51	0.380	1	--	--	--	--	--	--	--	--	82	113.6	113.1
dataset_static_nofloortexture_002	0.347	0.77	0.347	0	0.330	17.96	0.330	0	--	--	--	--	--	--	--	--	89	115.2	114.7

trans: green < 1 m amber >= 1 m red = the run broke down grey = no ground truth, so the number grades nothing

'--' under a system means it was not run on that dataset. 'clean' is the longest jump-free segment (translational); 'jmp' counts steps implying >10 m/s. 'rot' is not colour-graded.